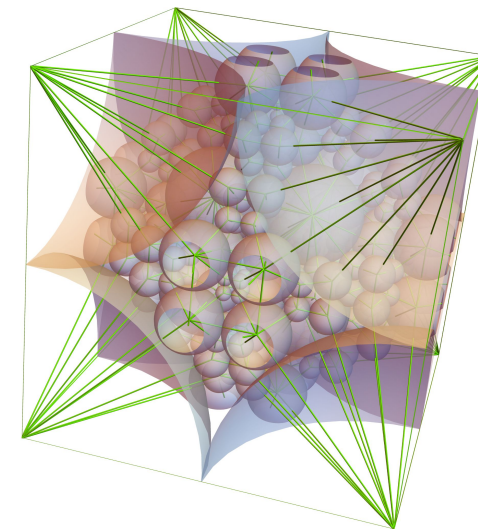
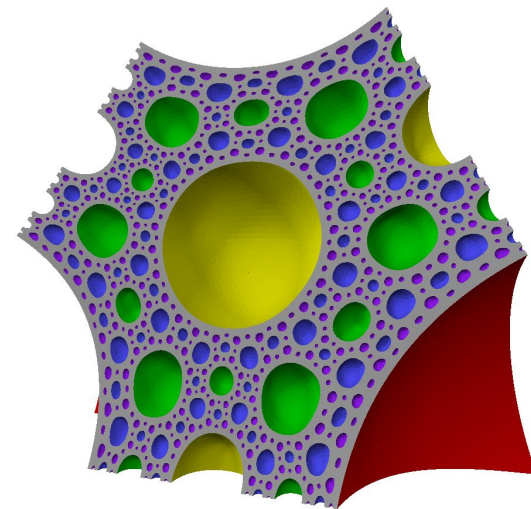


Automatisation de conception de structures fractales à l'aide du modèle BC-IFS



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DE BOURGOGNE

Boris Bordeaux, Christian Gentil, Lionel Garnier

LIB – Université de Bourgogne

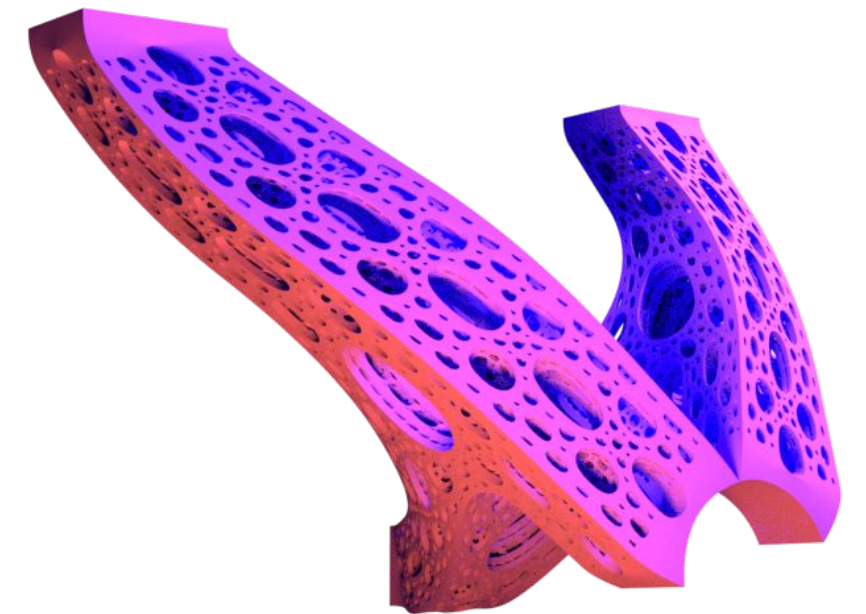
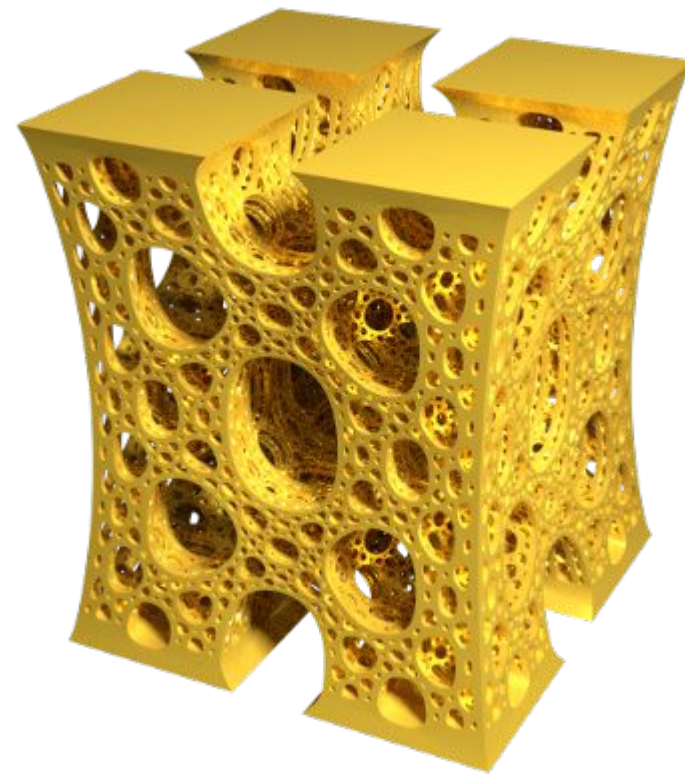
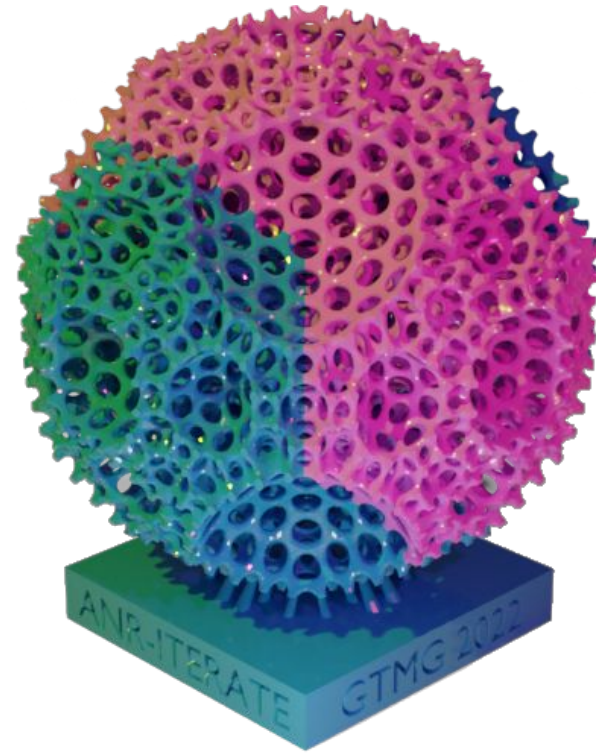
Projet ANR ITERATE – ANR-20-CE46-0011 (2021-2025)

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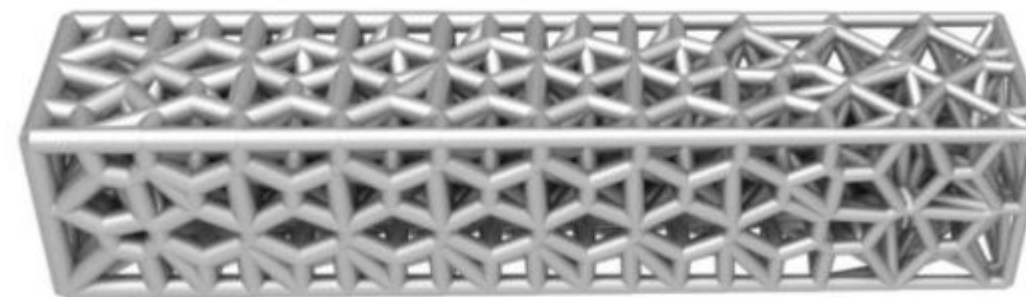
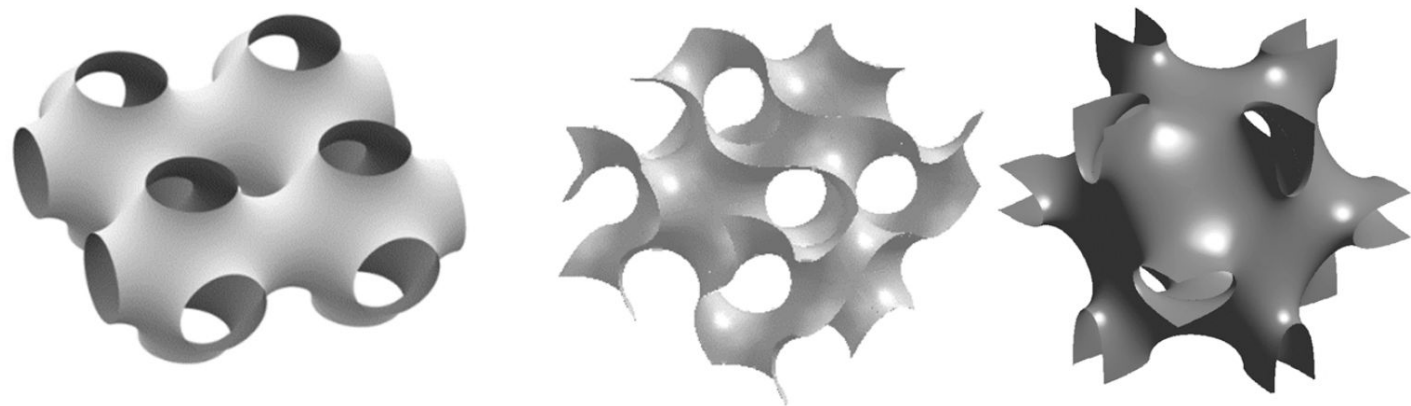
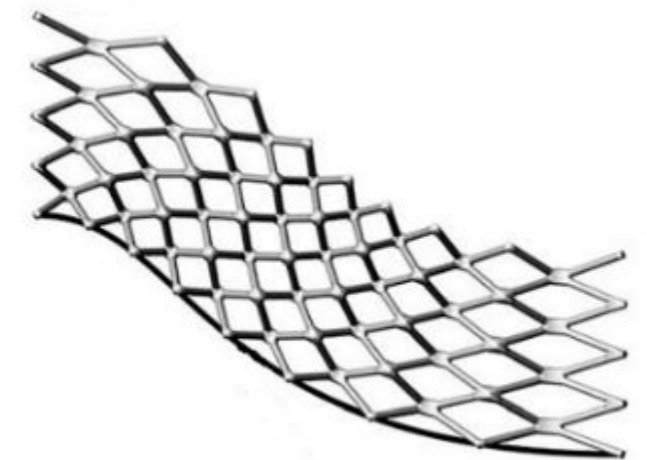
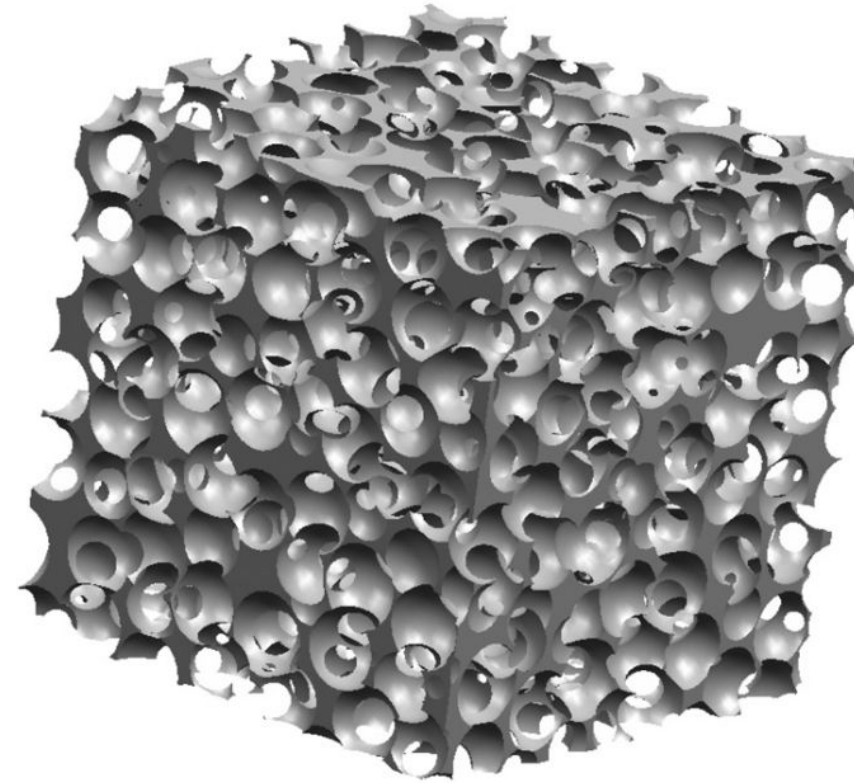
25 mai 2023, Séminaire LIB - Dijon

COLLÈGE DOCTORAL
VOTRE DOCTORAT
EN BOURGOGNE-
FRANCHE-COMTÉ

UBFC
UNIVERSITÉ
BOURGOGNE FRANCHE-COMTÉ



- Mousses
 - Aléatoire
- Structures lattices
 - Topologie régulière
 - Variation de géométrie
- Surfaces minimales
 - Triplement périodique
 - Géométrie contrainte



- Multi échelle
- Déterministe
- Répond à plusieurs propriétés physiques
 - Absorption mécanique
 - Thermique
 - Acoustique
- Généralise les structures lattices



- Algorithme de subdivision automatique en 2D
 - Modèle utilisé
 - Notations
 - Algorithme
 - Délai de subdivision
 - Conclusion
- Empilements polytopaux de sphères
 - Explications
 - Représentation avec le modèle BC-IFS
 - Extraction des règles de subdivision
 - Conclusion
- Conclusion

■ IFS

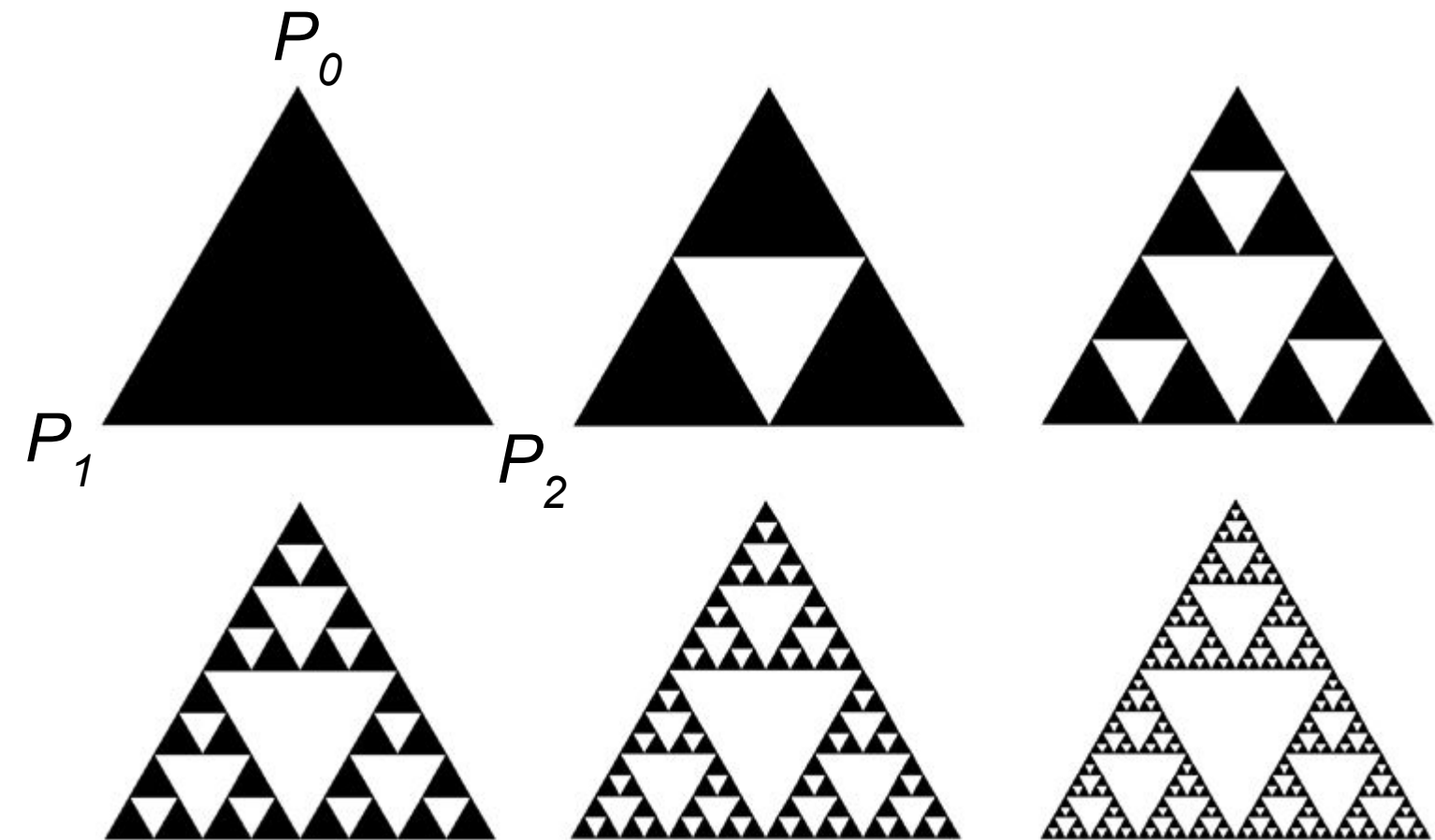
- Transformations contractantes
- Points de contrôles

$$T_0 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$T_1 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

$$T_2 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$S = T_0(S) \cup T_1(S) \cup T_2(S)$$



■ IFS

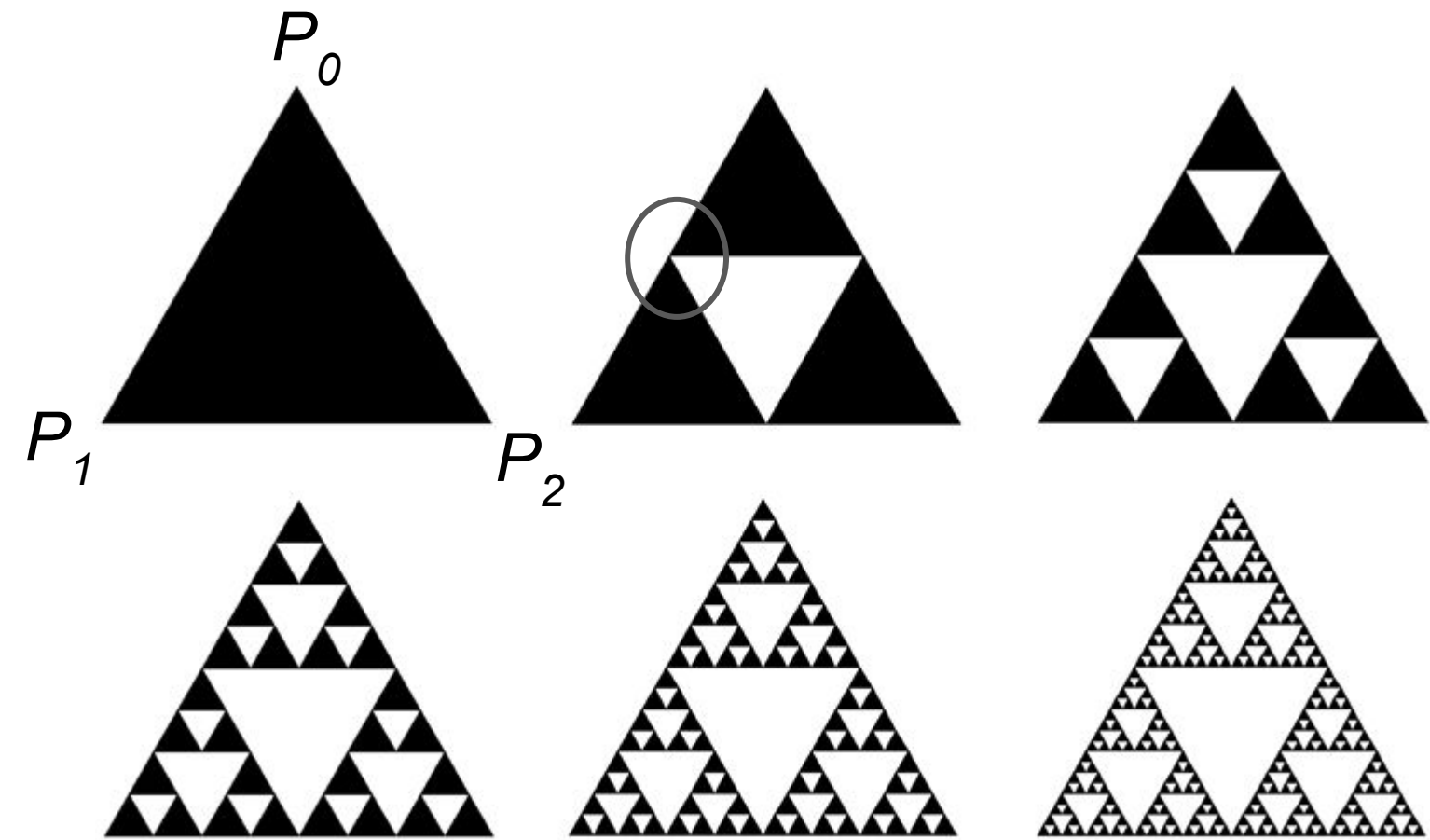
- Transformations contractantes
- Points de contrôles

$$T_0 = \begin{pmatrix} \boxed{\frac{1}{2}} & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

$$T_1 = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

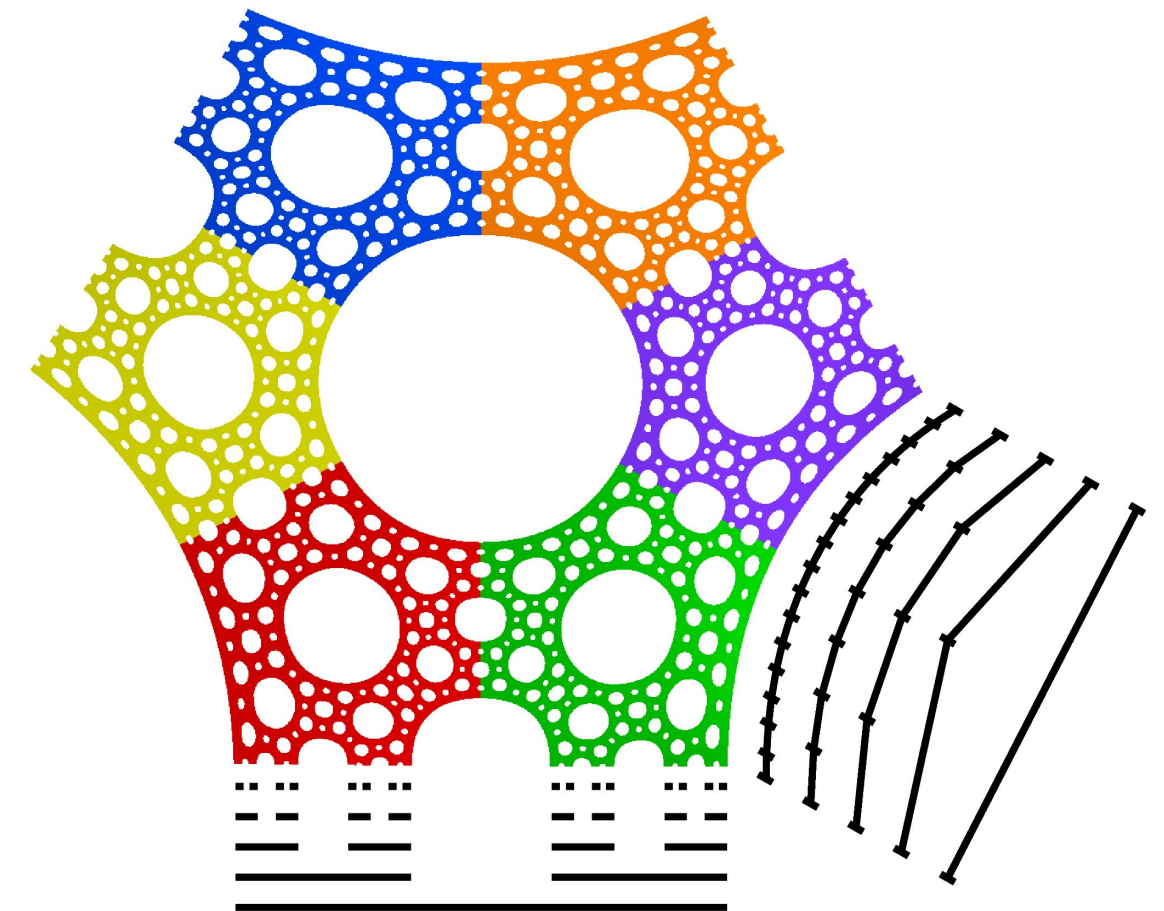
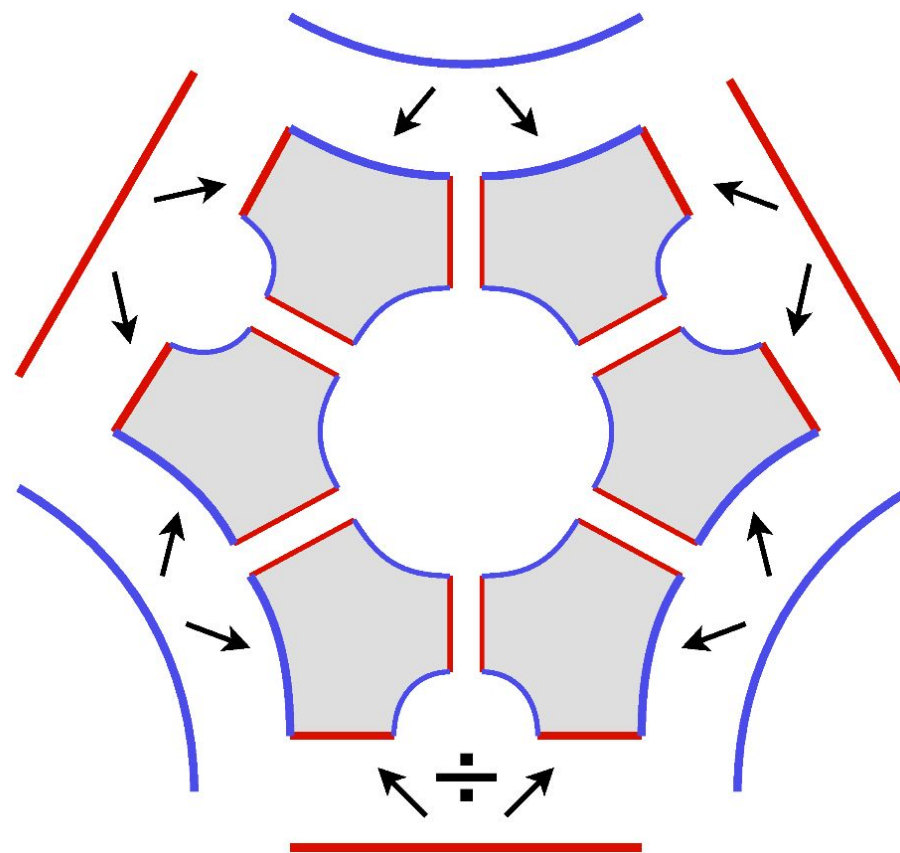
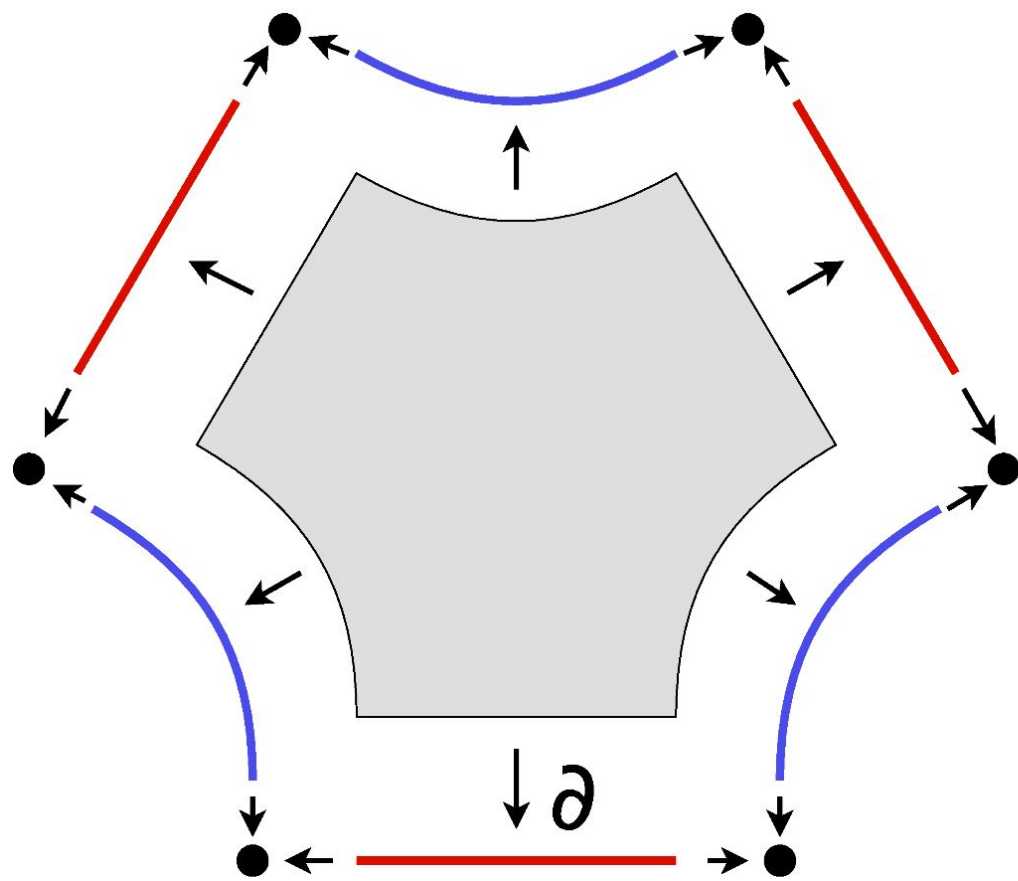
$$T_2 = \begin{pmatrix} 1 & \boxed{\frac{1}{2}} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

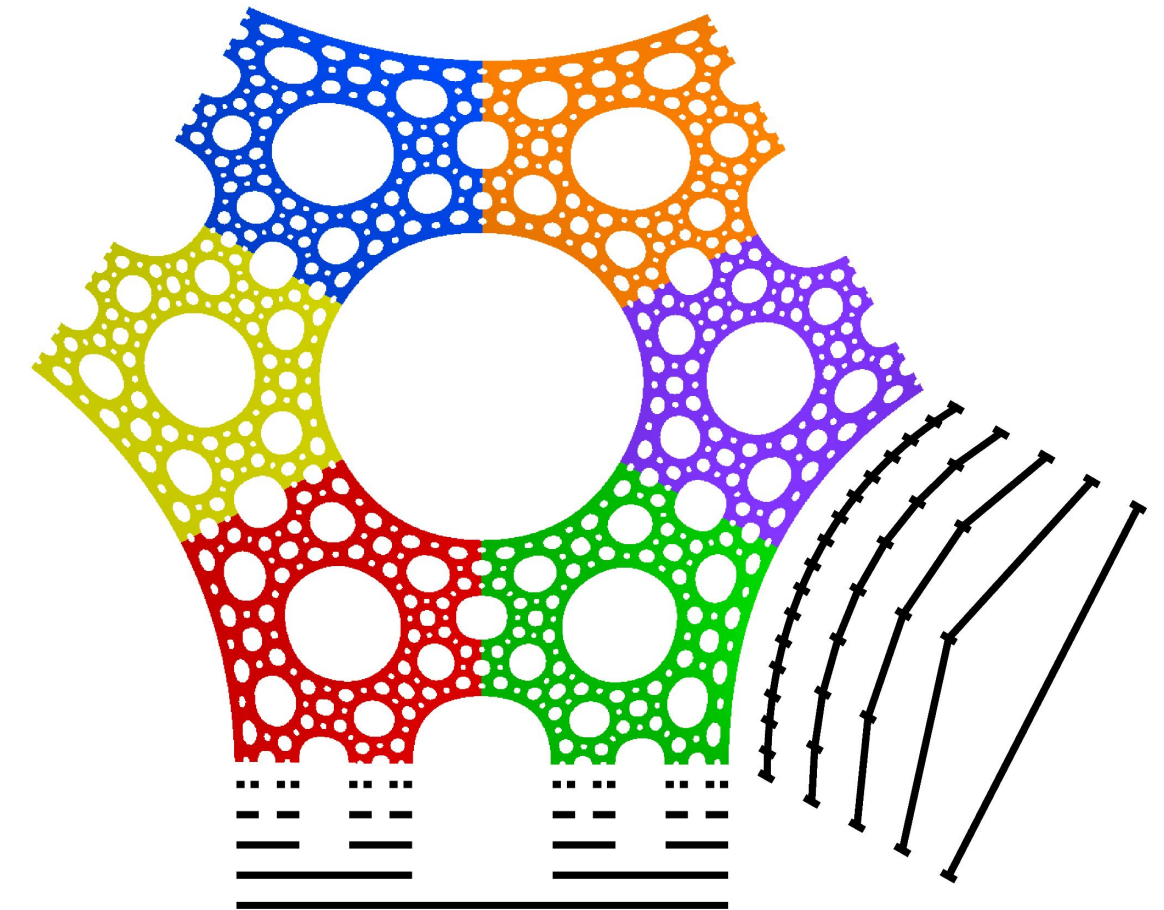
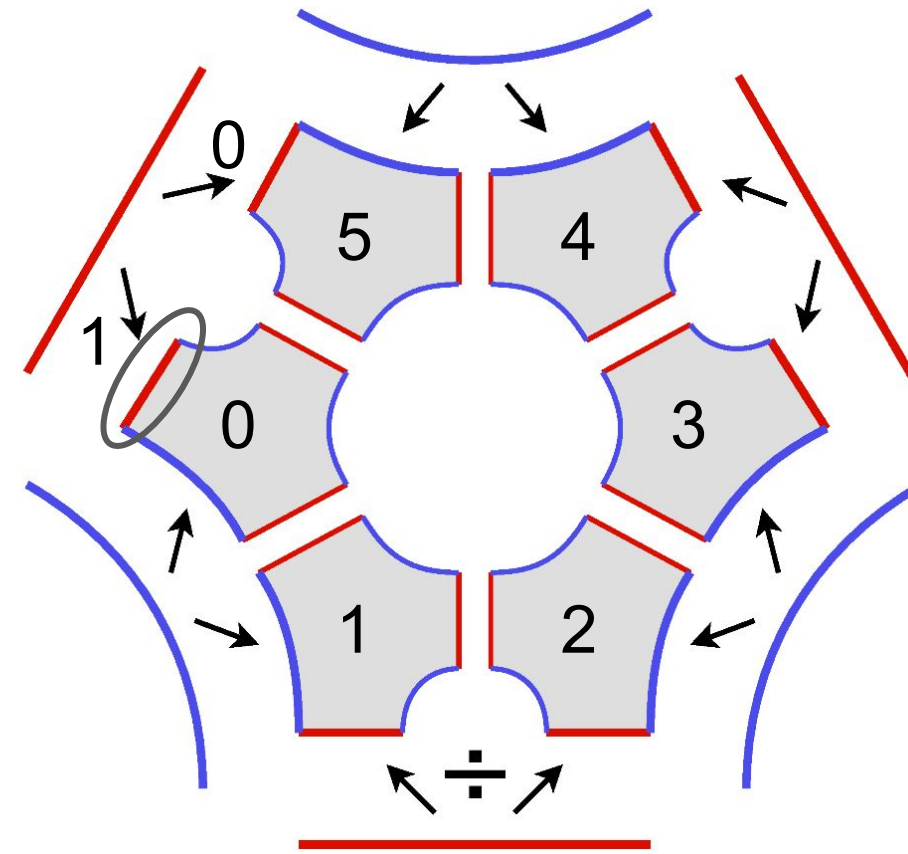
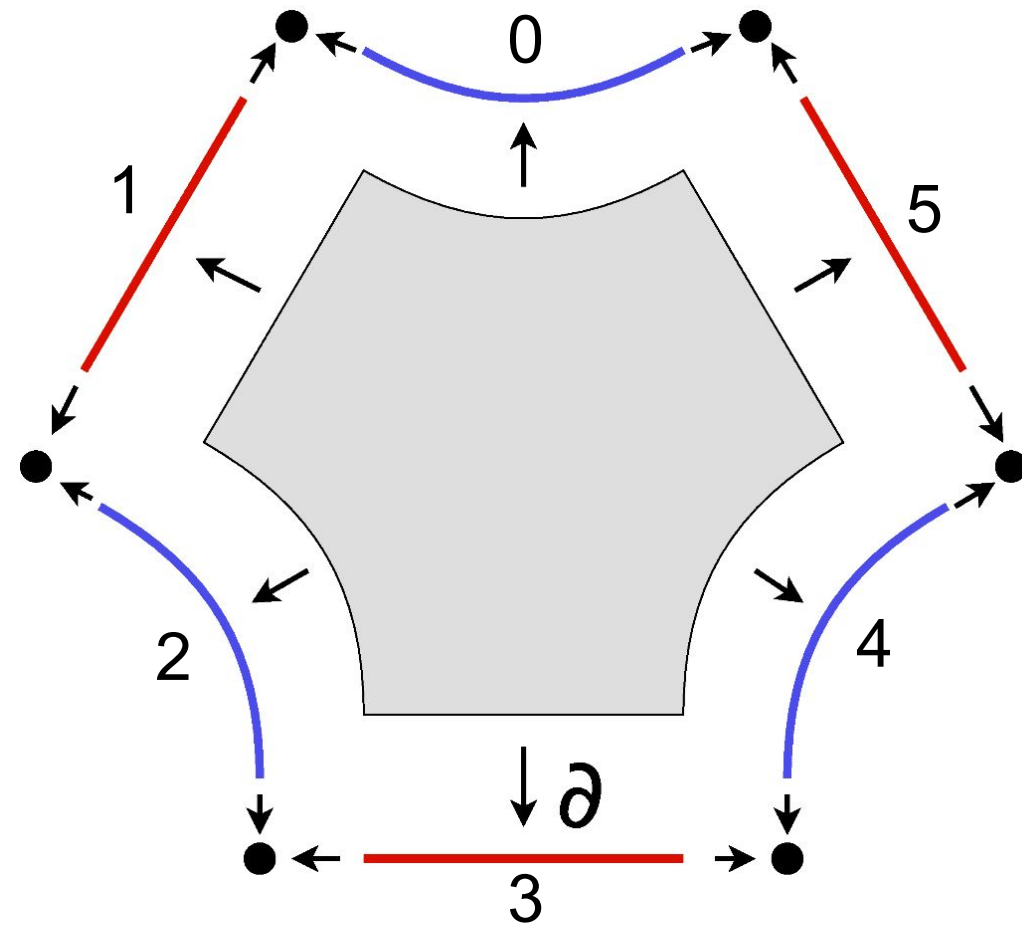
$$S = T_0(S) \cup T_1(S) \cup T_2(S)$$



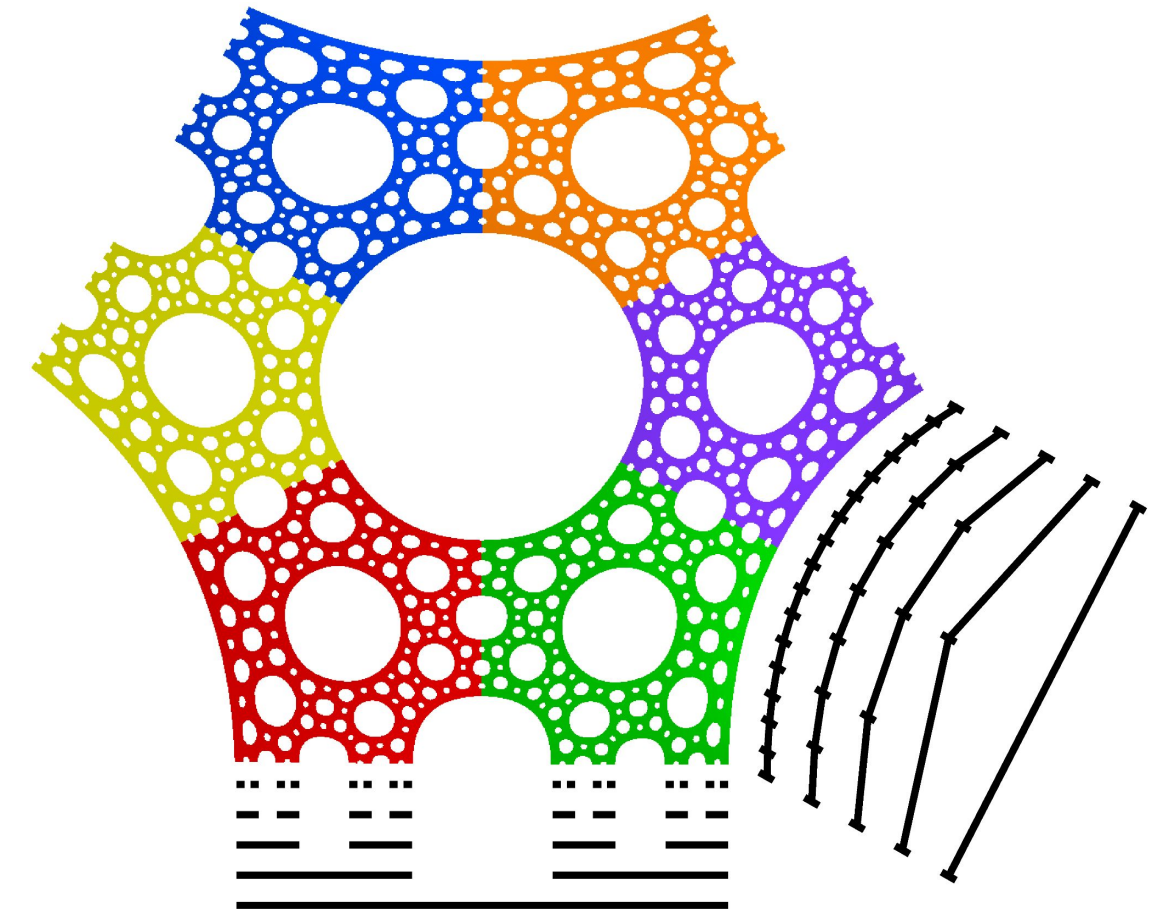
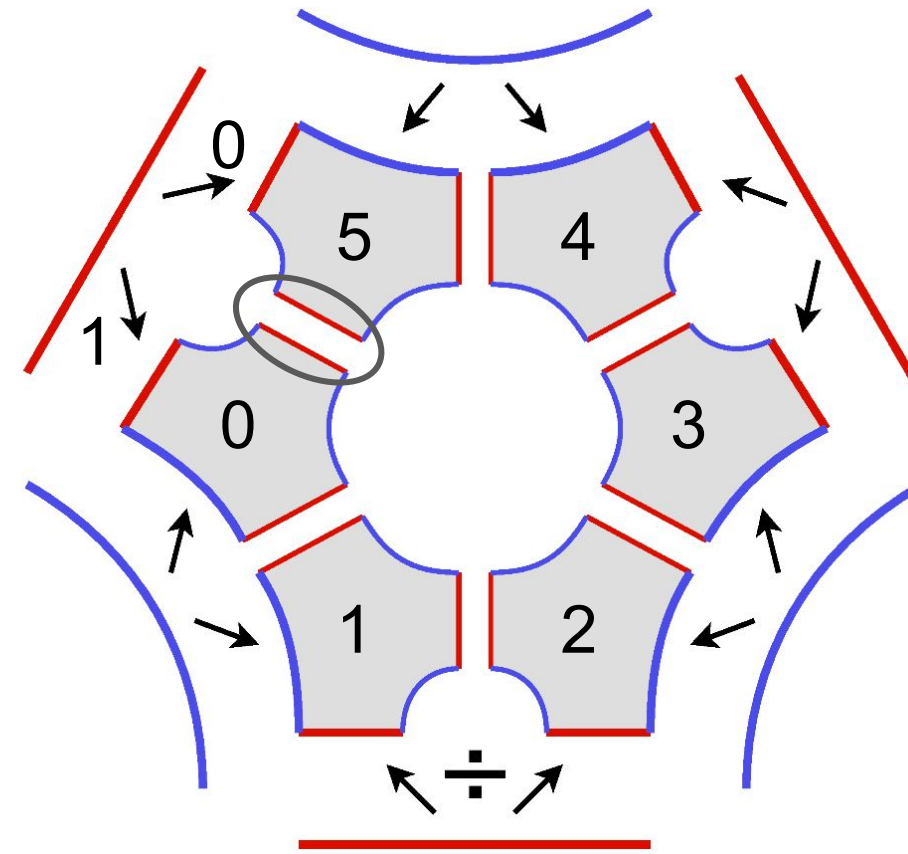
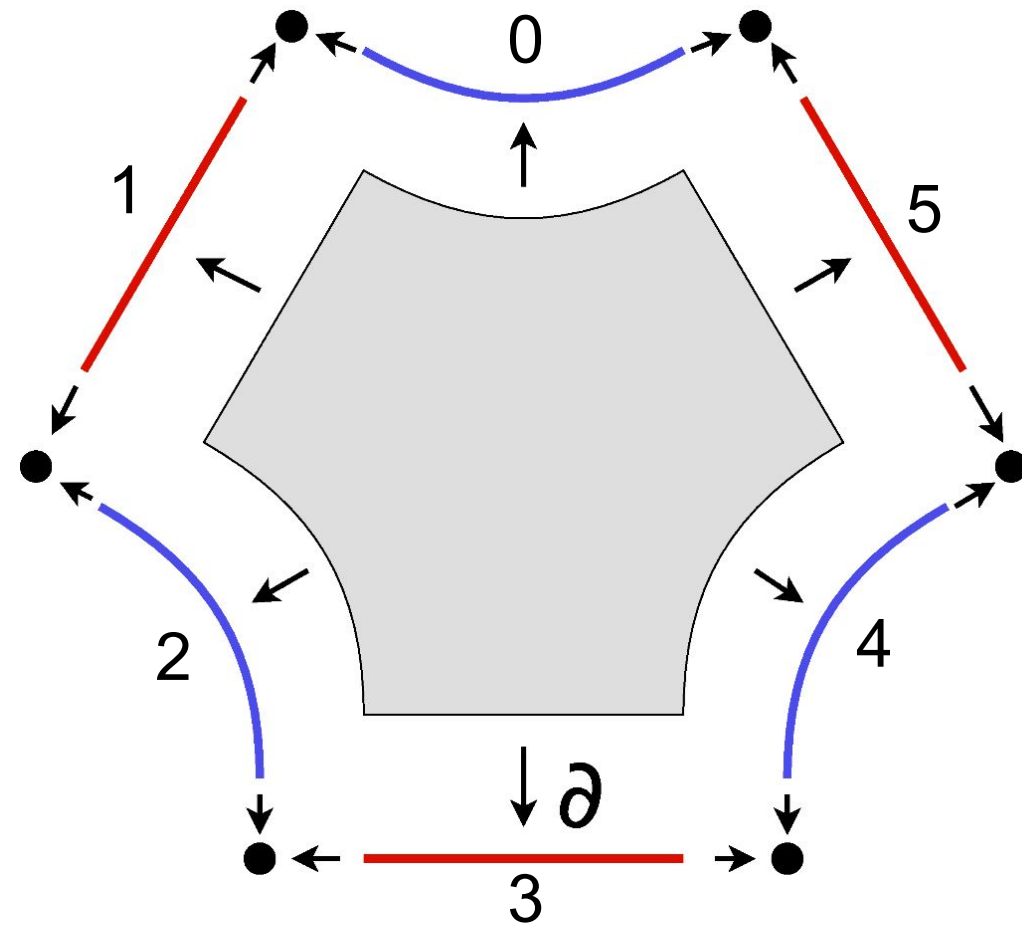
■ BC-IFS

- Décomposition cellulaire
- Contrôle de la topologie fractale
- Topologie cohérente



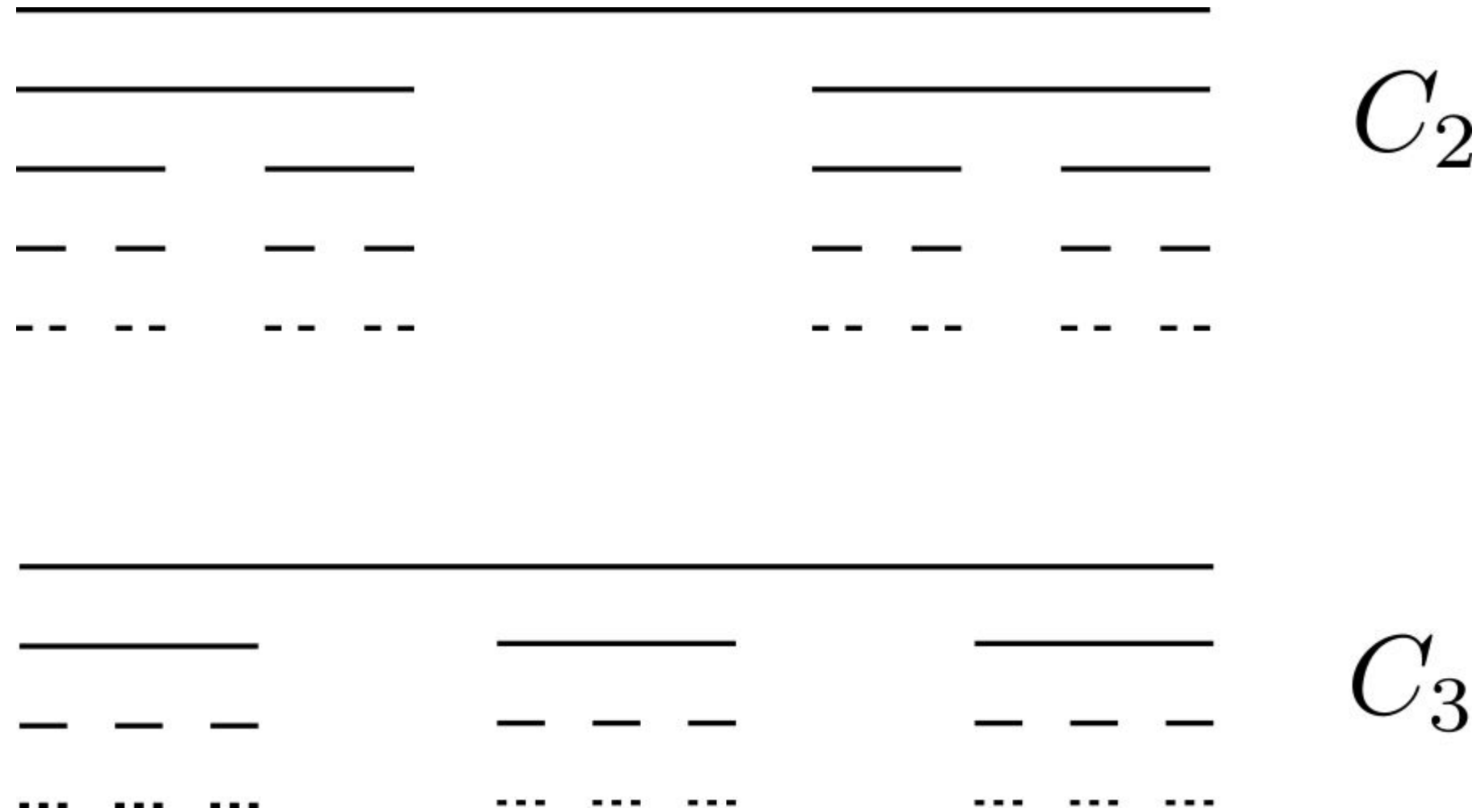


Incidence : $\text{Sub}(1) \text{ de } \text{Bord}(1) = \text{Bord}(1) \text{ de } \text{Sub}(0)$

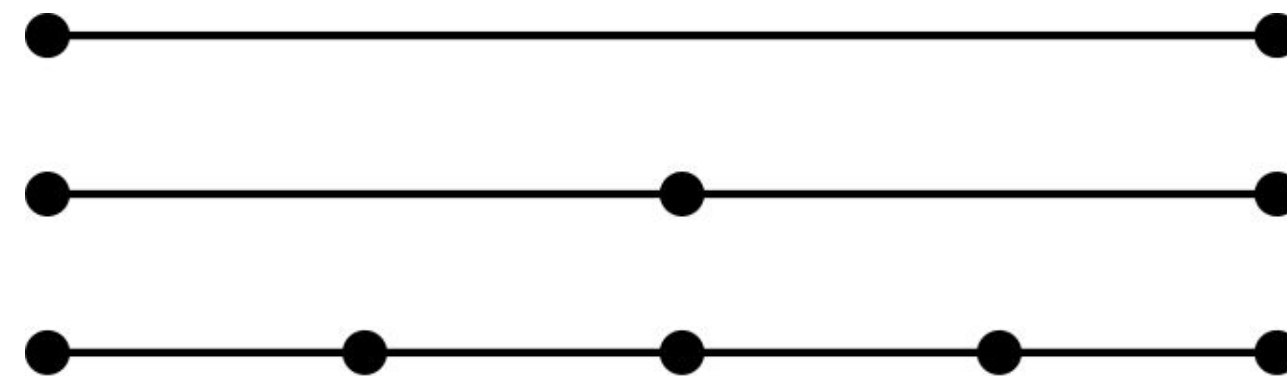


Adjacence : $\text{Bord}(5) \text{ de Sub}(0) = \text{Bord}(3) \text{ de Sub}(5)$

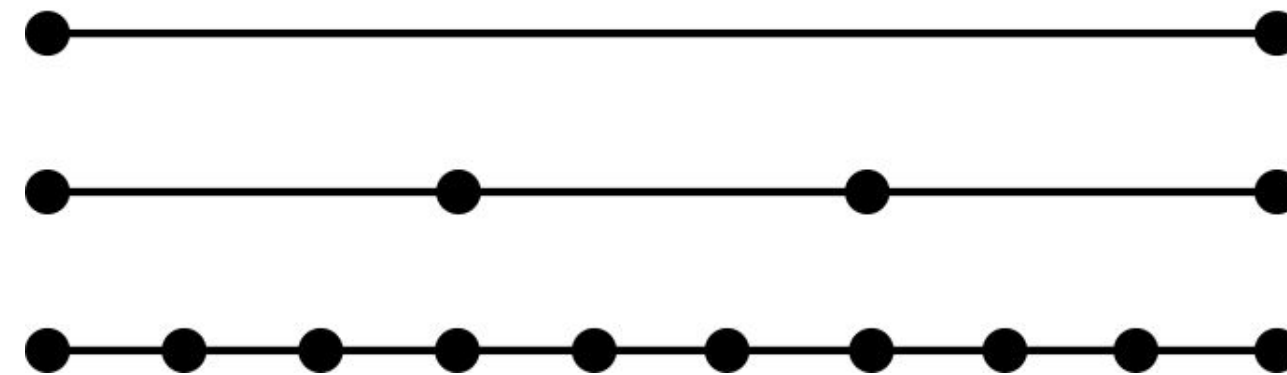
- Bords de type Cantor
 - Subdivisions non connexes de type Cantor



- Bords de type Courbe
 - Subdivisions connexes de type Courbe



B_2

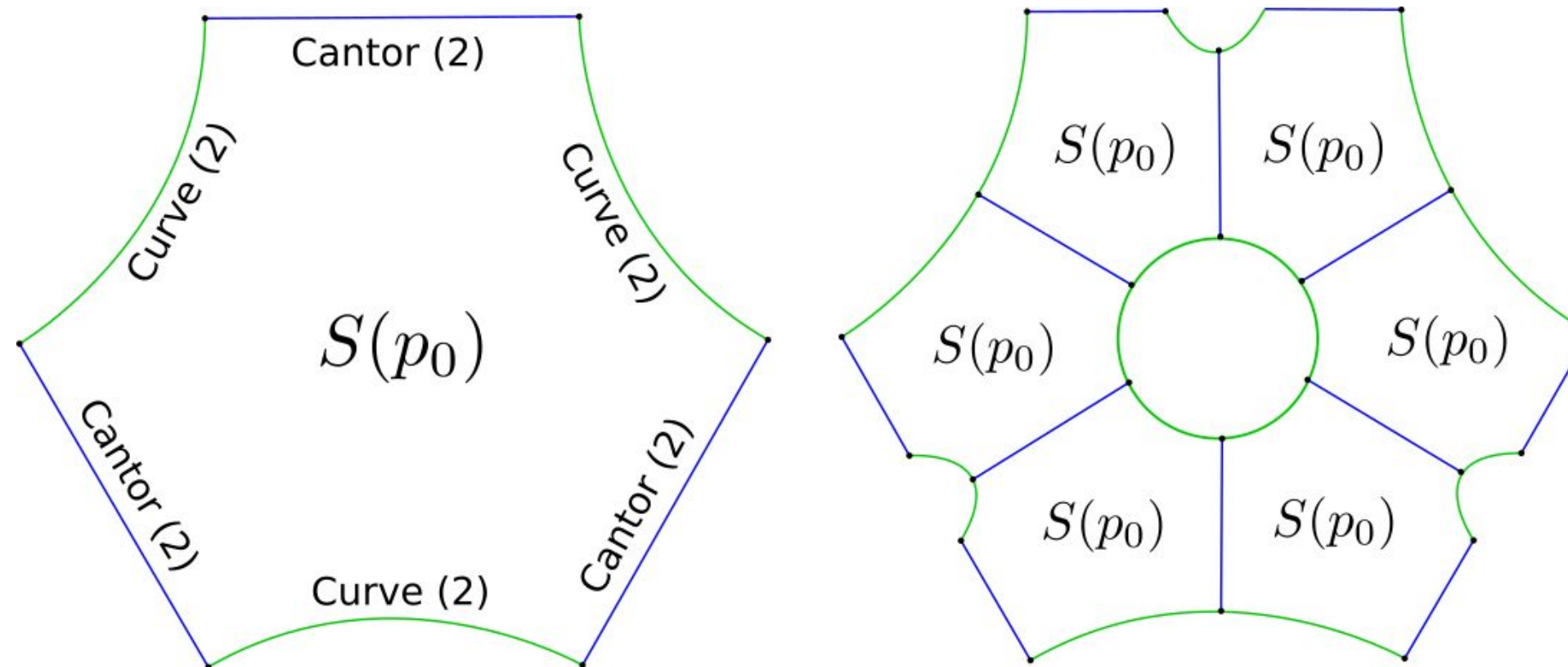


B_3

- Faces définies par leur bords
 - Ordre important à une permutation circulaire près

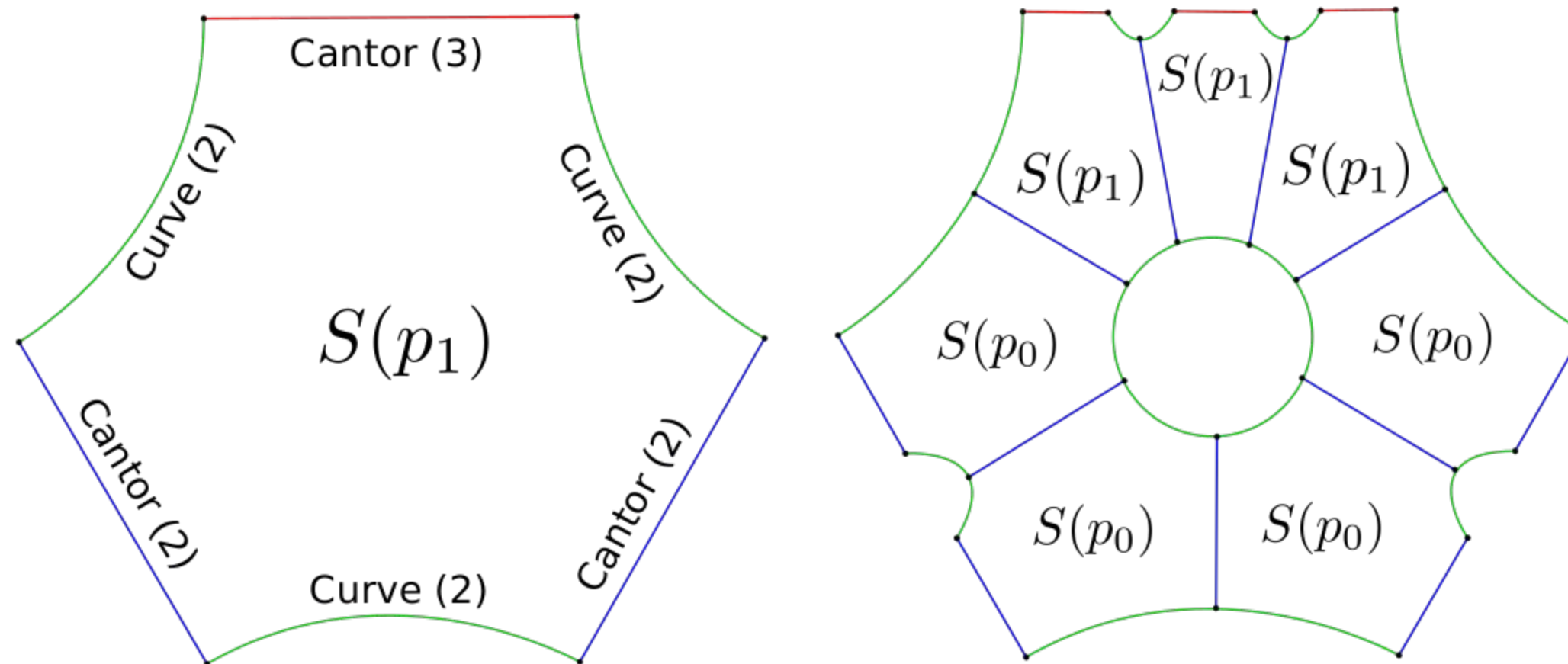
$$S(p), p = \{X_n, X_n, X_n, \dots\}$$

$$S(p_0), p_0 = \{C_2, B_2, C_2, B_2, C_2, B_2\}$$

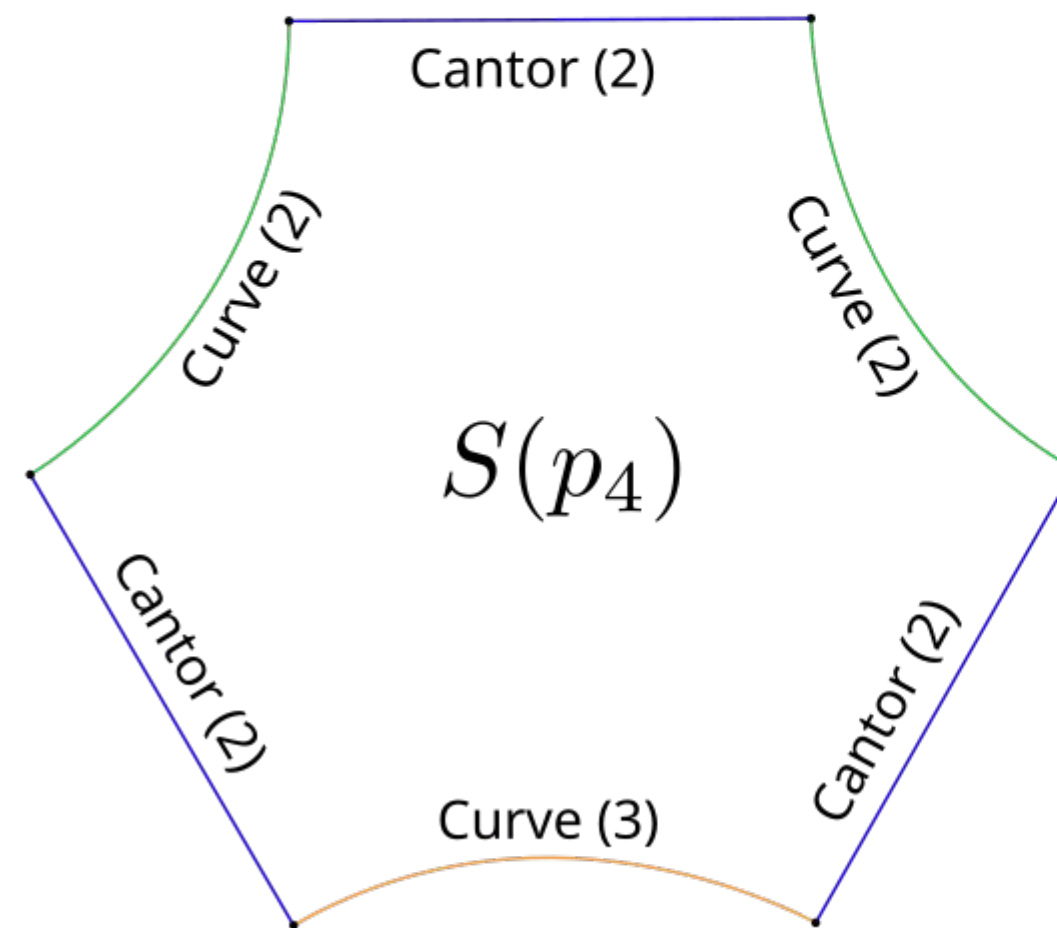


$$p_0 = \{C_2, B_2, C_2, B_2, C_2, B_2\}$$

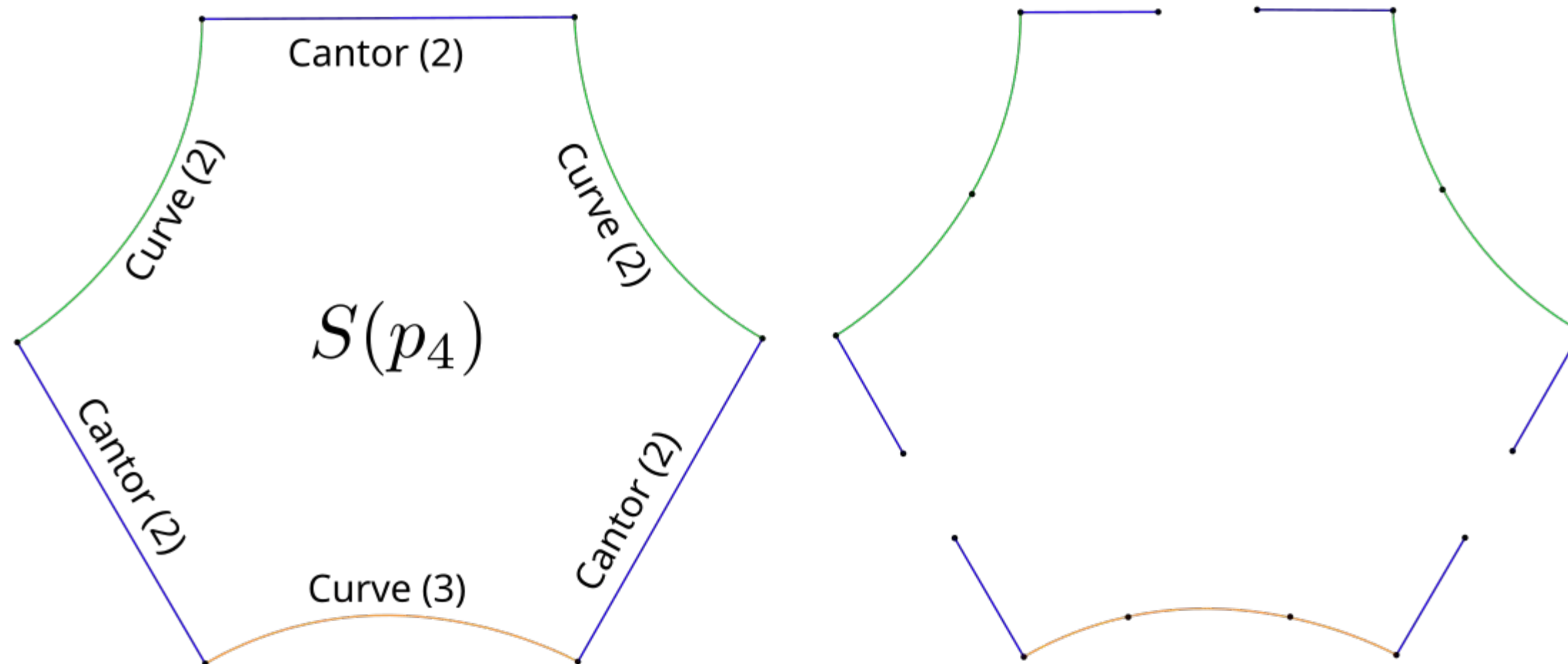
$$p_1 = \{C_3, B_2, C_2, B_2, C_2, B_2\}$$



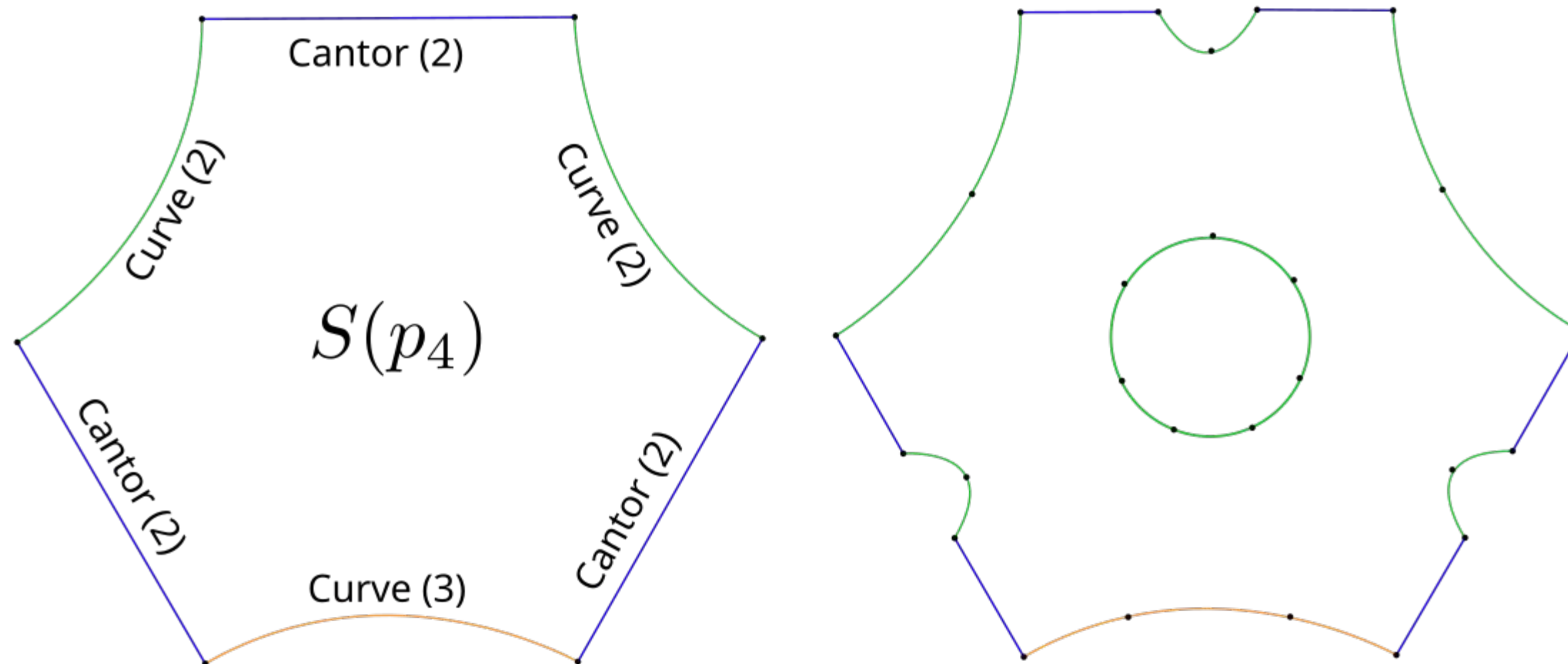
$$p_4 = \{C_2, B_2, C_2, B_3, C_2, B_2\}$$



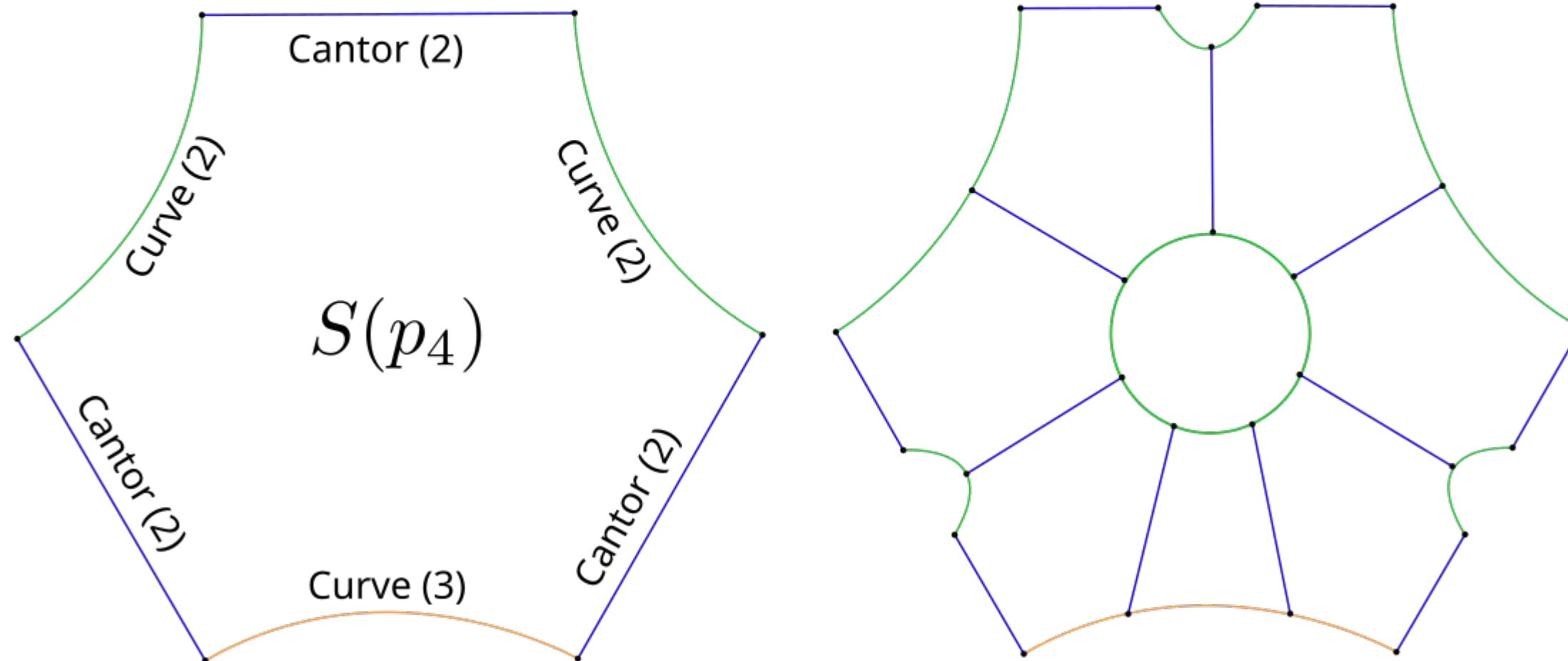
$$p_4 = \{C_2, B_2, C_2, B_3, C_2, B_2\}$$



$$p_4 = \{C_2, B_2, C_2, B_3, C_2, B_2\}$$



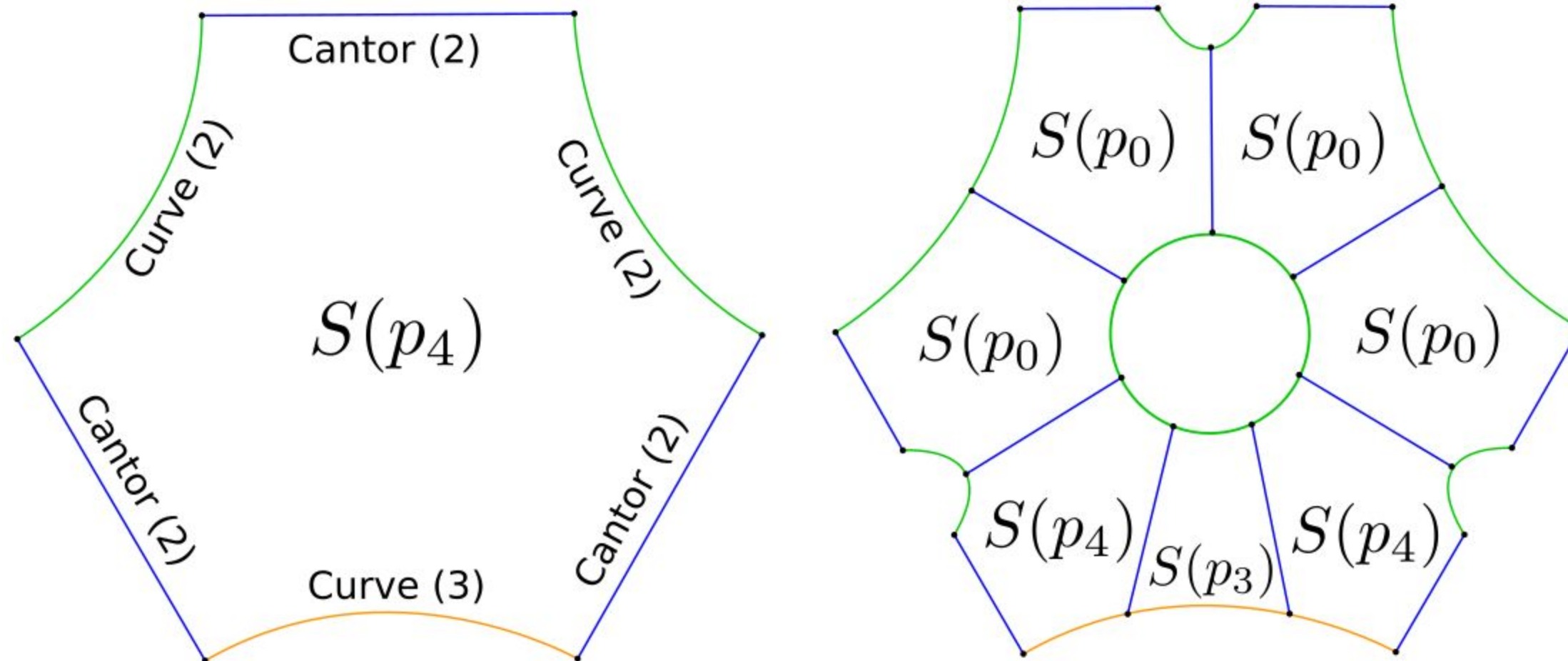
$$p_4 = \{C_2, B_2, C_2, B_3, C_2, B_2\}$$



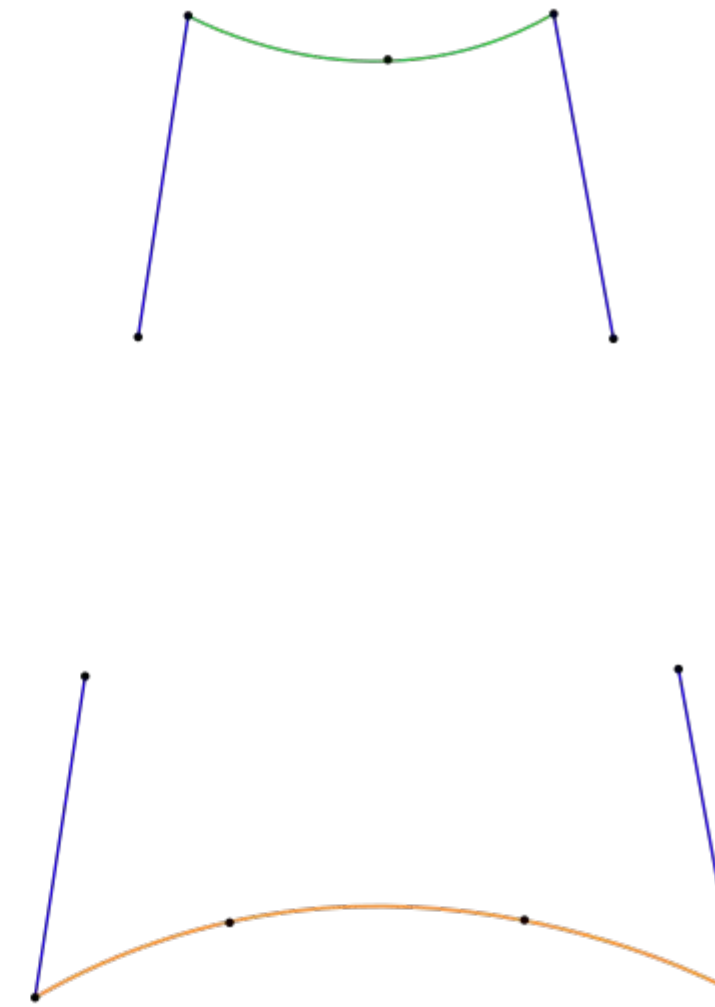
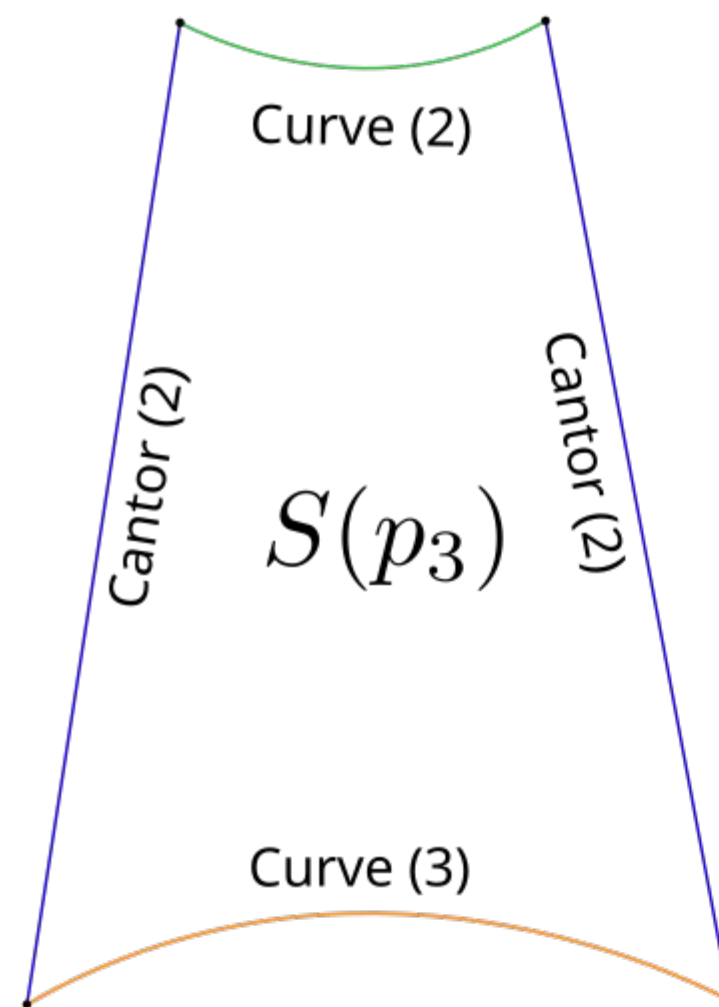
$$p_0 = \{C_2, B_2, C_2, B_2, C_2, B_2\}$$

$$p_3 = \{B_2, C_2, B_3, C_2\}$$

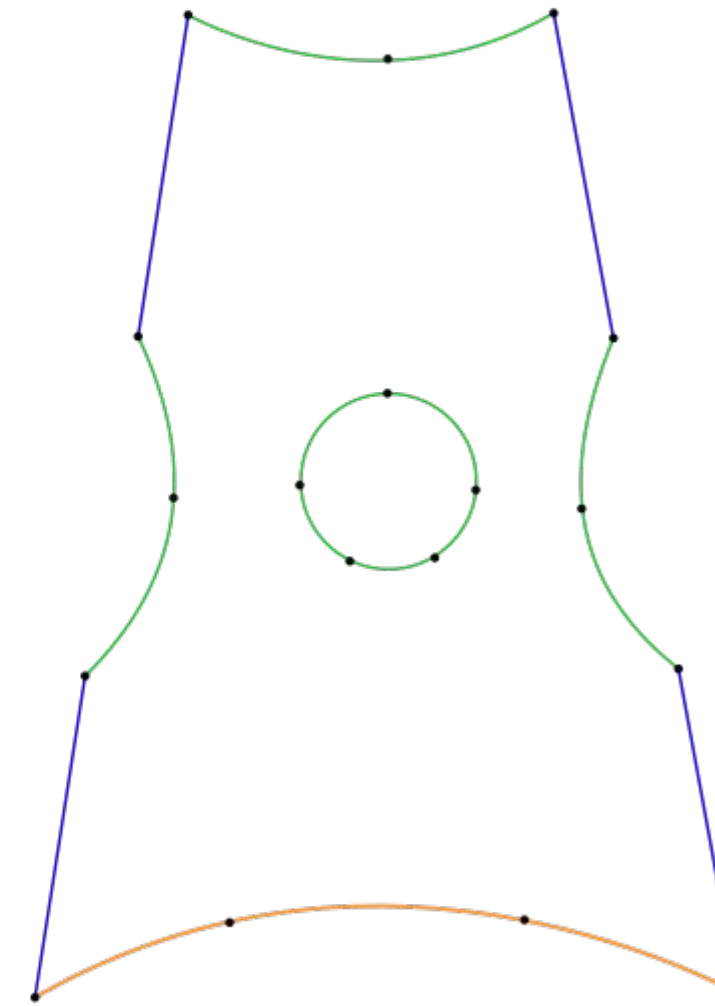
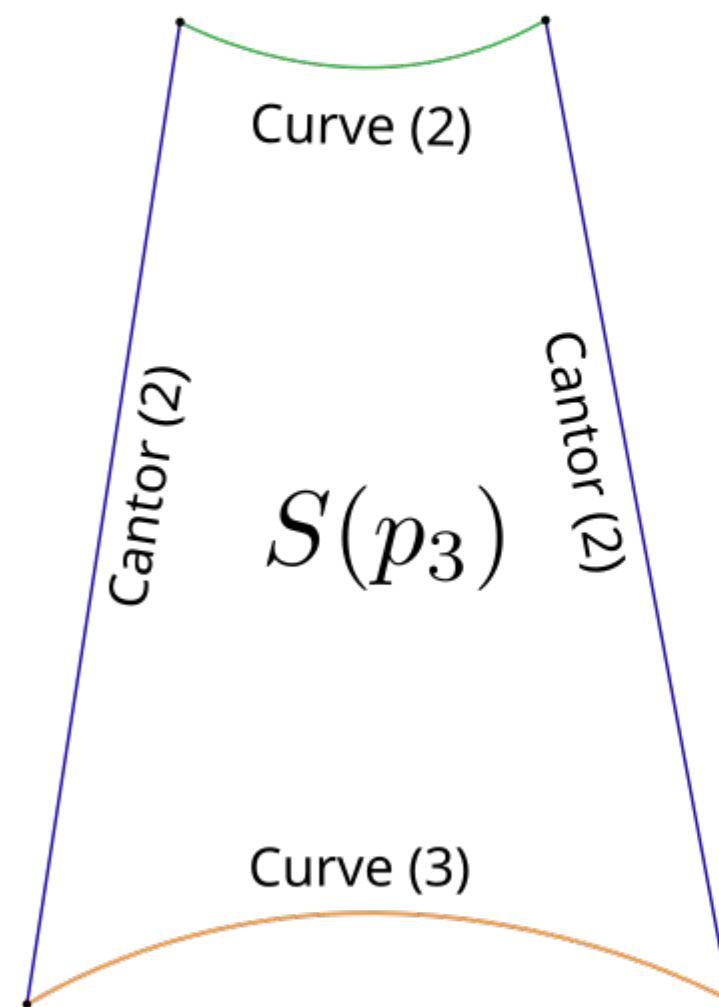
$$p_4 = \{C_2, B_2, C_2, B_3, C_2, B_2\}$$



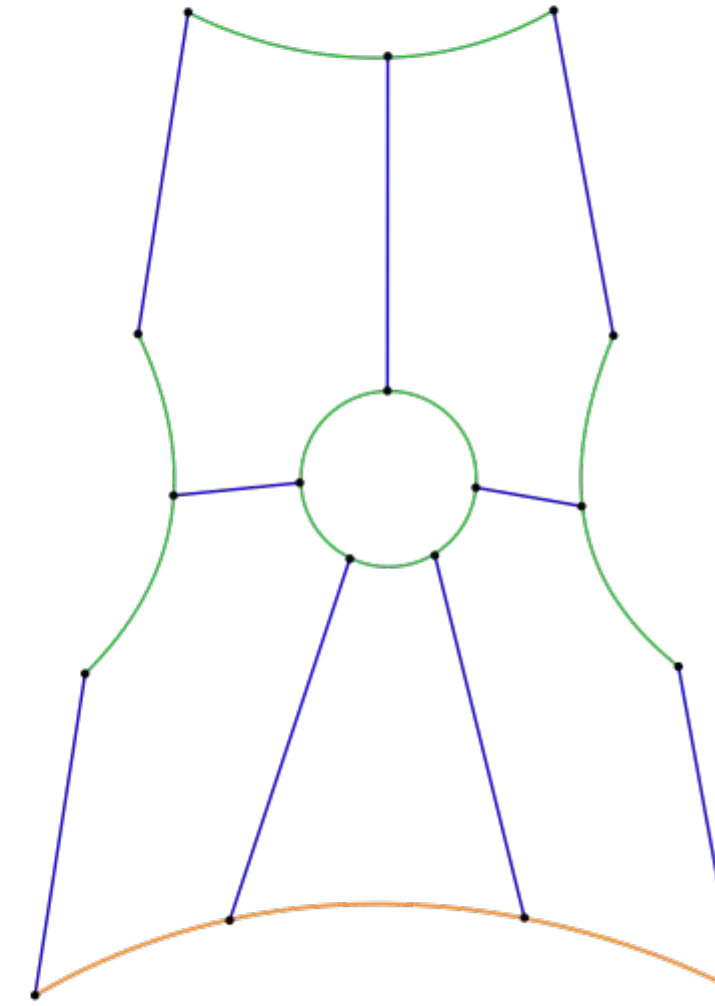
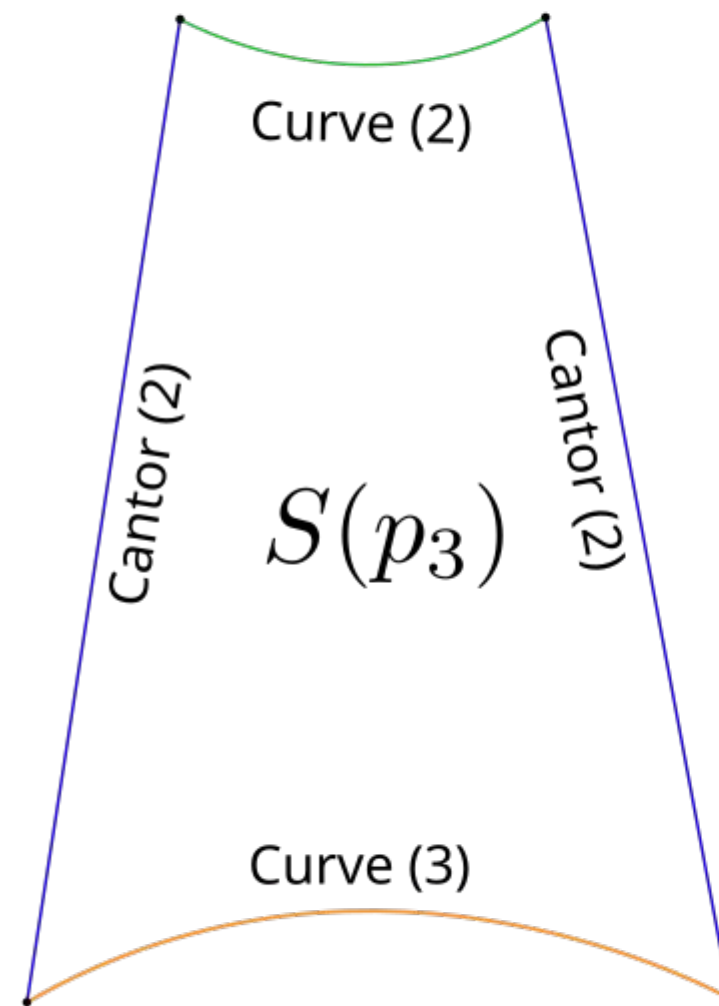
$$p_3 = \{B_2, C_2, B_3, C_2\}$$



$$p_3 = \{B_2, C_2, B_3, C_2\}$$



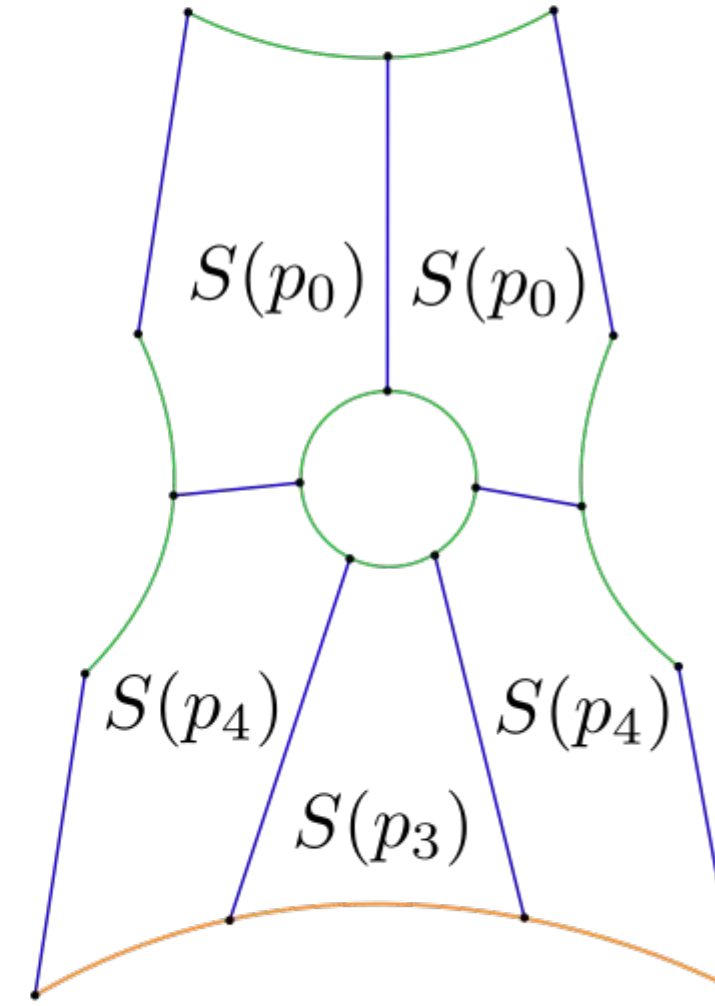
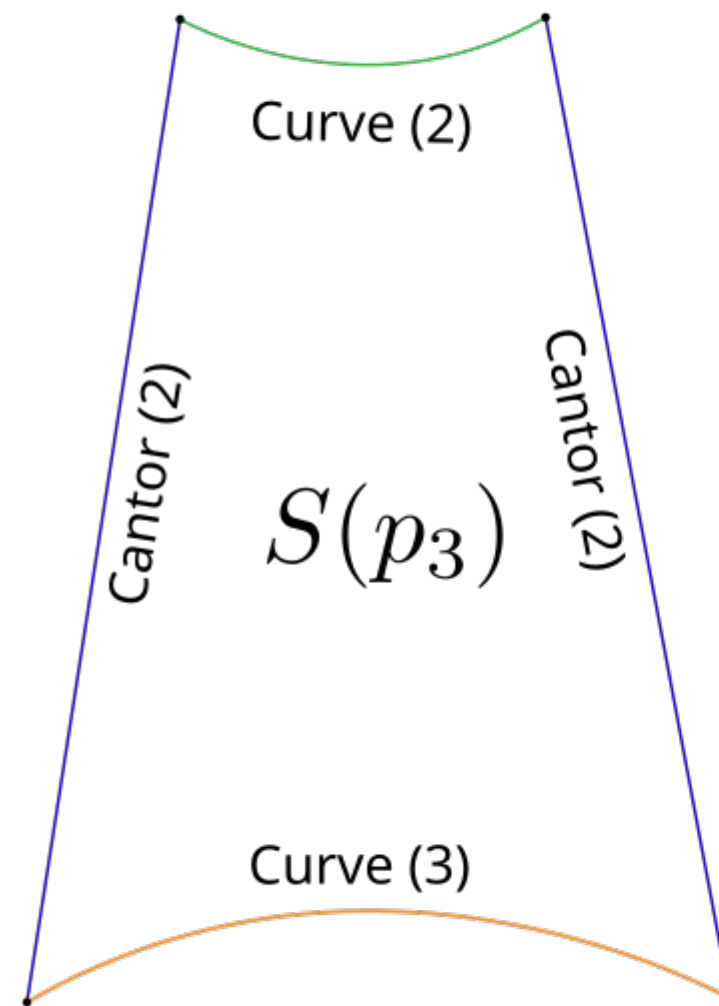
$$p_3 = \{B_2, C_2, B_3, C_2\}$$

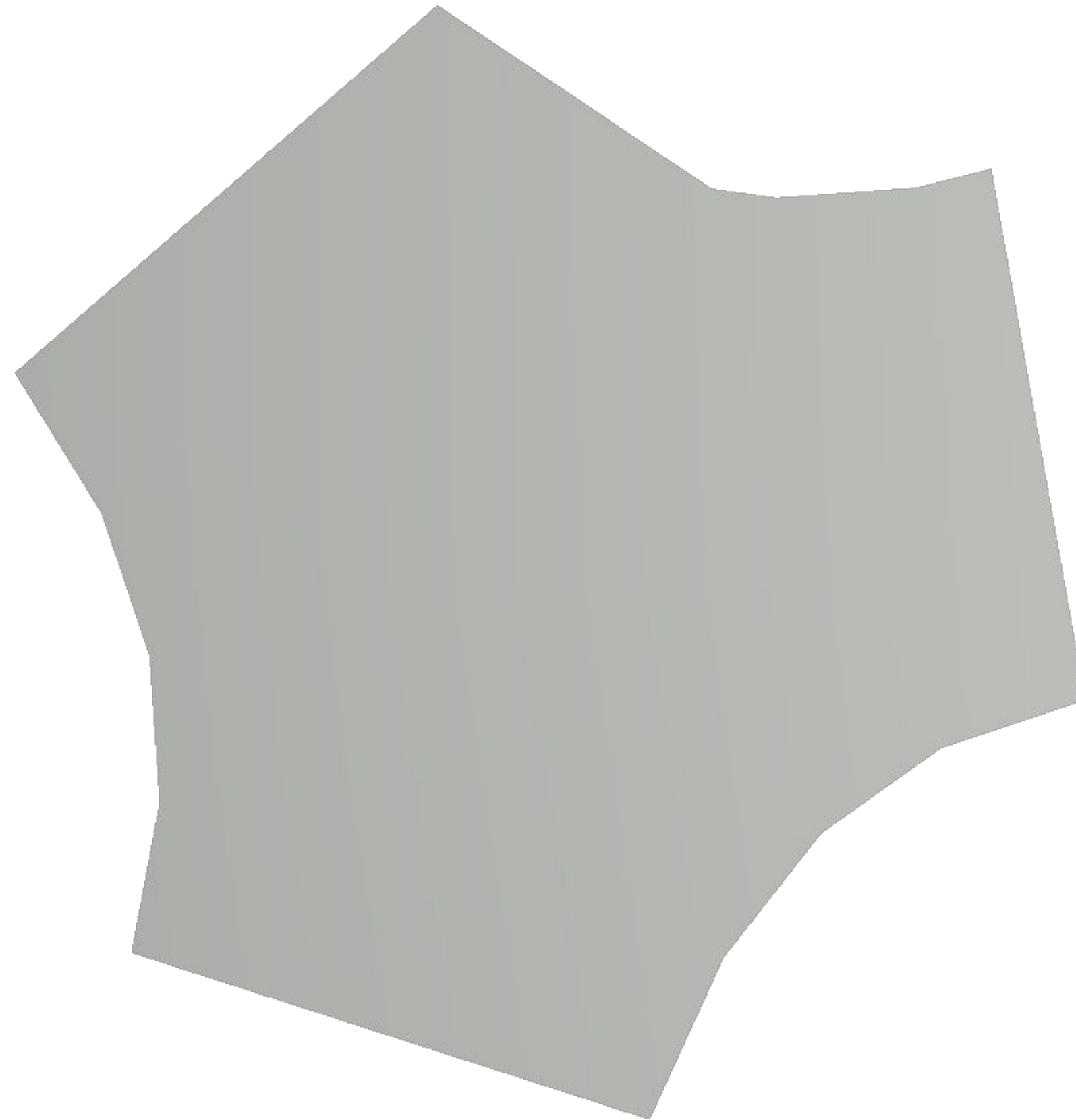


$$p_0 = \{C_2, B_2, C_2, B_2, C_2, B_2\}$$

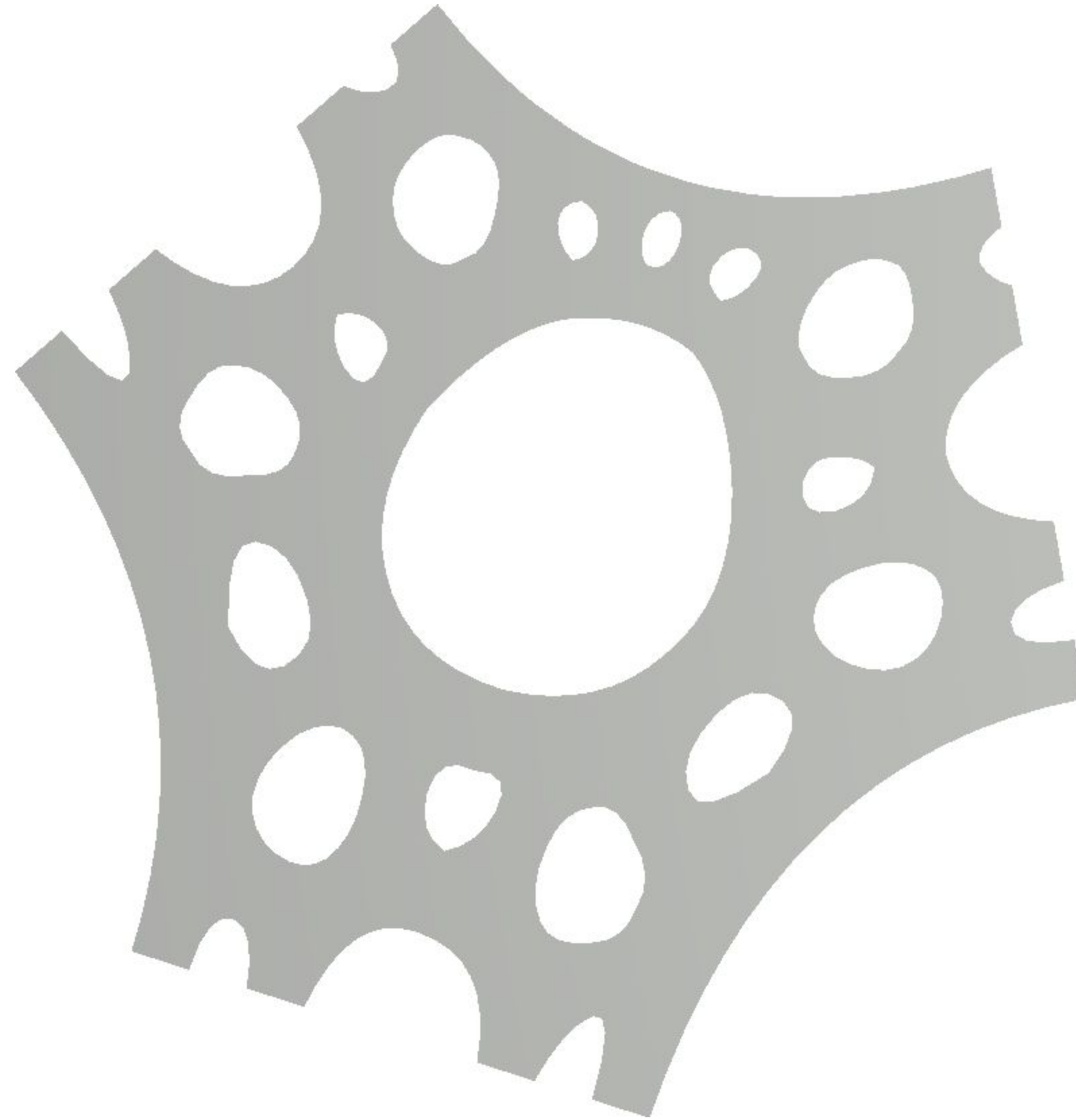
$$p_4 = \{C_2, B_2, C_2, B_3, C_2, B_2\}$$

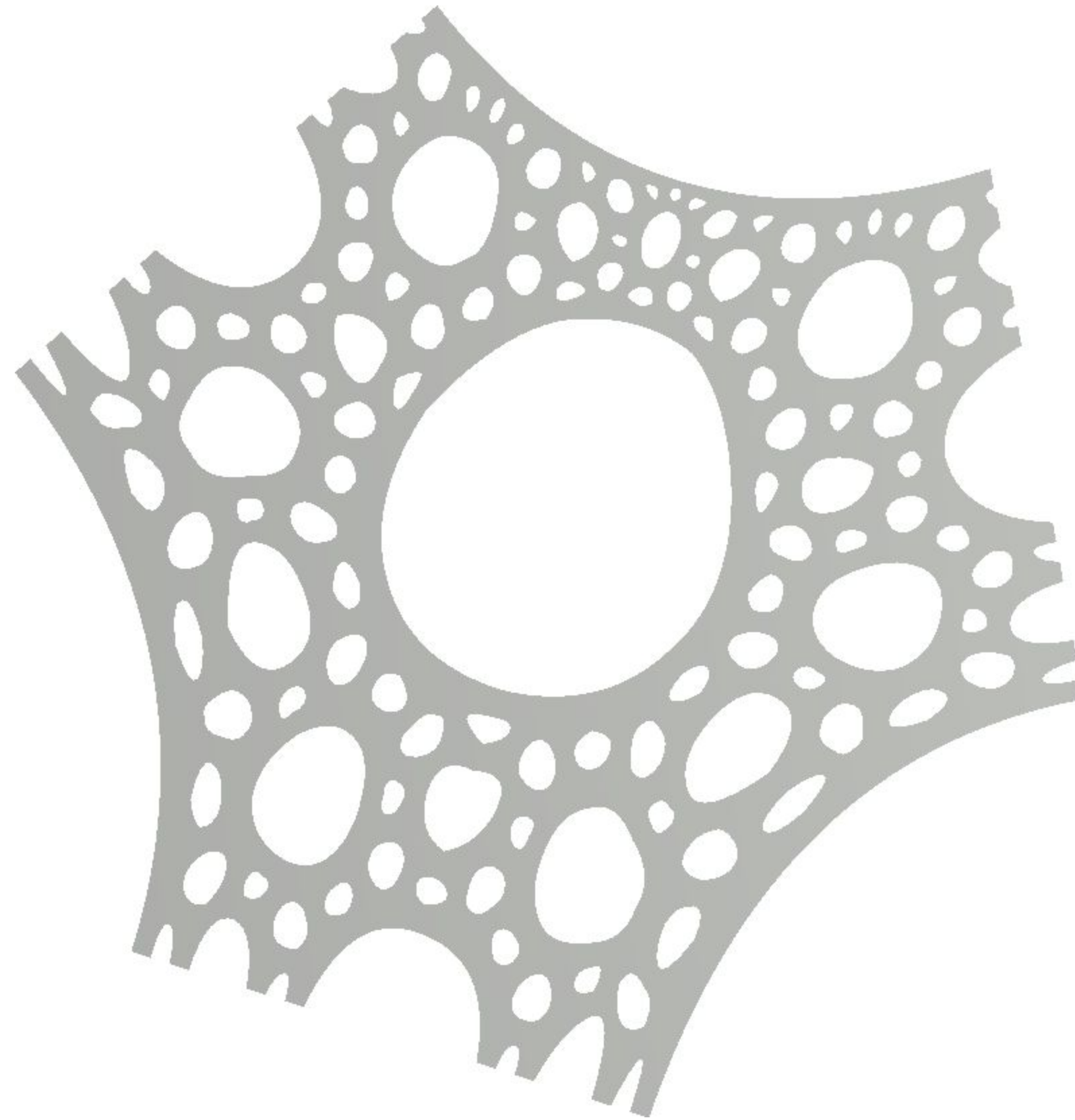
$$p_3 = \{B_2, C_2, B_3, C_2\}$$

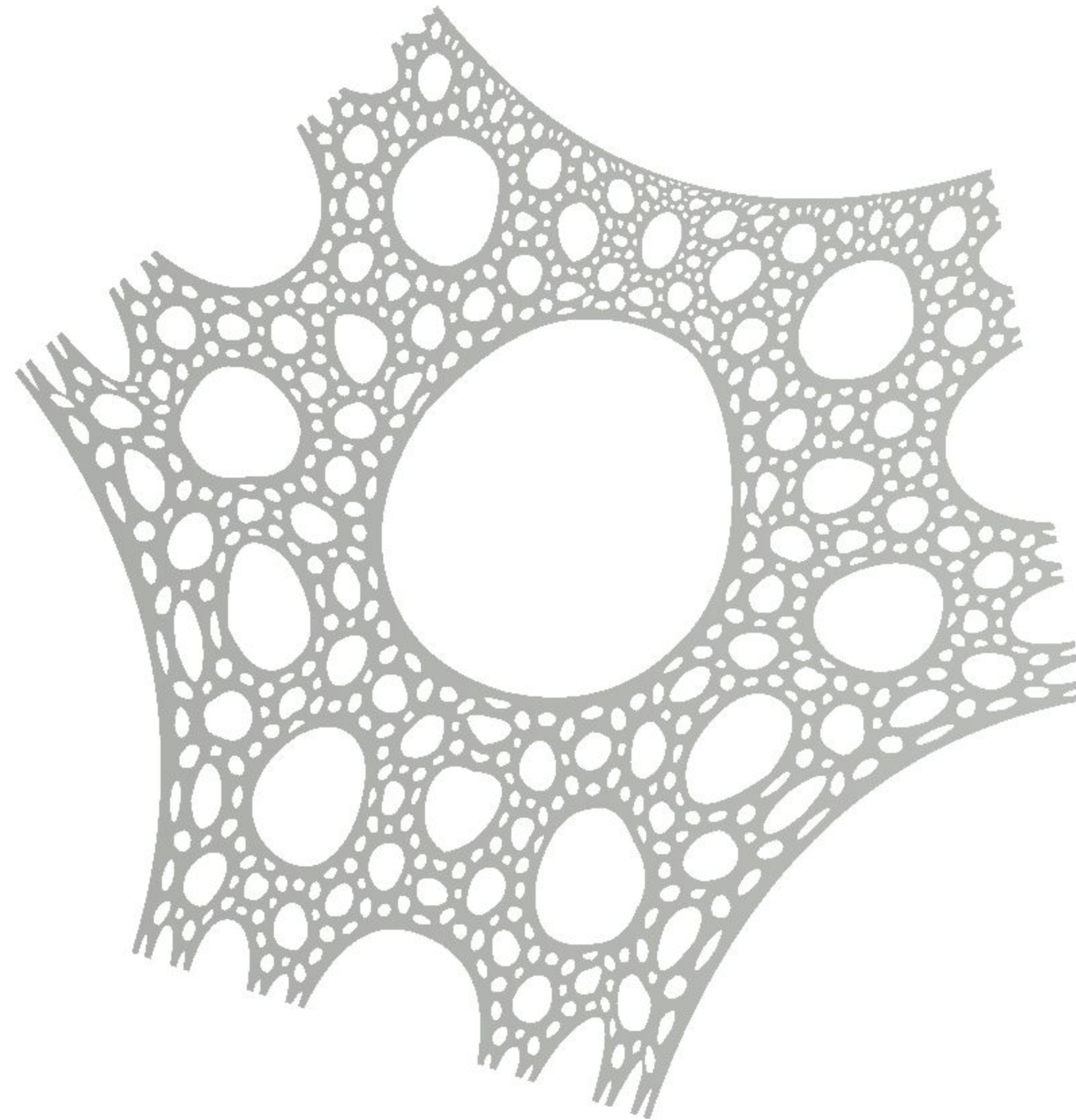




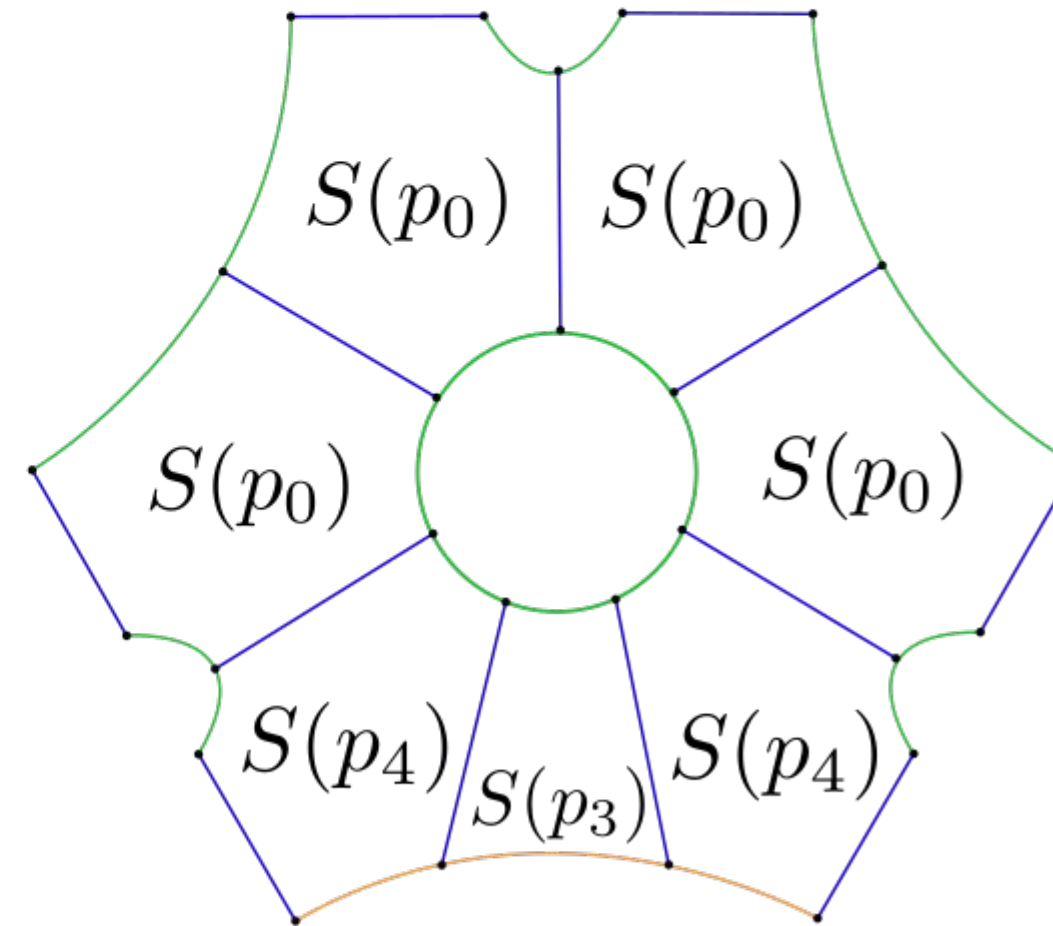




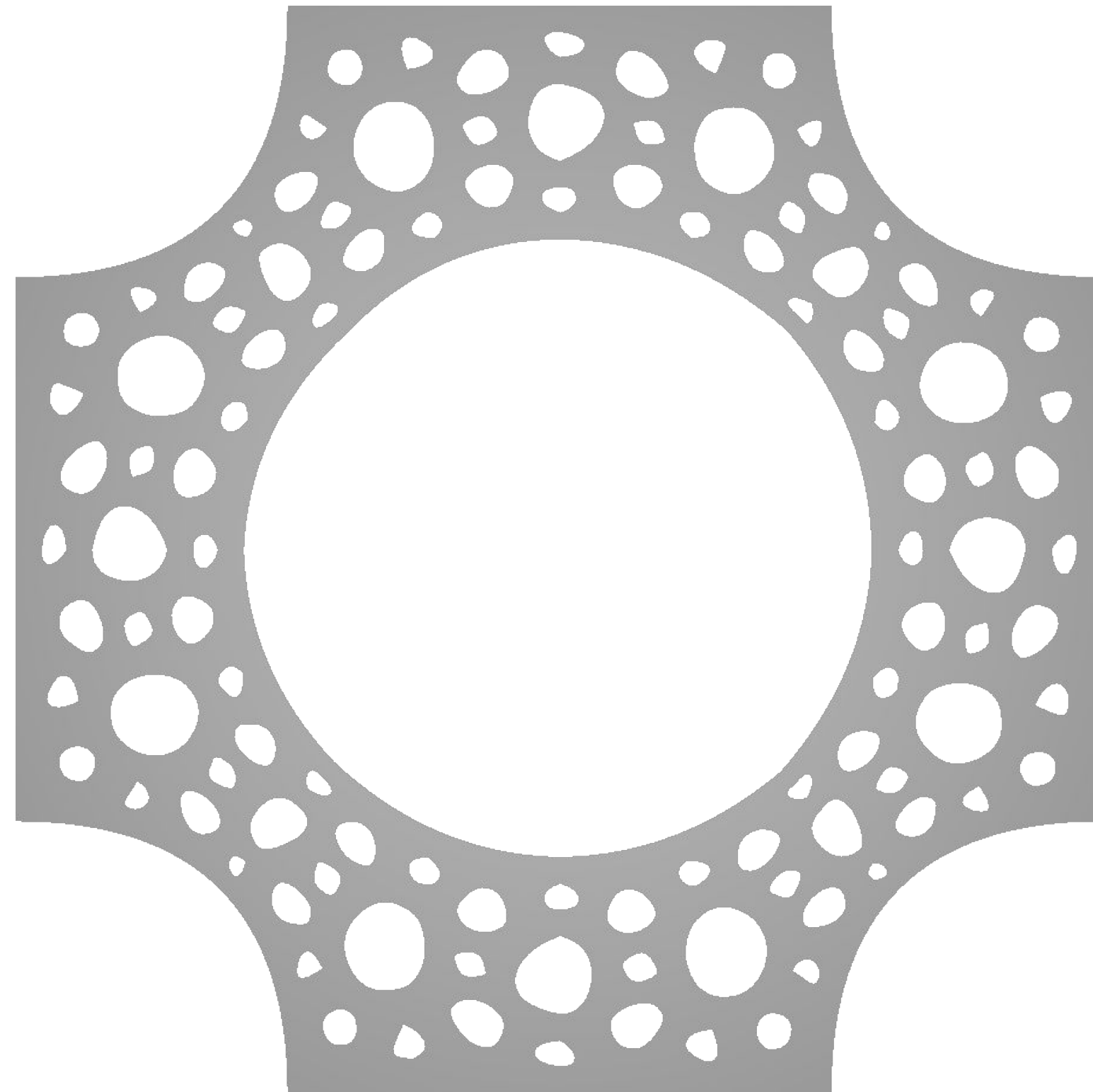




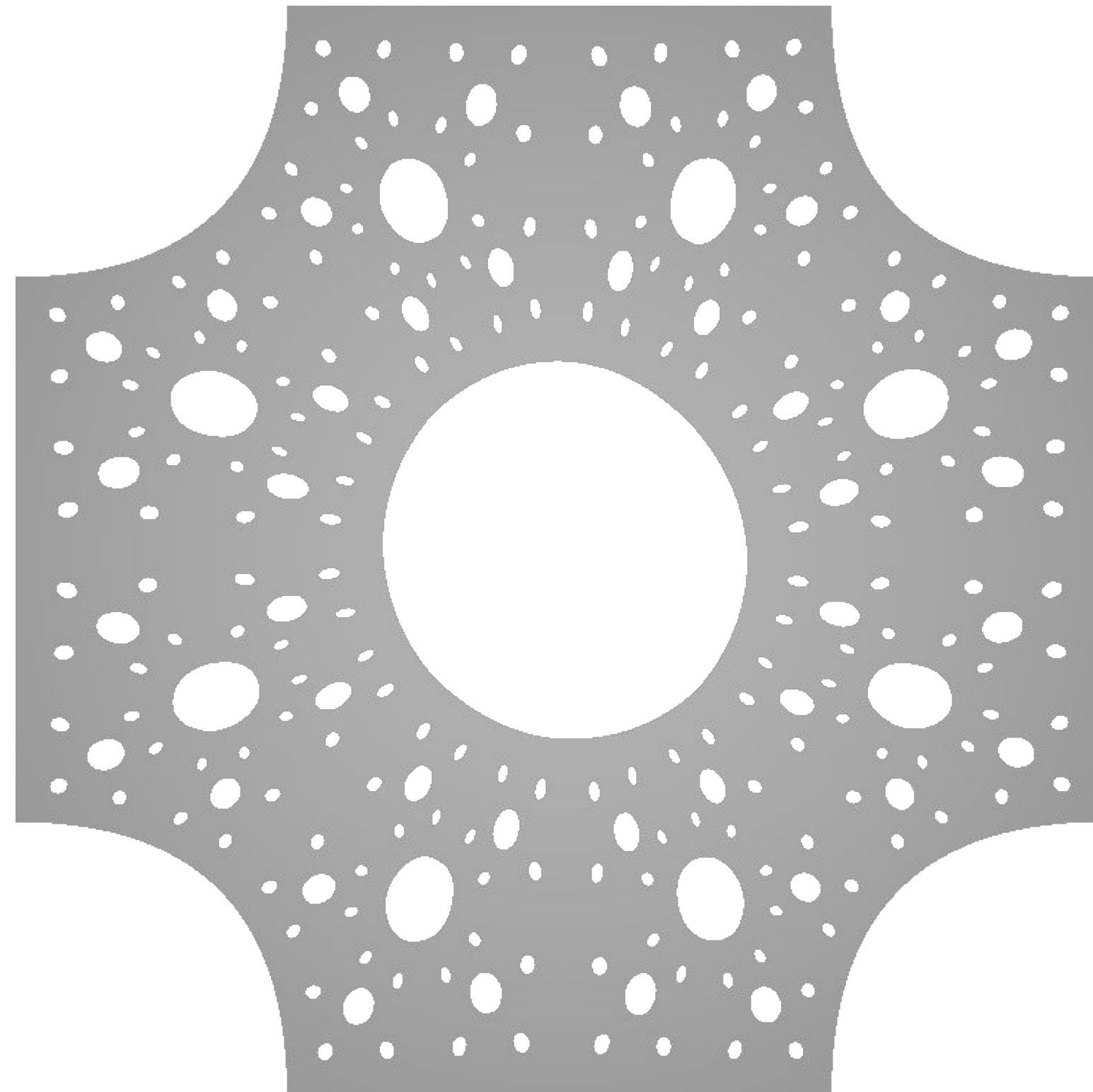
- Bords de raccord
- Bords de lacune centrale
- Bords entre les Cantors



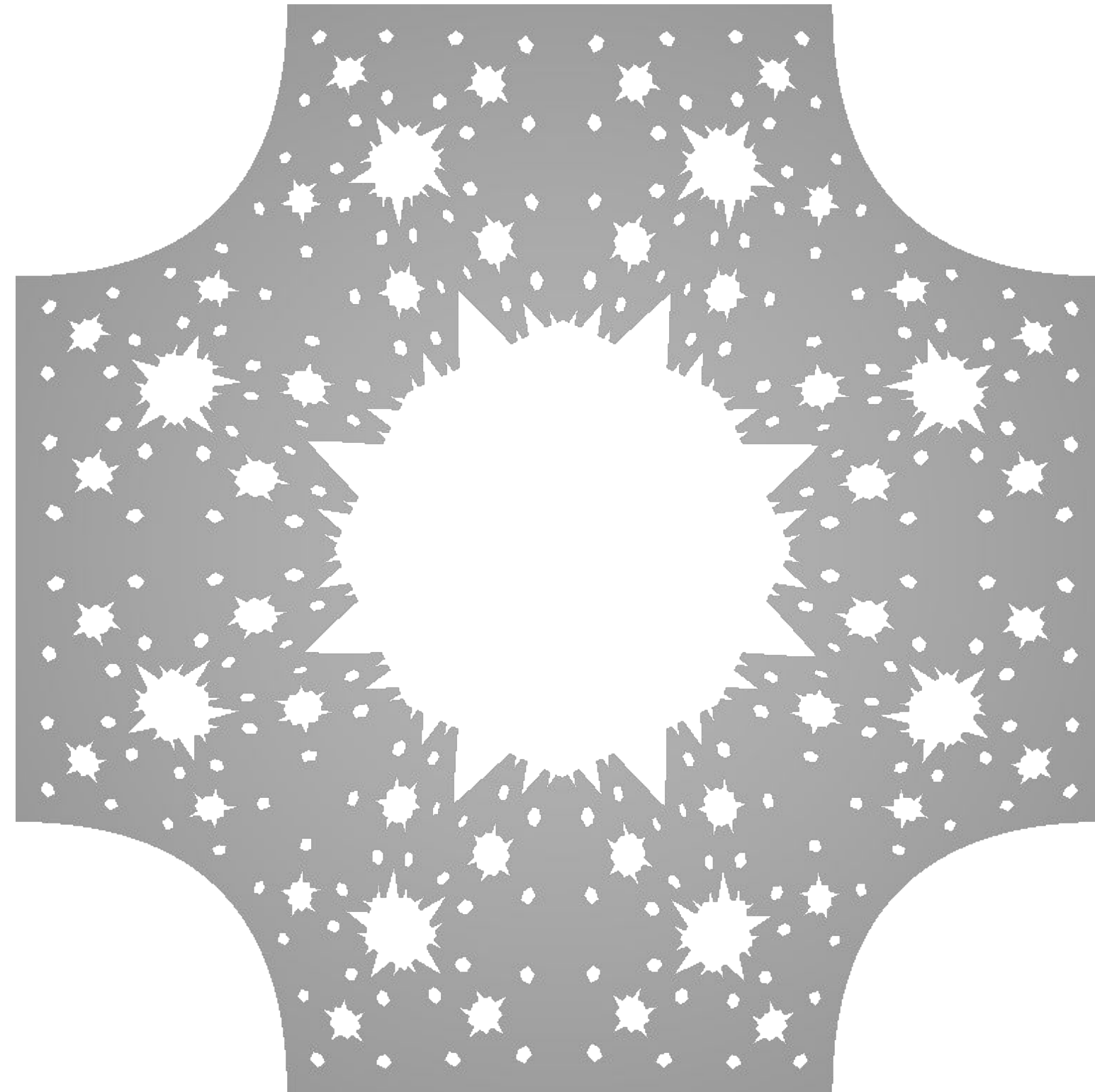
- Bords de raccord : C_2
- Bords de lacune centrale : B_2
- Bords entre les Cantors : B_2



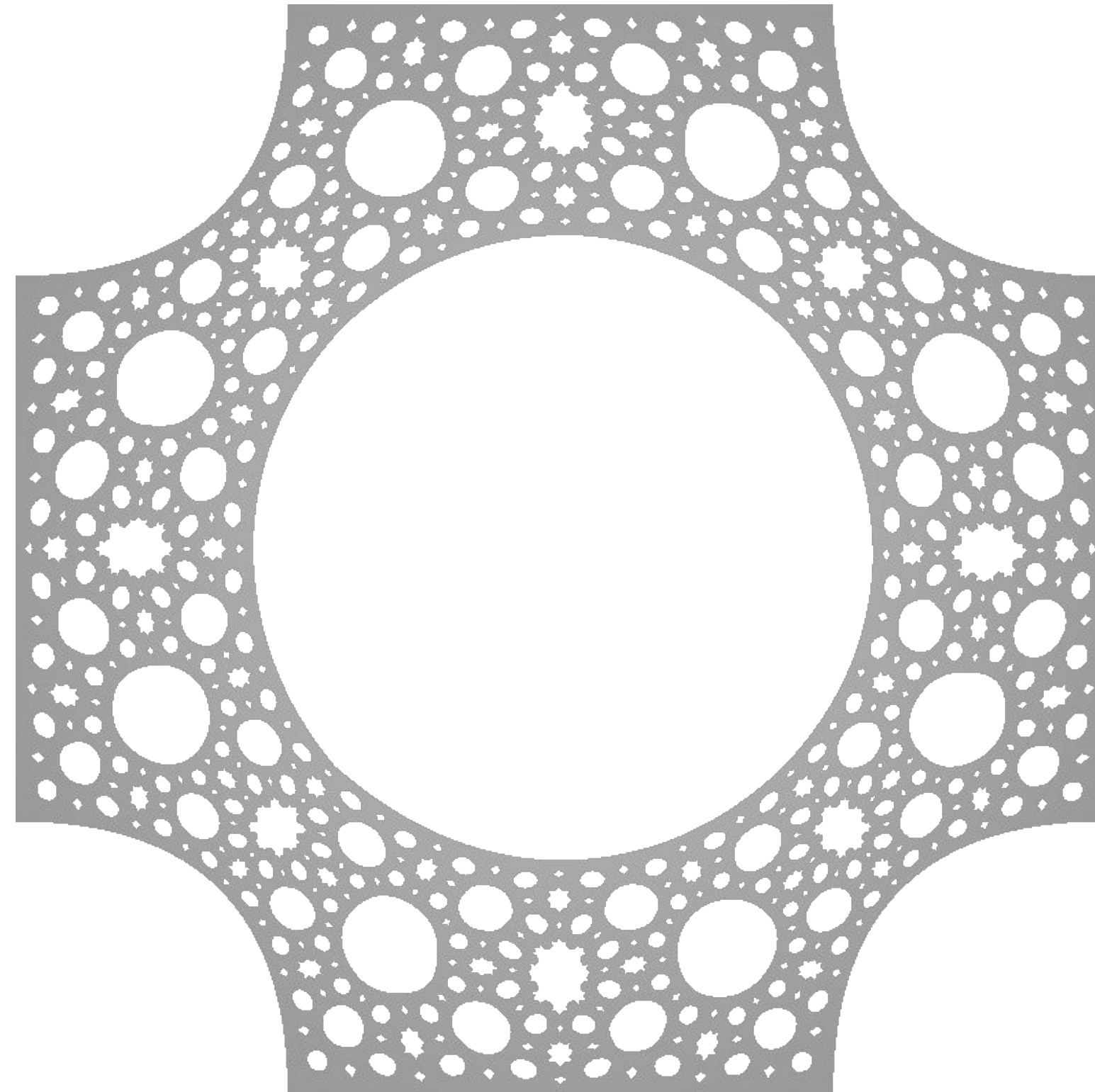
- Bords de raccord : B_2
- Bords de lacune centrale : B_2
- Bords entre les Cantors : B_2



- Bords de raccord : B_2
- Bords de lacune centrale : C_2
- Bords entre les Cantors : B_2



- Bords de raccord : C_2
- Bords de lacune centrale : B_2
- Bords entre les Cantors : C_2



- Dans une structure
 - Assemblage de plusieurs cellules
- Simuler un délai de subdivision
- Adapter la lacunarité localement

■ Pour les bords

- Se subdivisent en un même bord avec un retard en moins

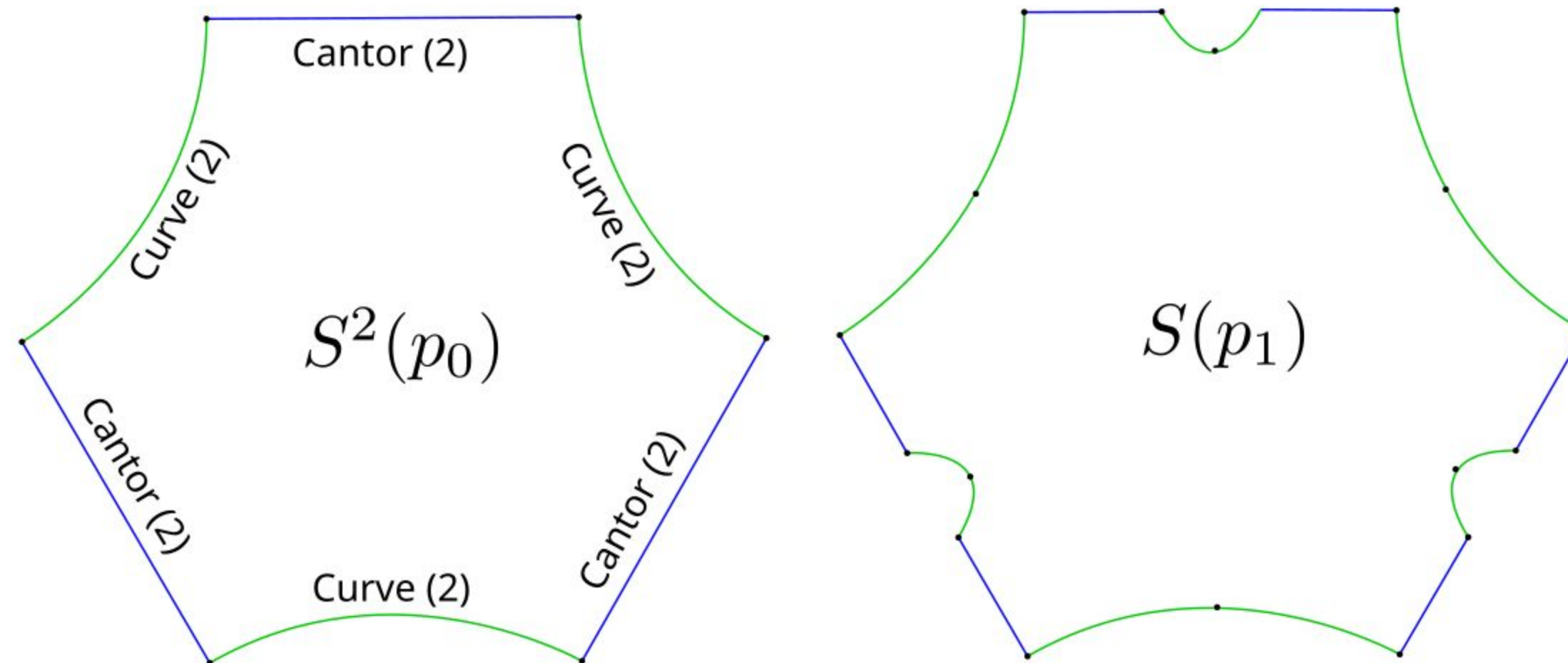
$$\begin{array}{l} \text{_____} \\ \text{_____} \end{array} \quad \begin{array}{l} C_2^3 \\ C_2^2 \end{array}$$

■ Pour les faces

- Se subdivisent en 1 seule face sans lacune
- Les bords seront subdivisés
- Cas particuliers des bords de type Cantor
 - Il faut connecter les subdivisions

$$\begin{array}{l} \text{_____} \\ \text{_____} \\ \text{---} \quad \text{---} \\ \text{---} \quad \text{---} \\ \text{---} \quad \text{---} \end{array} \quad \begin{array}{l} \text{_____} \\ \text{_____} \\ \text{---} \quad \text{---} \\ \text{---} \quad \text{---} \\ \text{---} \quad \text{---} \end{array} \quad C_2$$

$$p_0 = \{C_2, B_2, C_2, B_2, C_2, B_2\}$$

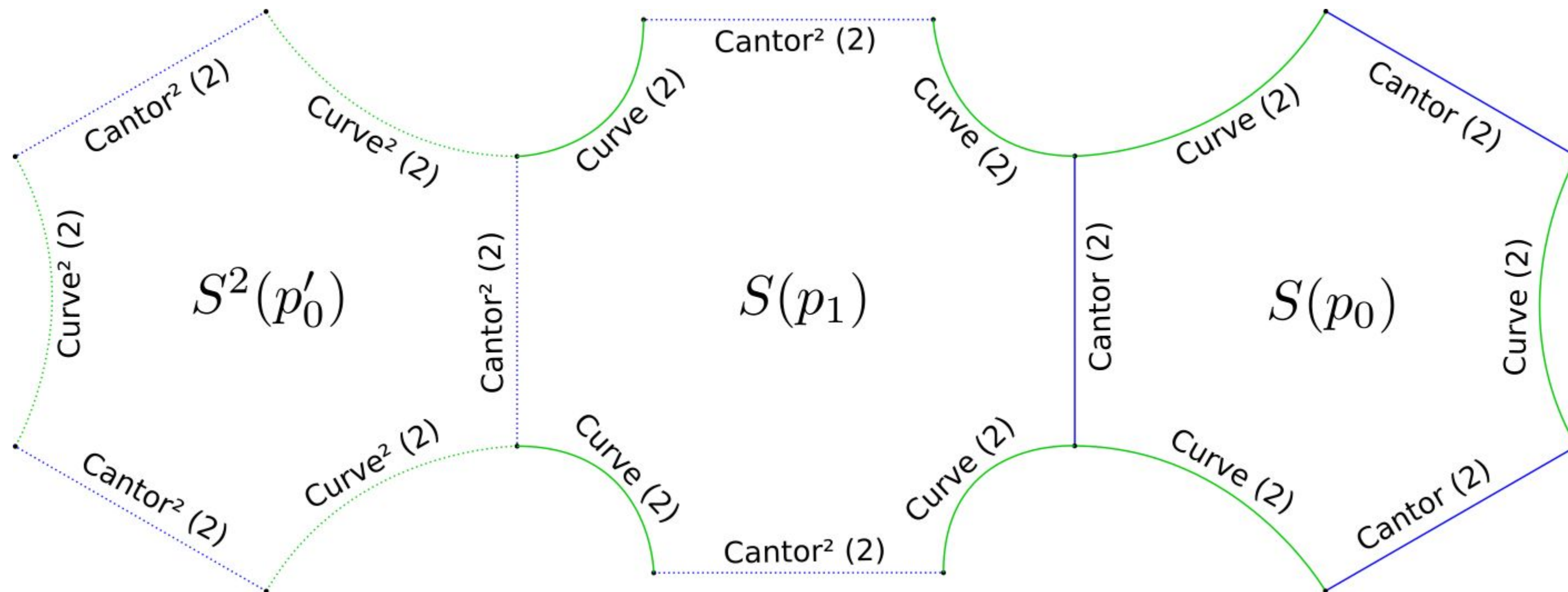


$$p_1 = \{C_2, B_2, B_2, C_2, B_2, B_2, C_2, B_2, B_2, C_2, B_2, B_2, C_2, B_2, B_2, C_2, B_2, B_2\}$$

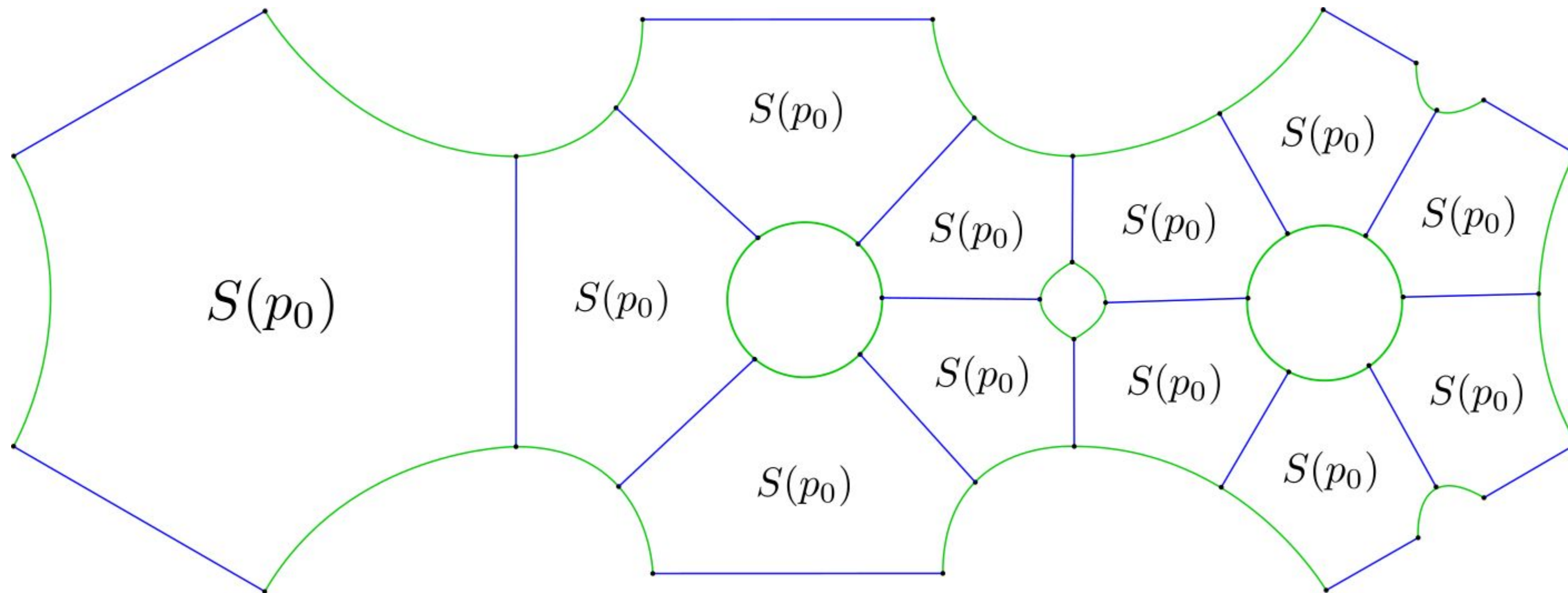
$$p_0 = \{C_2, B_2, C_2, B_2, C_2, B_2\}$$

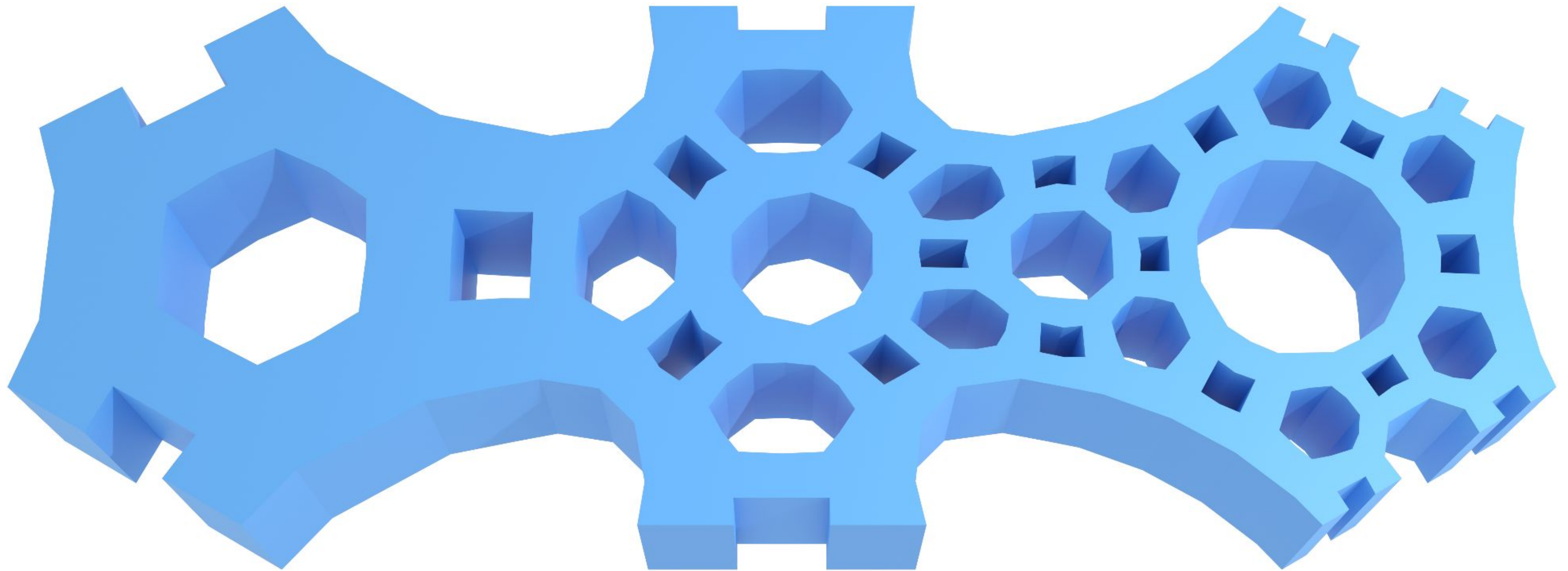
$$p'_0 = \{C_2^2, B_2^2, C_2^2, B_2^2, C_2^2, B_2^2\}$$

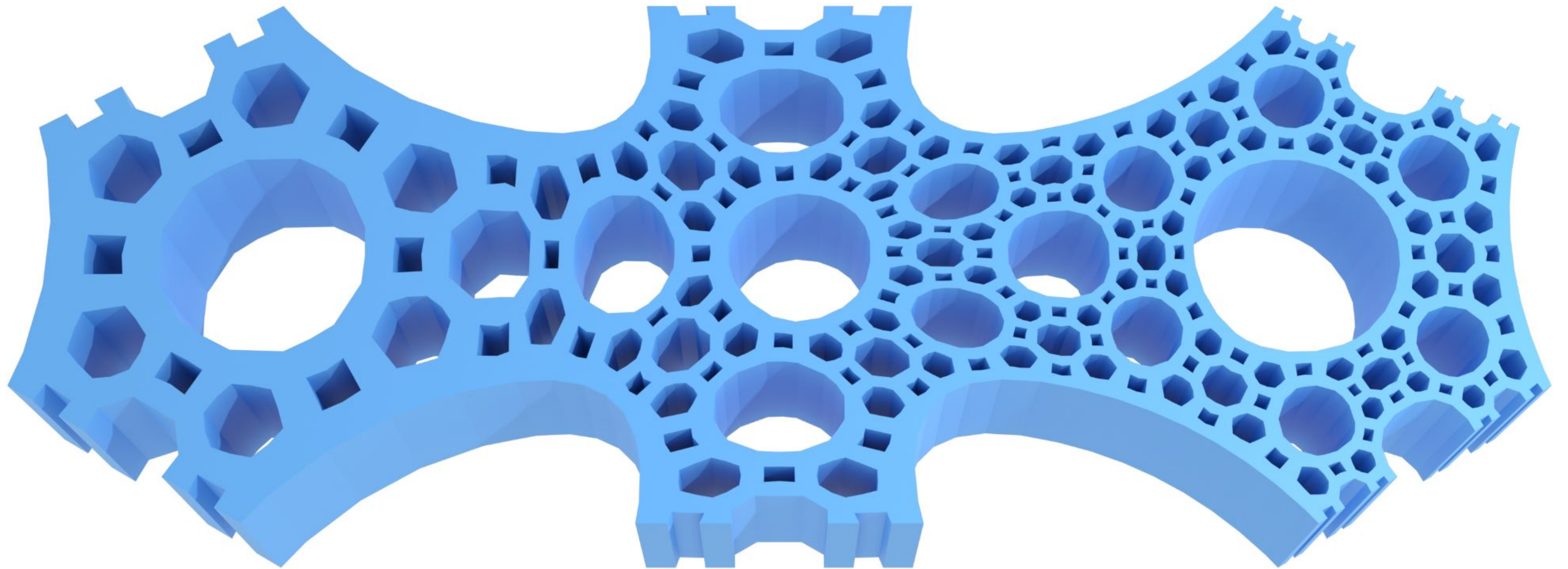
$$p_1 = \{C_2^2, B_2, C_2^2, B_2, C_2^2, B_2, C_2, B_2\}$$

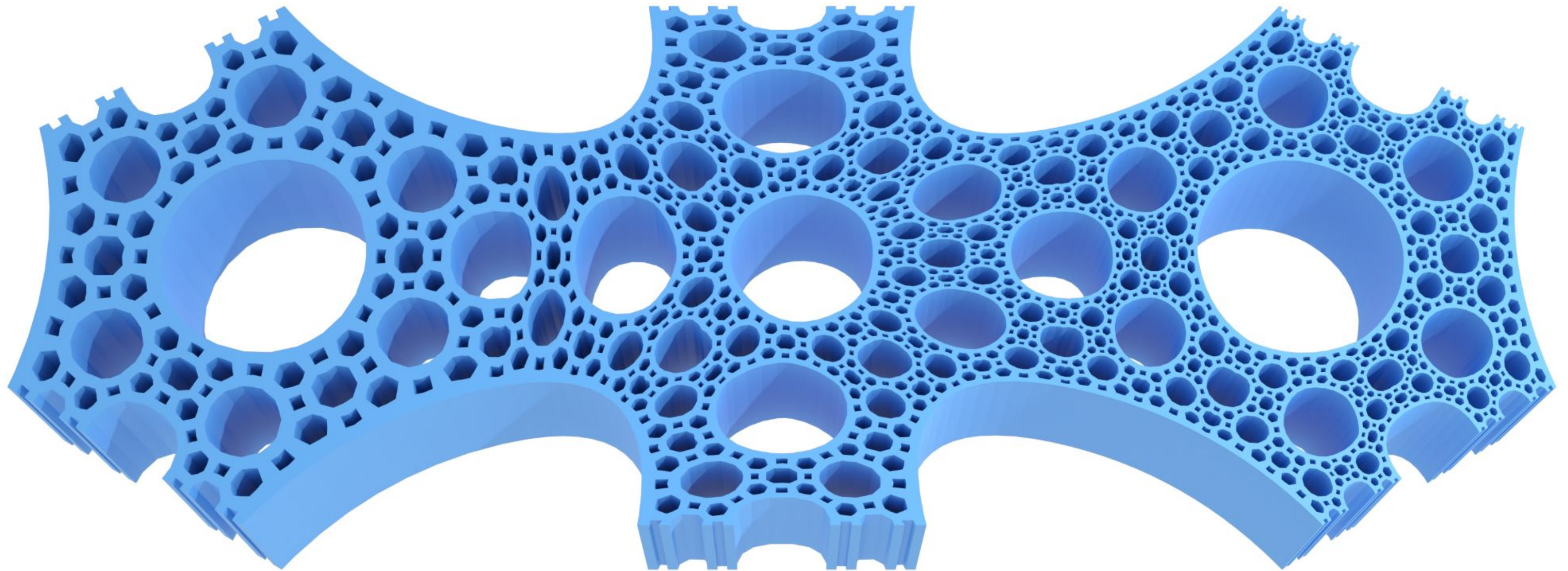


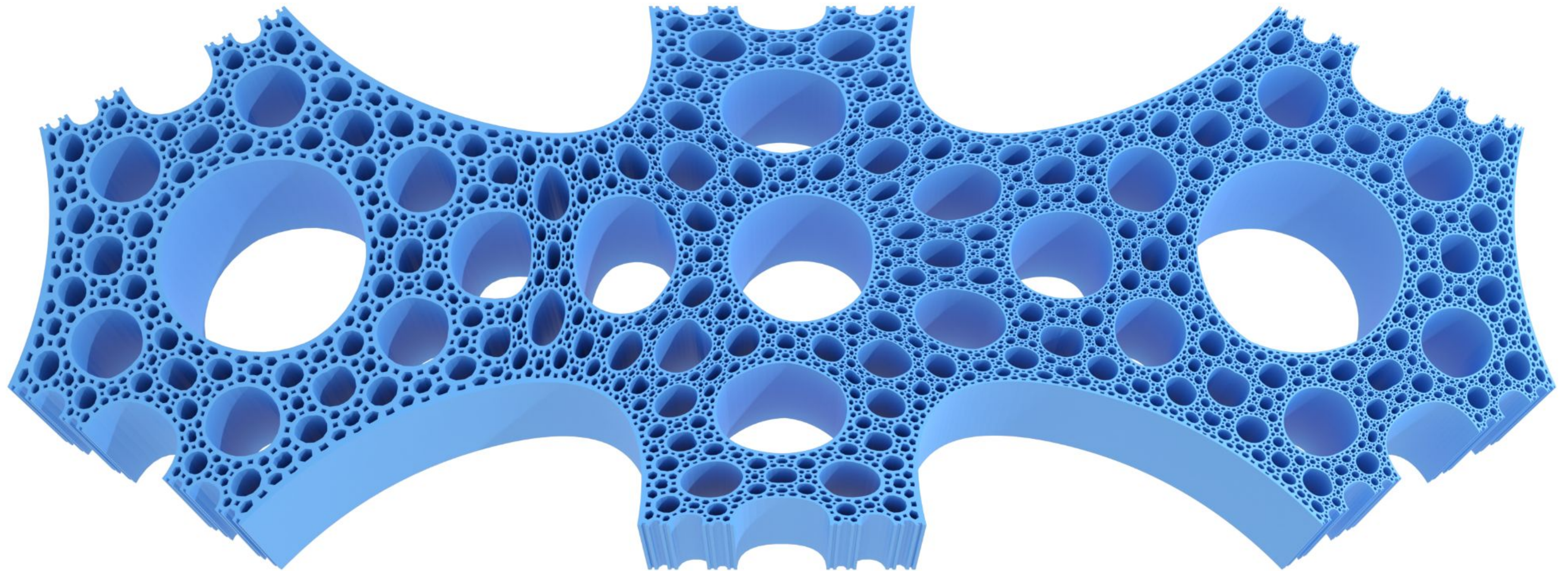
$$p_0 = \{C_2, B_2, C_2, B_2, C_2, B_2\}$$











- Algorithme de subdivision d'une face 2D
- Retard pour différer la subdivision
 - Adapte la lacunarité localement
- Travaux futurs
 - Passage à la 3D

Empilements polytopaux de sphères

Boris Bordeaux¹, Iván Rasskin², Christian Gentil¹, Jorge Luis Ramírez Alfonsín³, Lionel Garnier¹

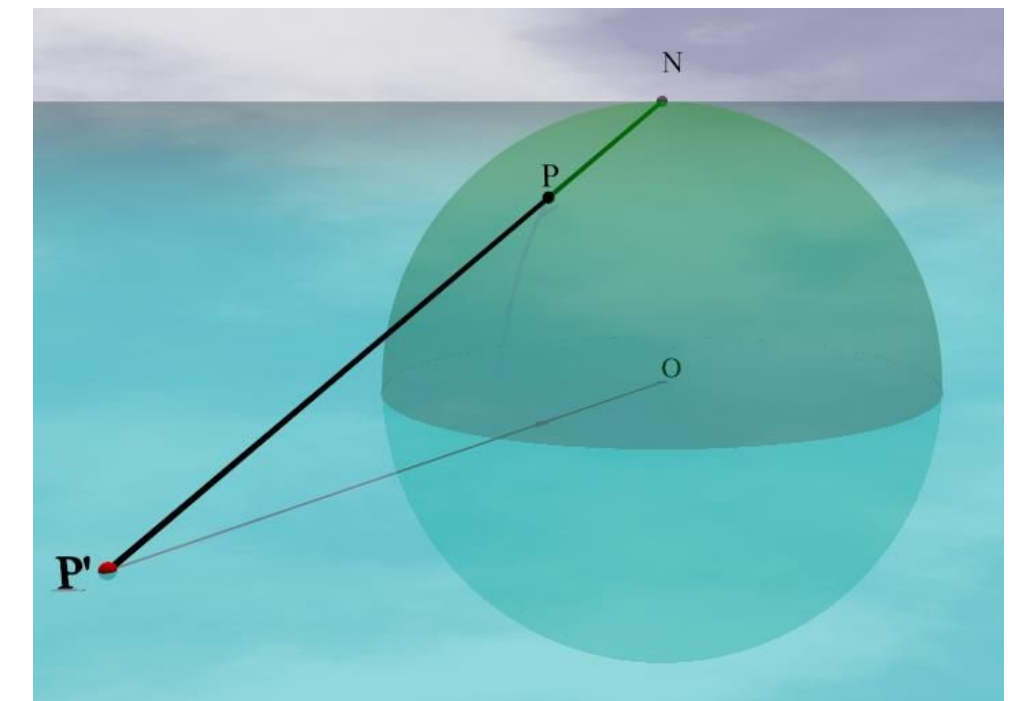
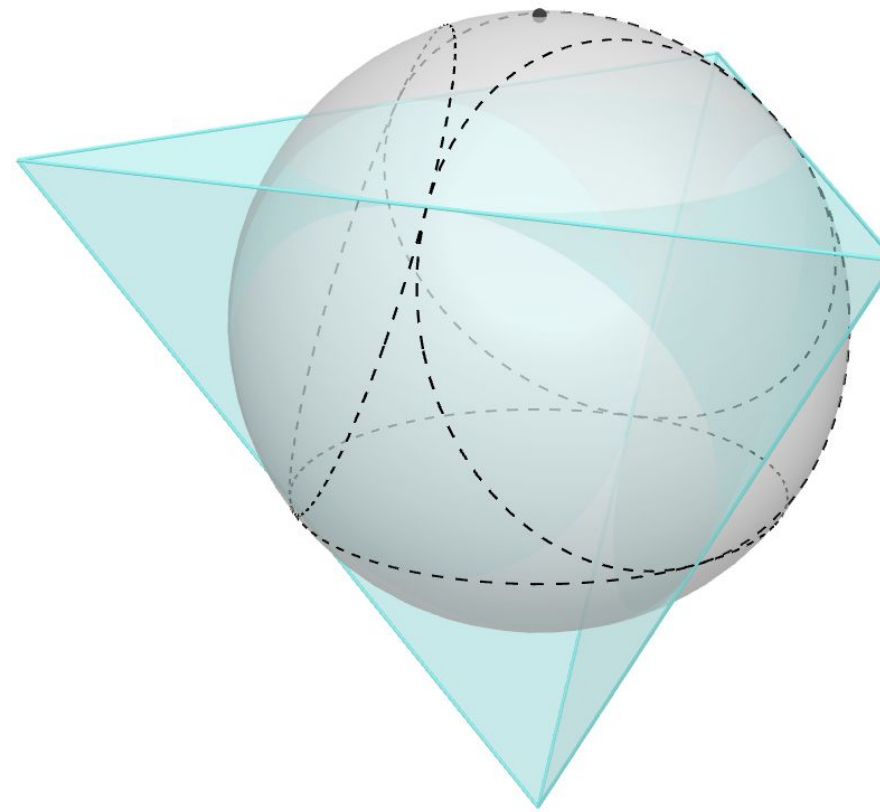
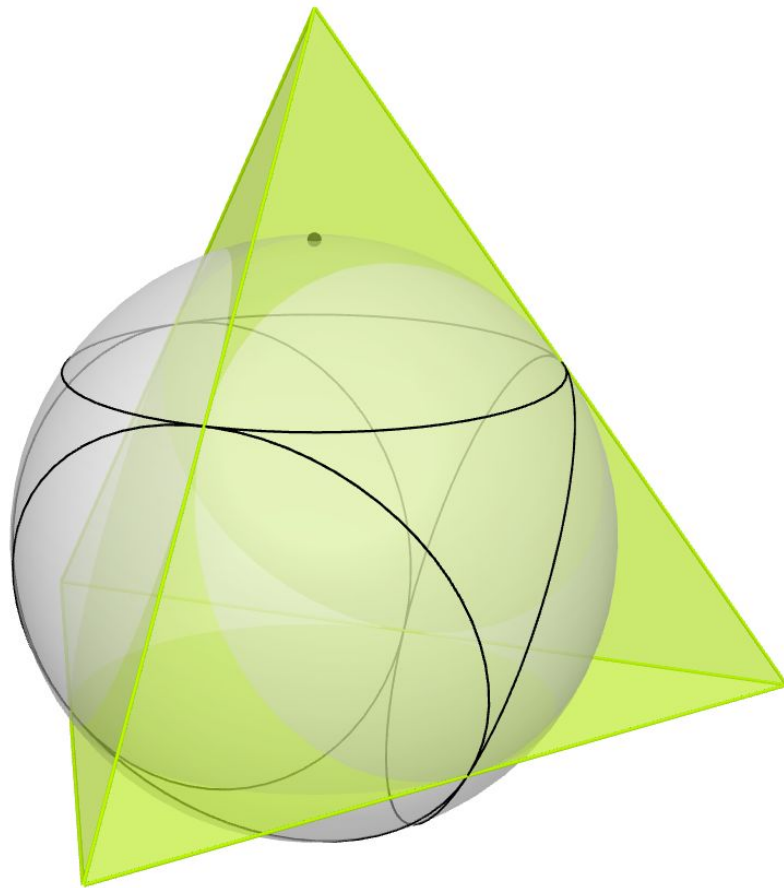
¹*LIB – Université de Bourgogne*

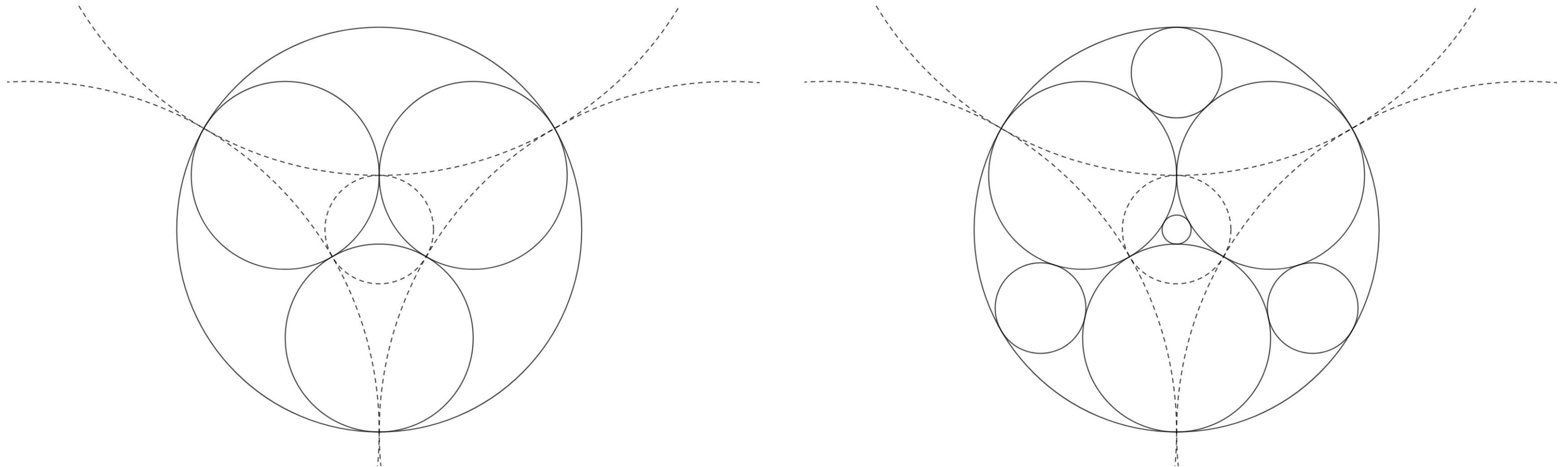
²*AZT – Université Technique de Graz*

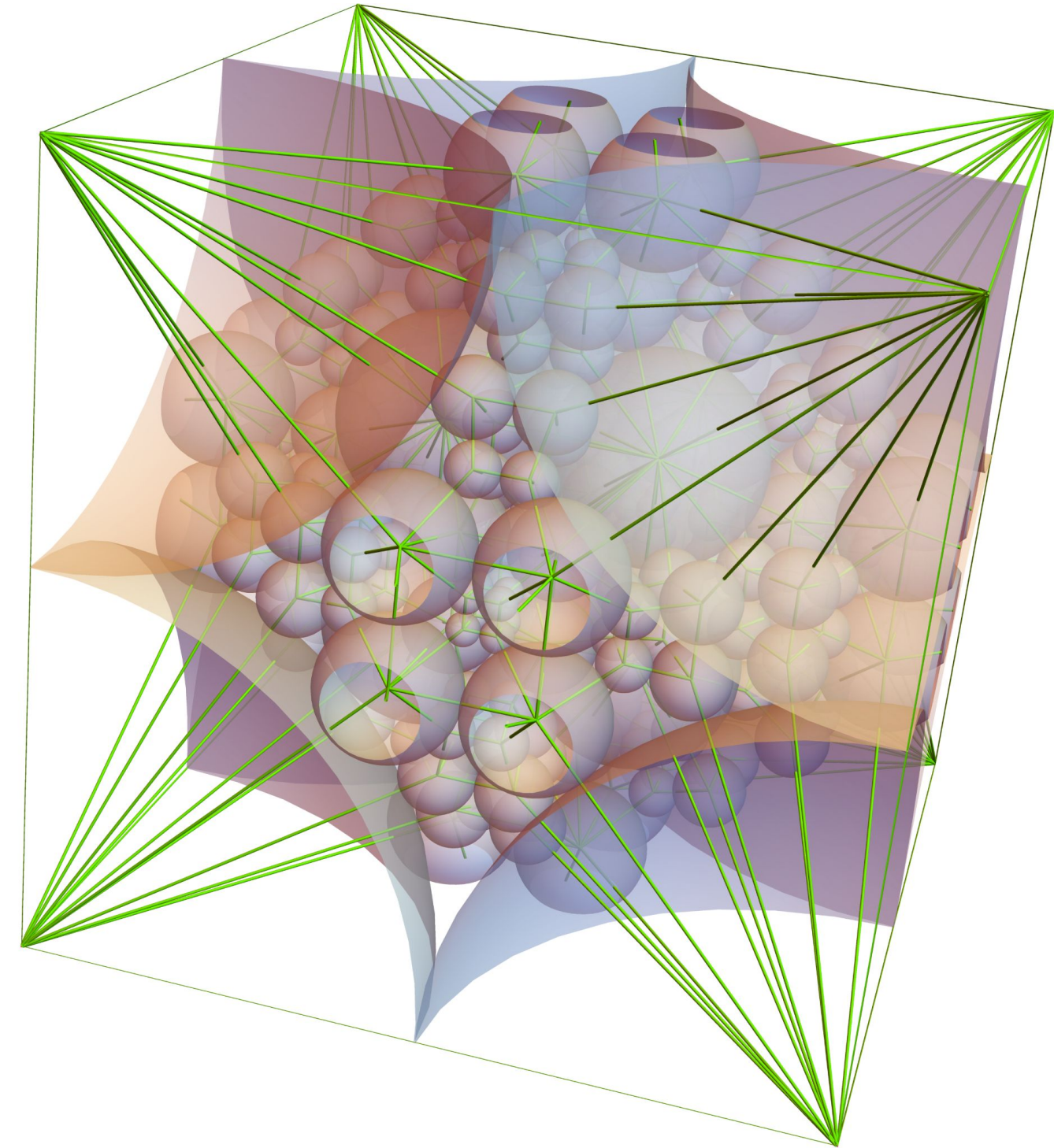
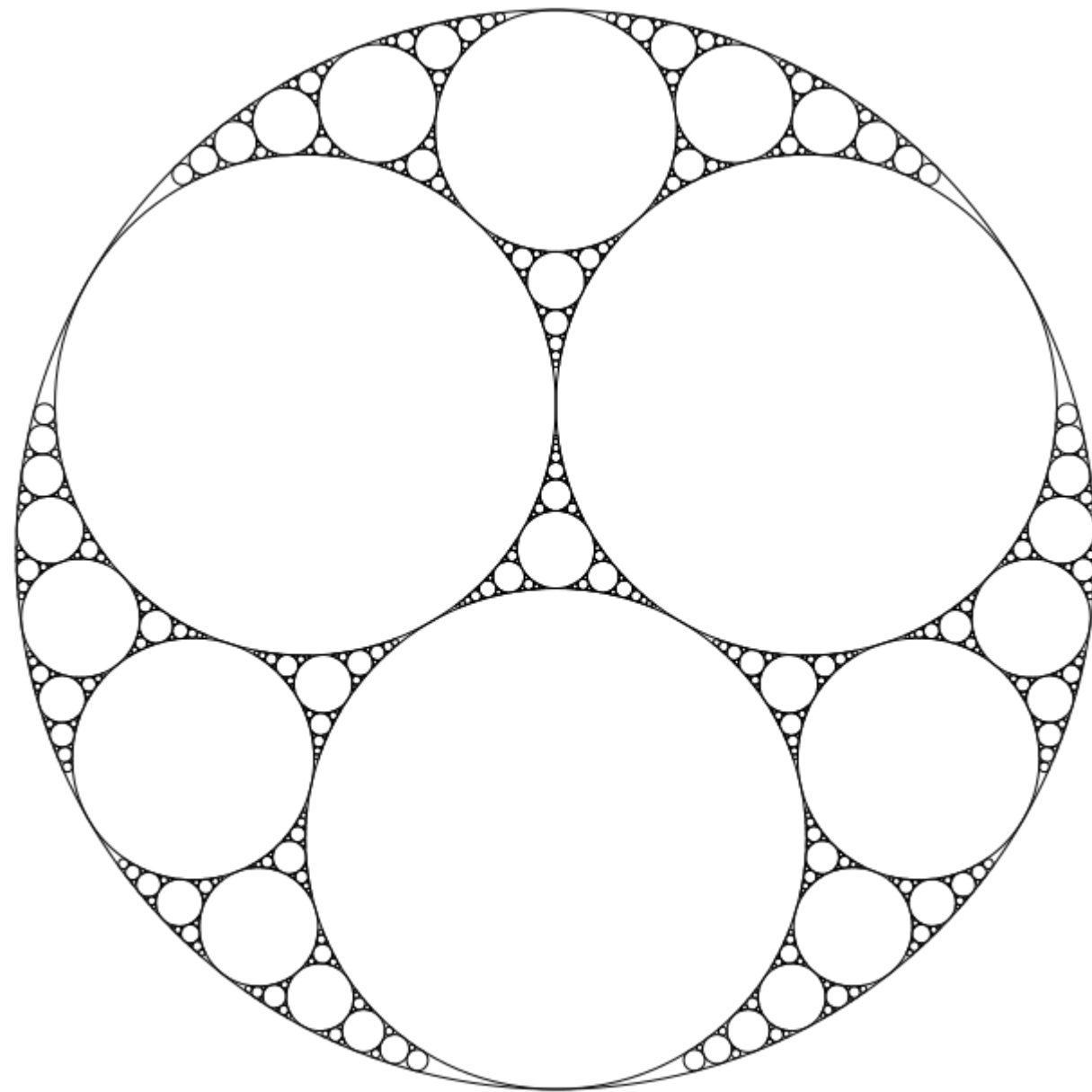
³*IMAG – Université de Montpellier*

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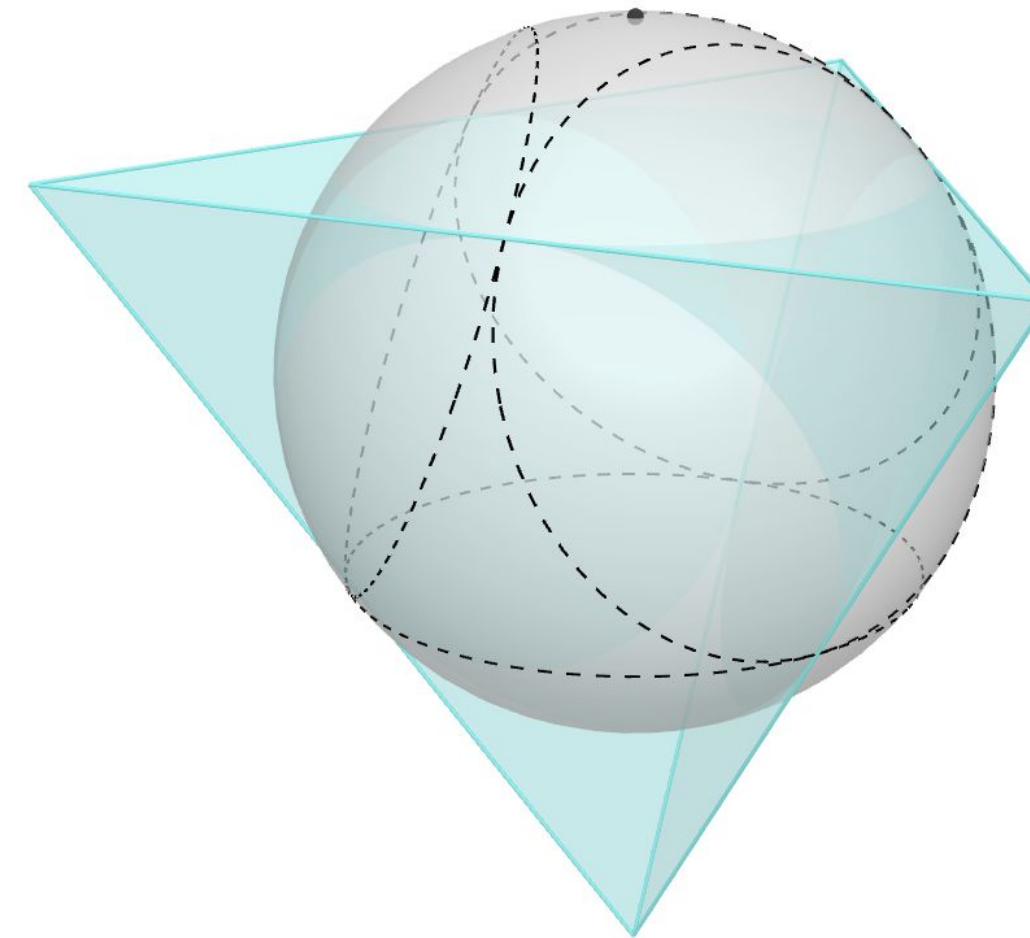
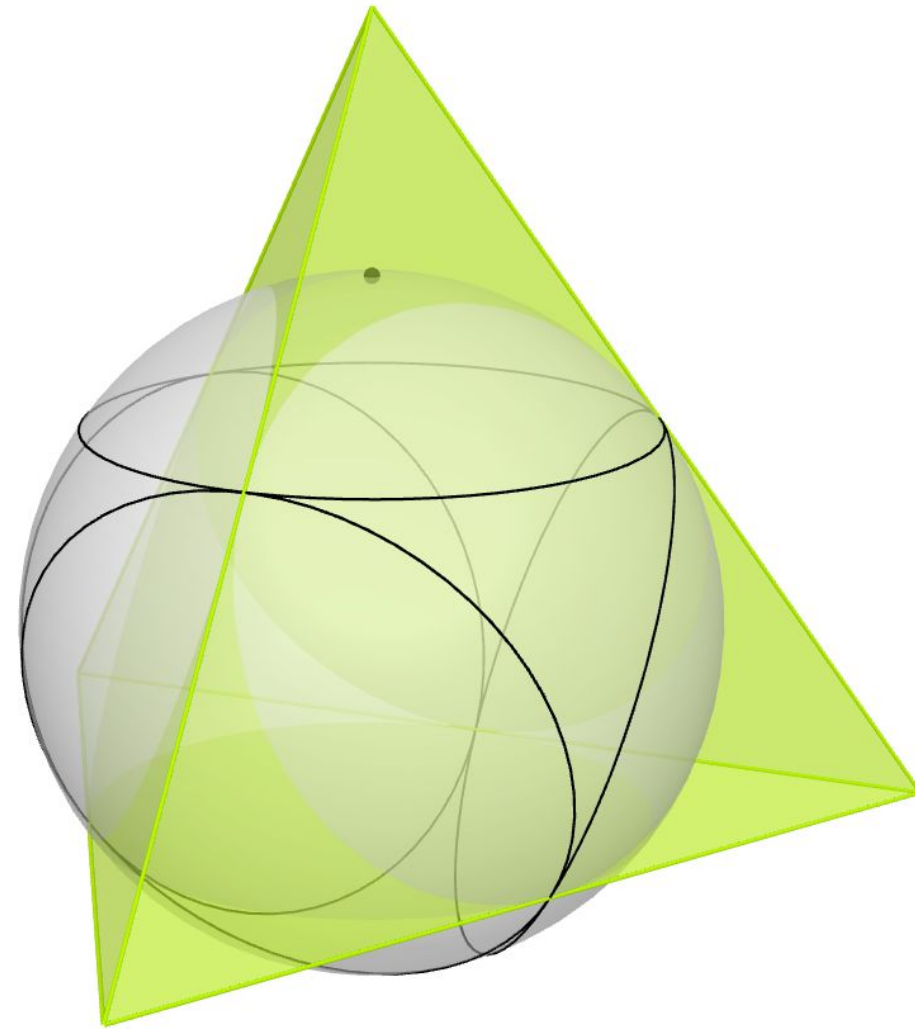


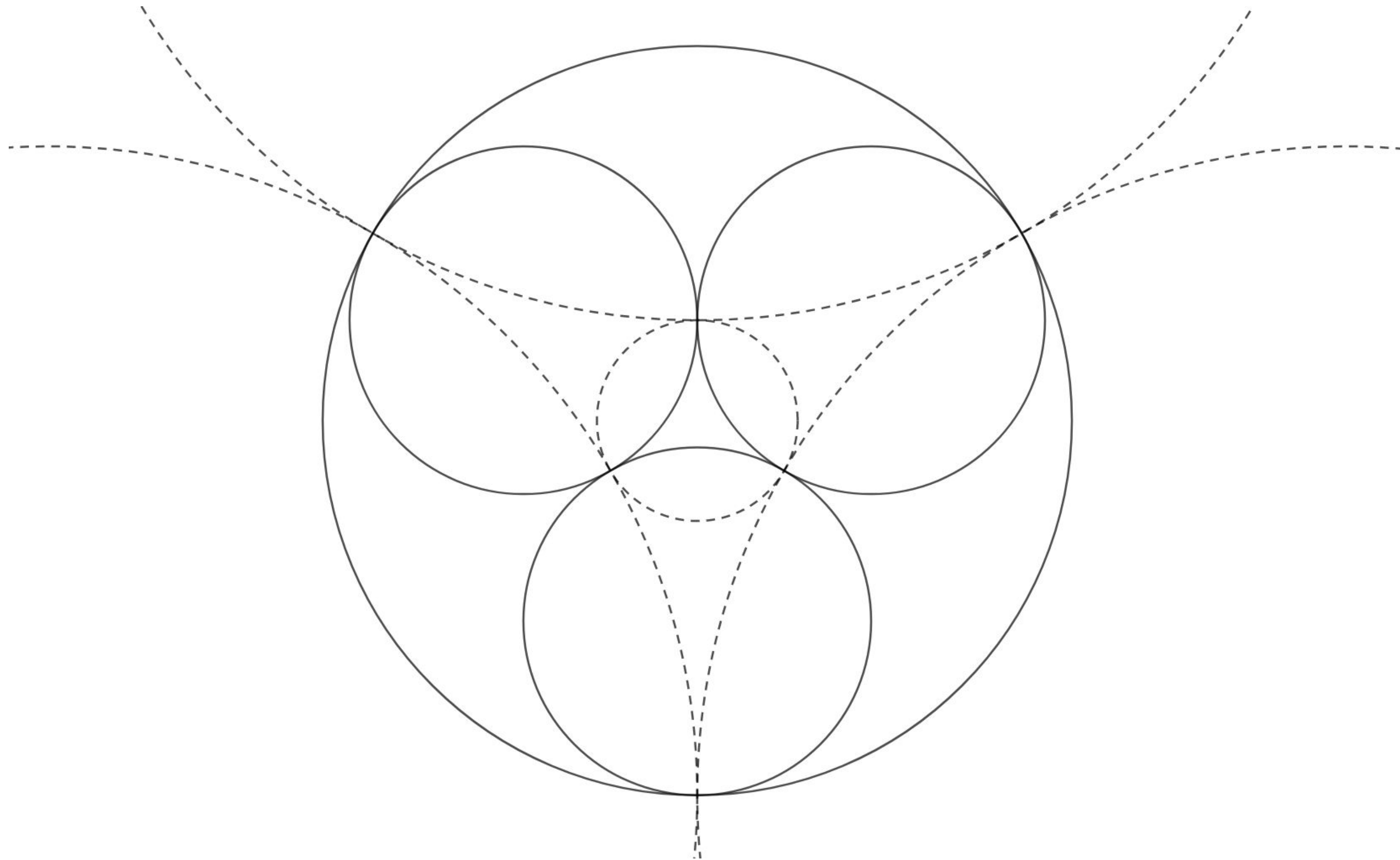


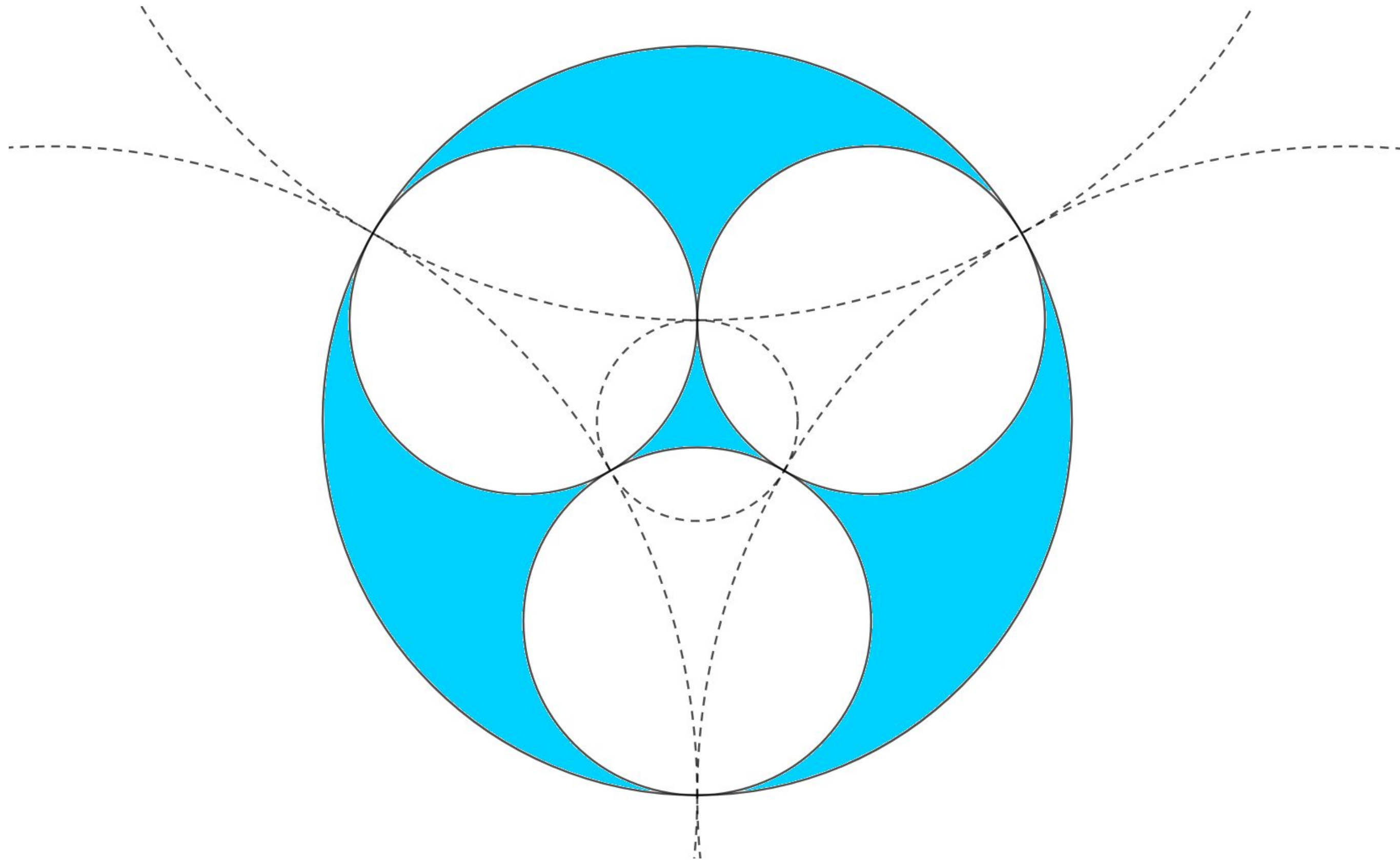
- Des empilements de sphères aux BC-IFS
- Extraction de règles de construction
- Conclusion

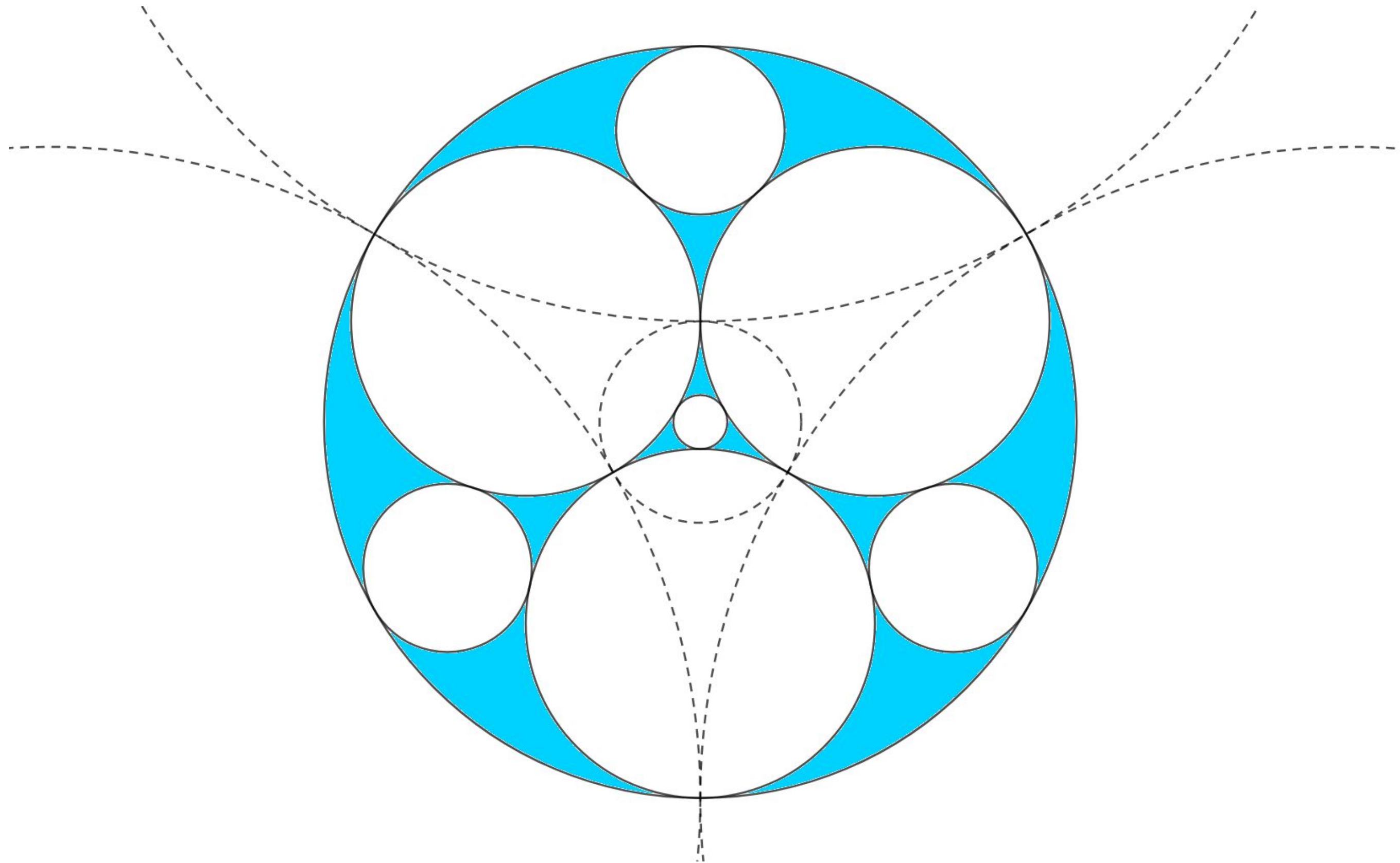
Des empilements de sphères aux BC-IFS

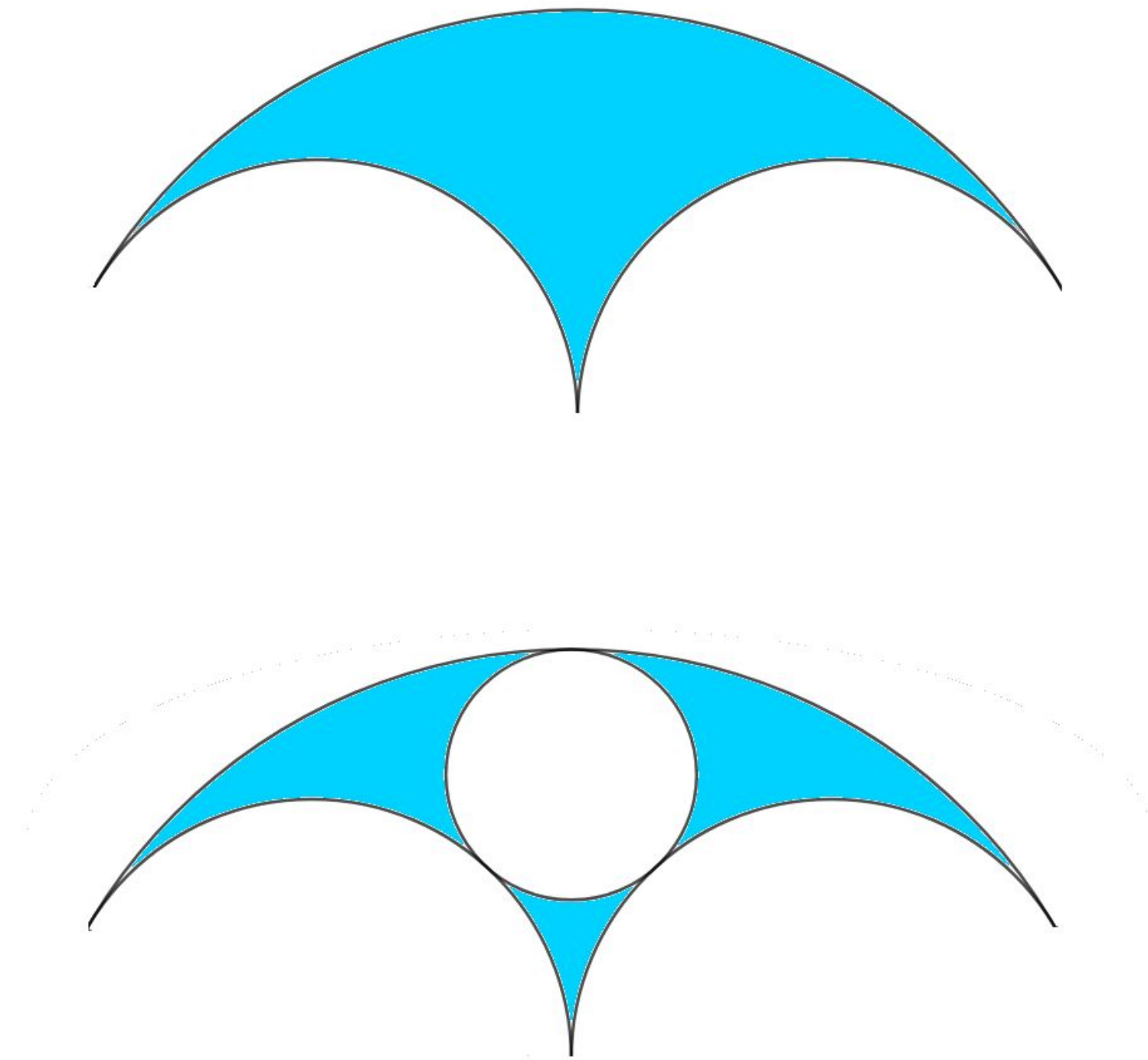


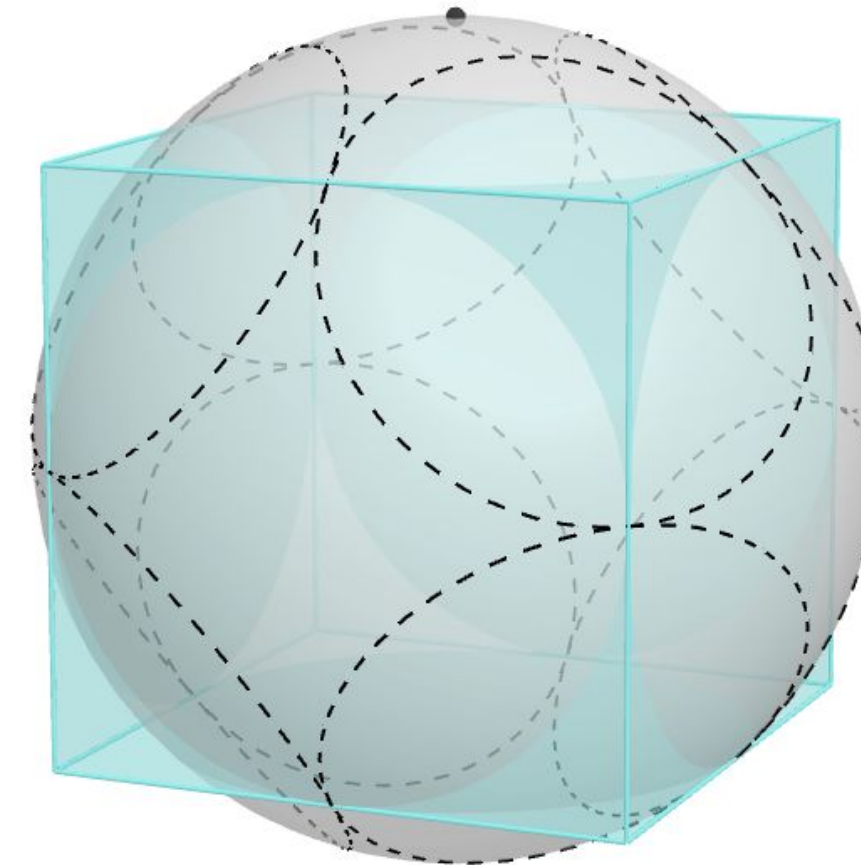
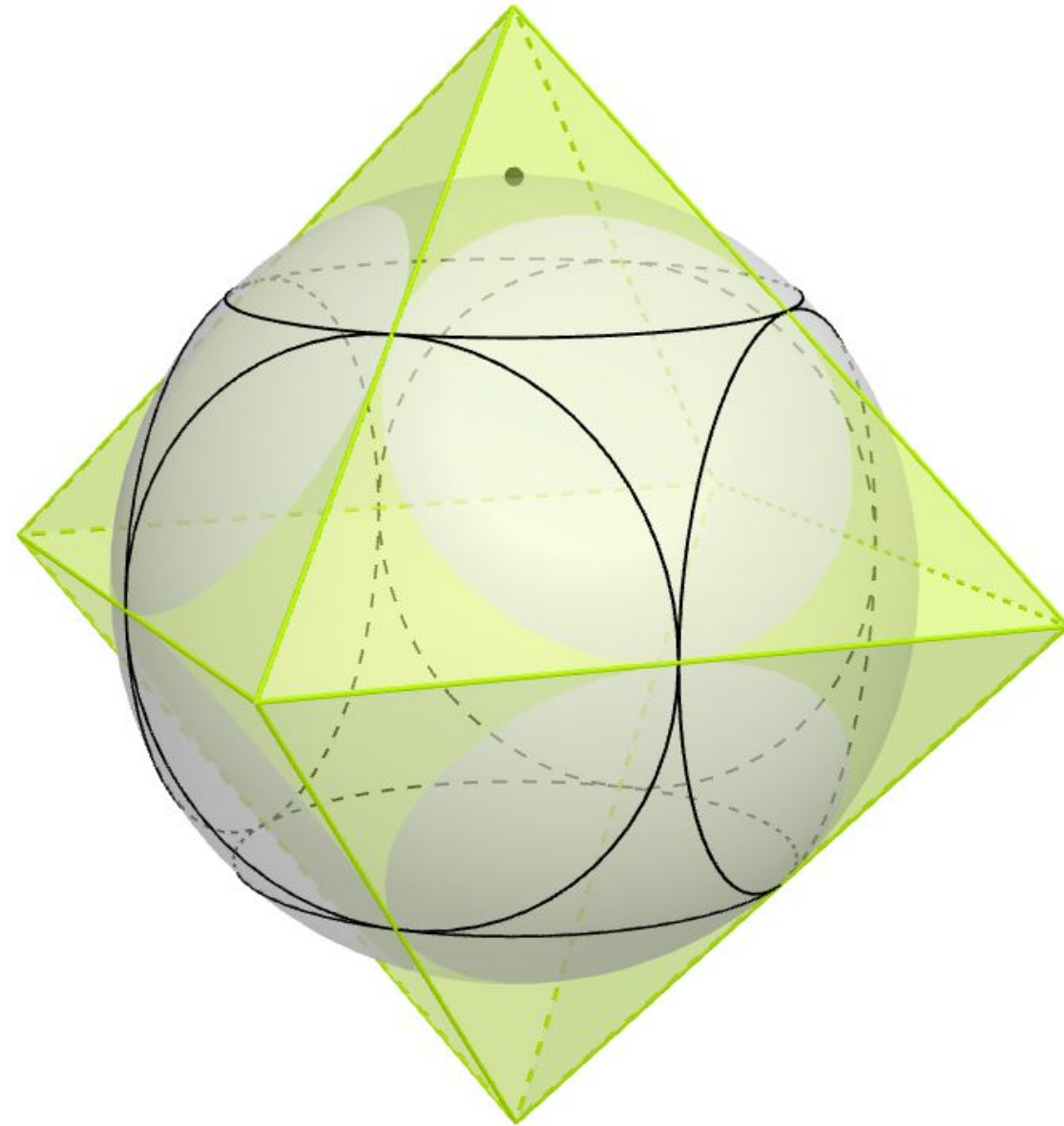


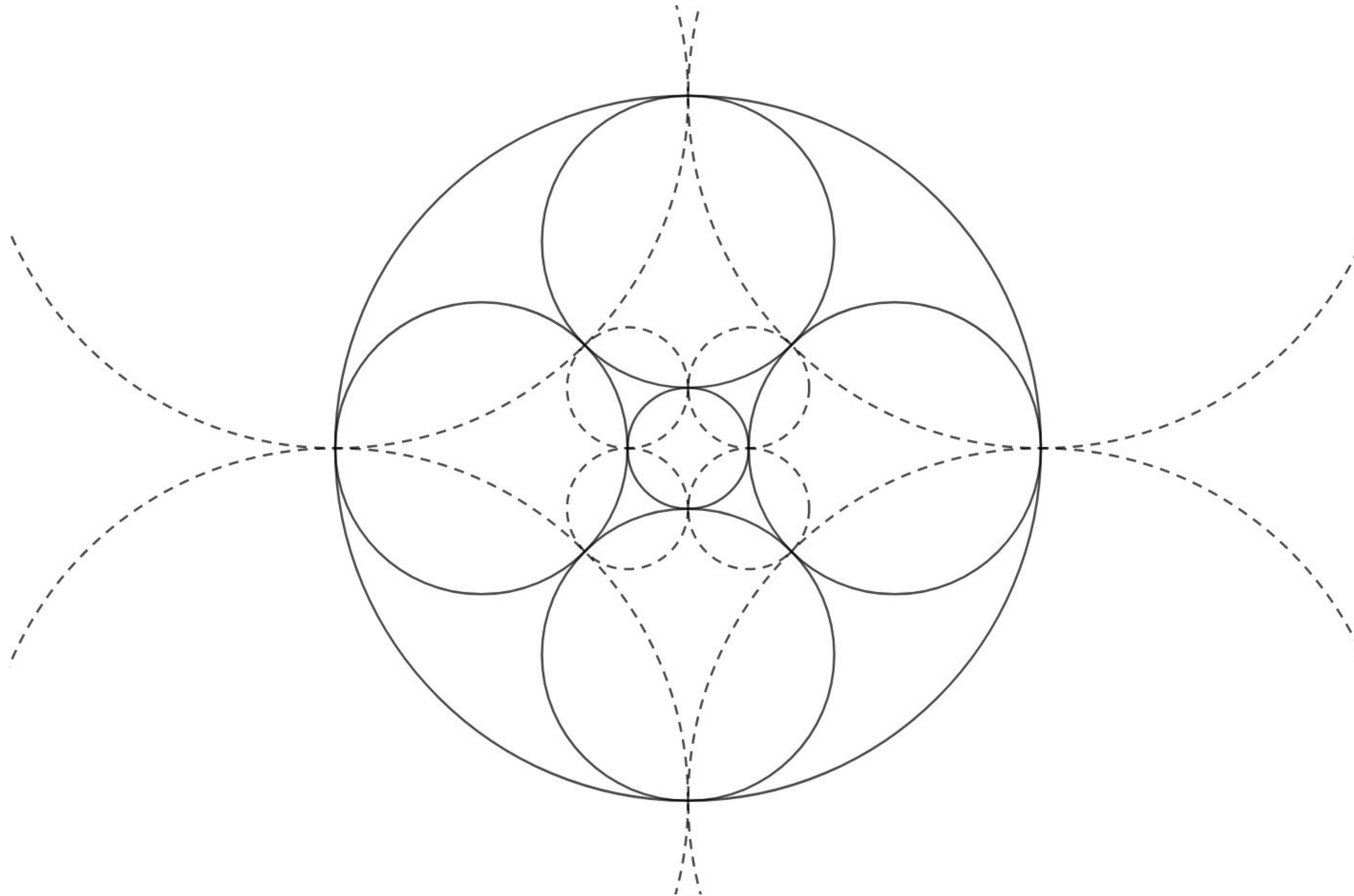


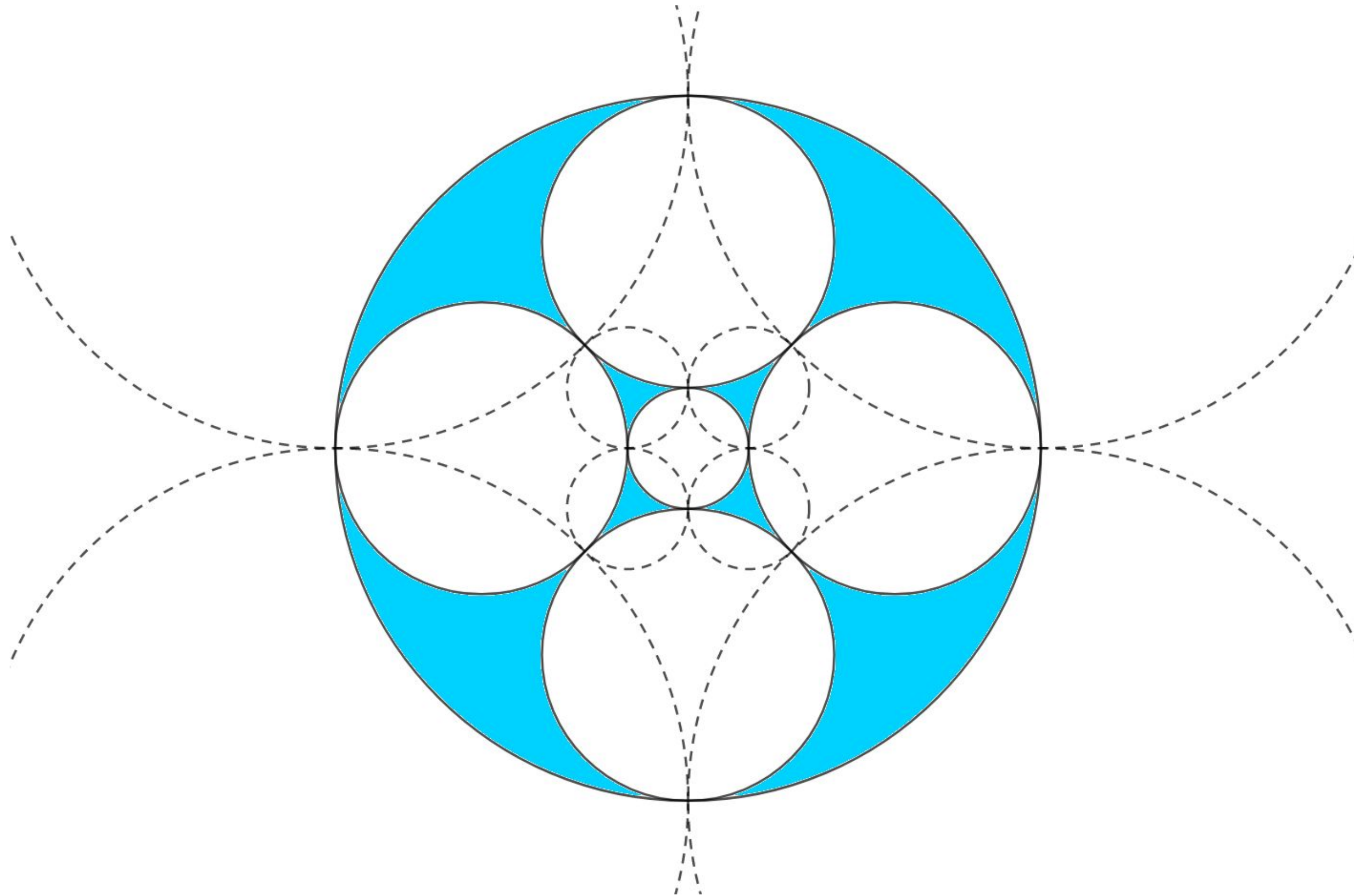


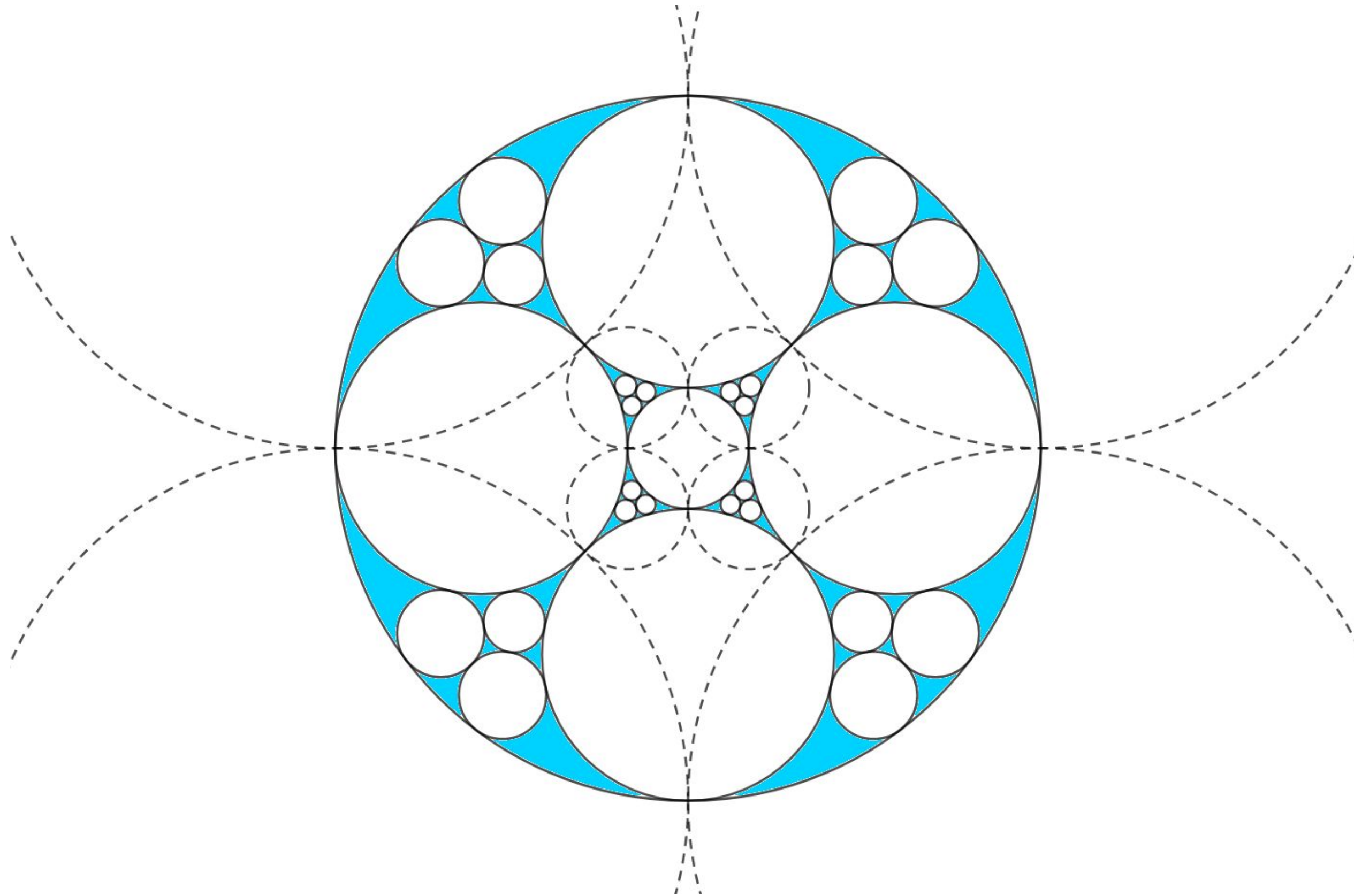


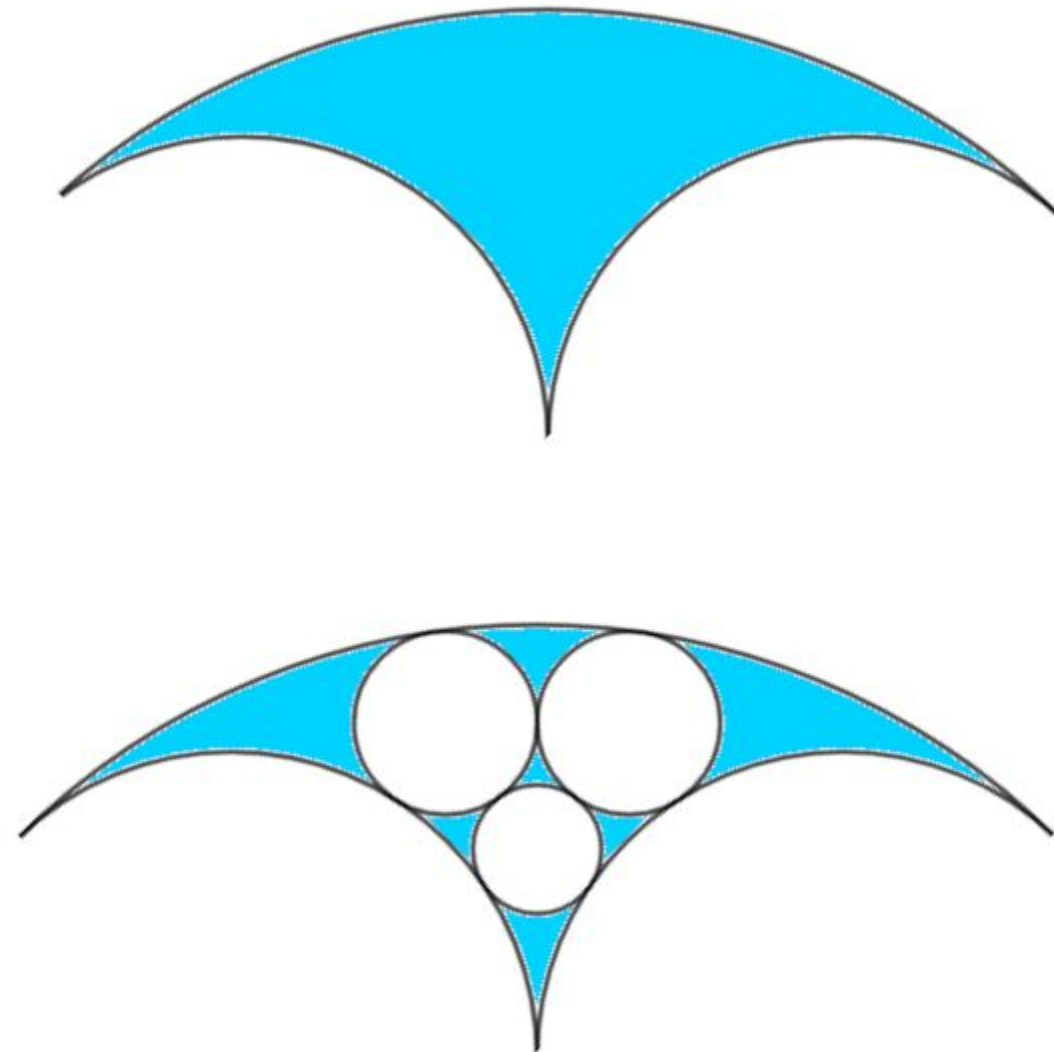












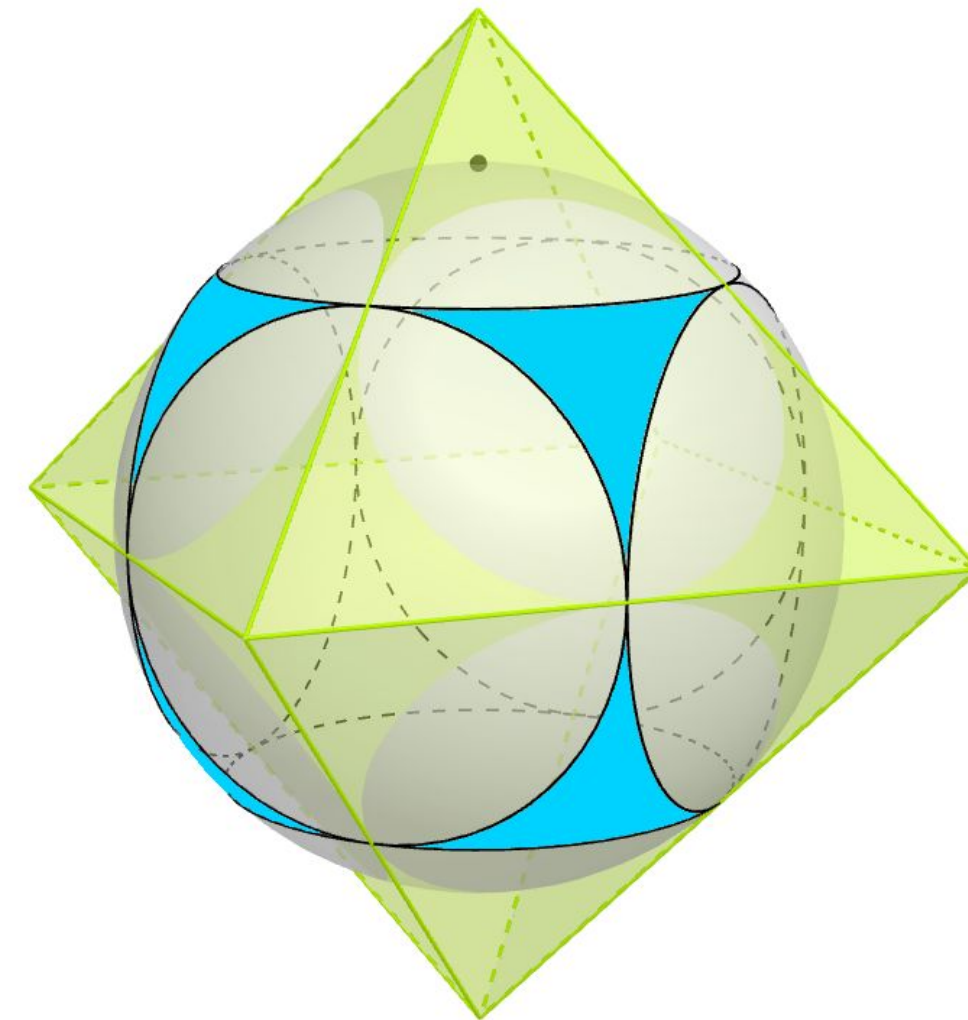
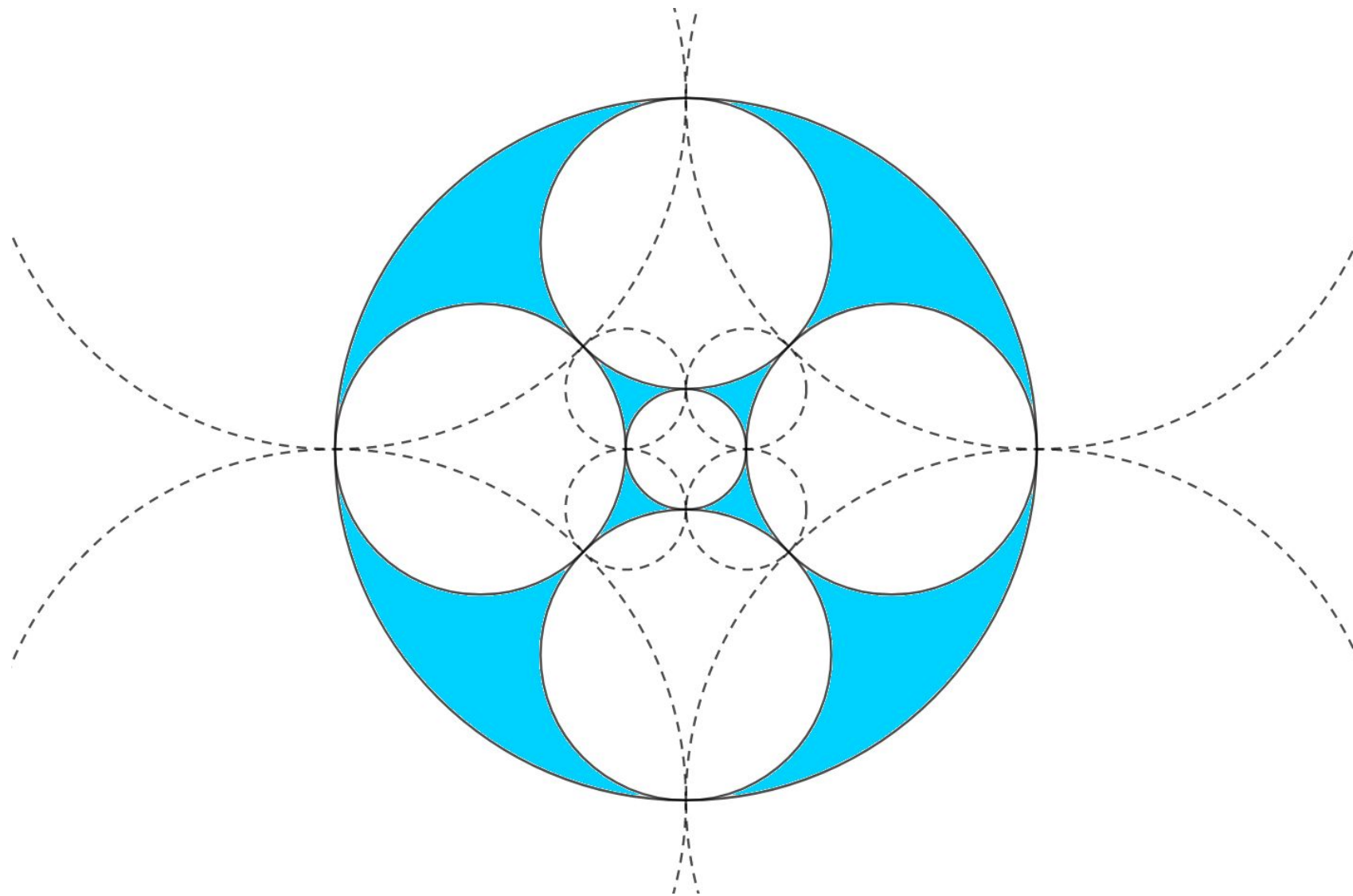
Extraction de règles de construction



- Combien de faces ?
- Comment sont-elles agencées ?
- Combien de bords par face ?
- Comment se subdivisent-elles ?
- Quelles contraintes d'adjacence ?
- Comment se subdivisent les bords ?
- Quelles contraintes d'incidence ?

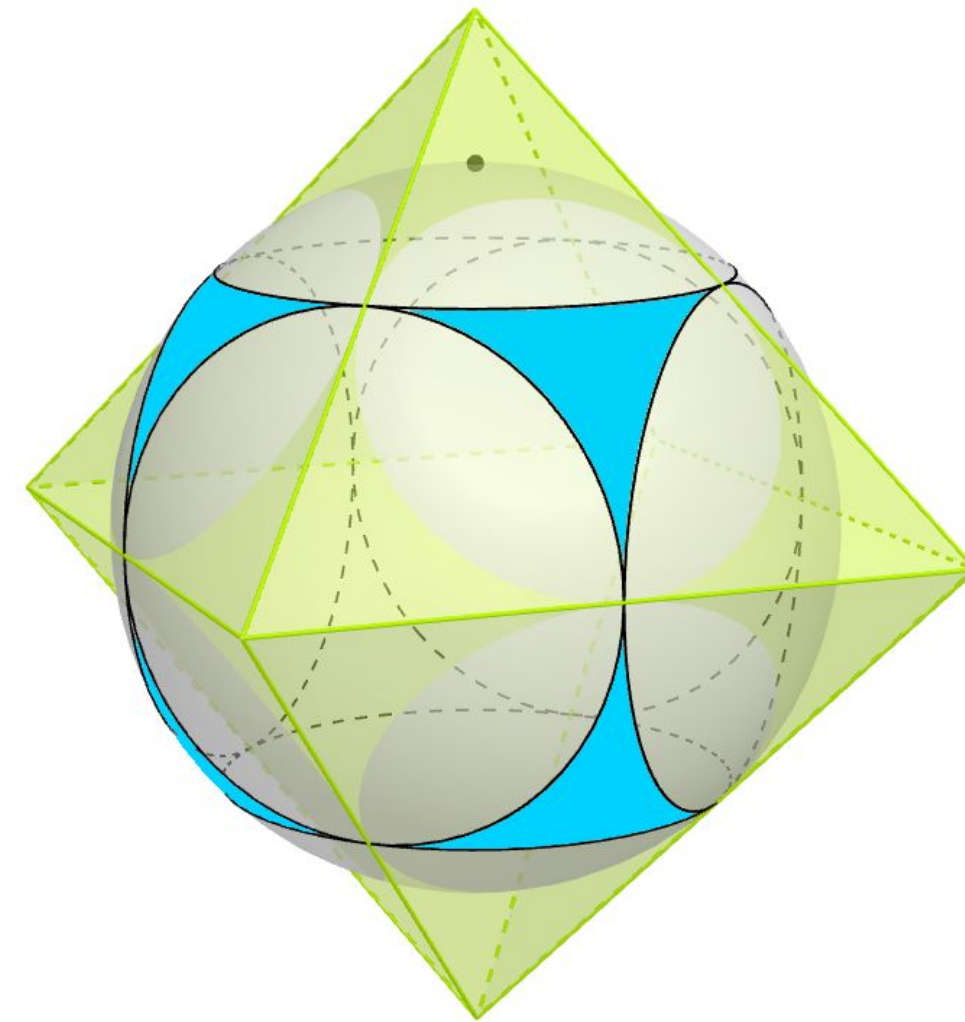
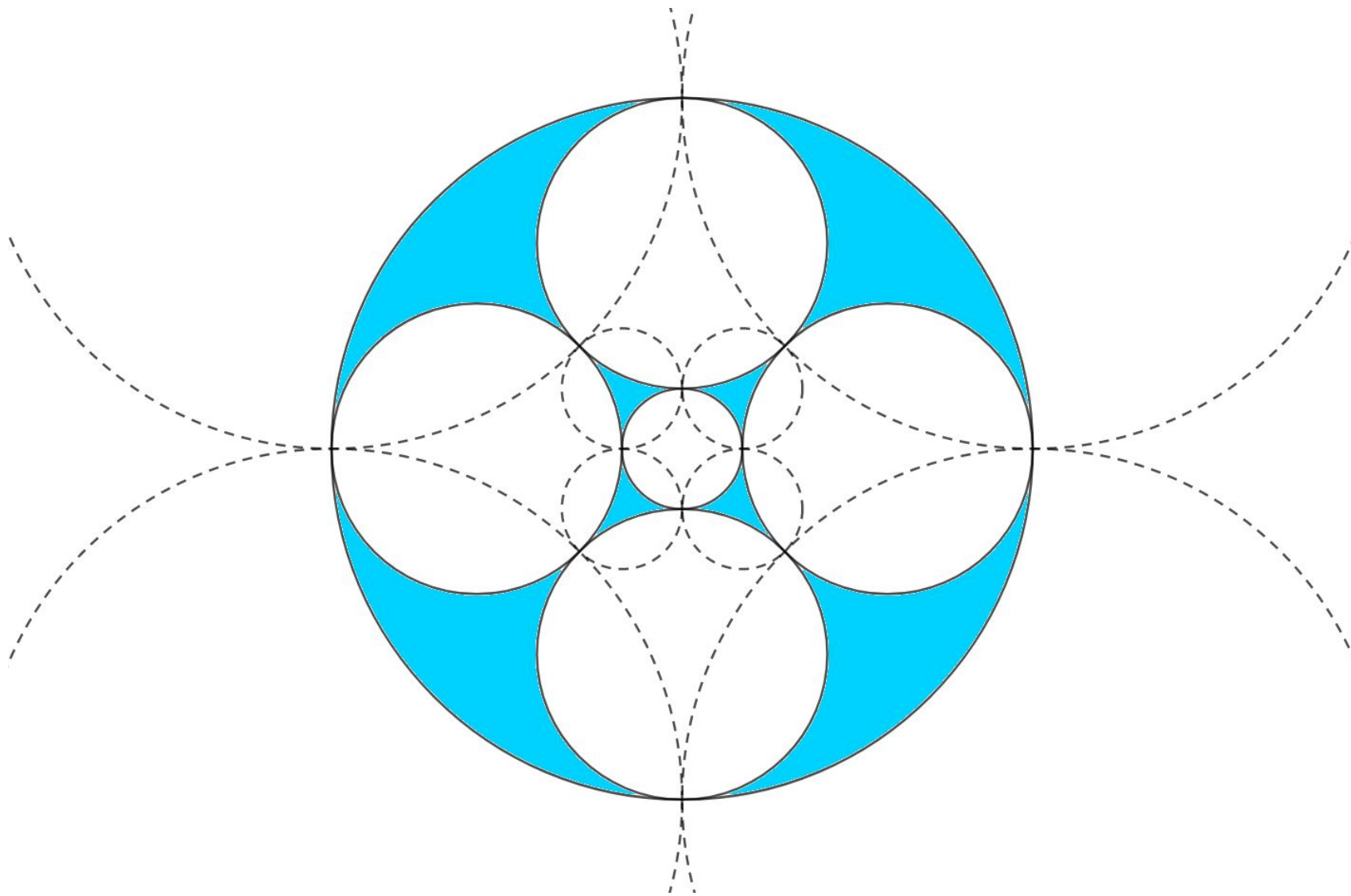
Combien de faces ?

Une face fractale par face du polyèdre



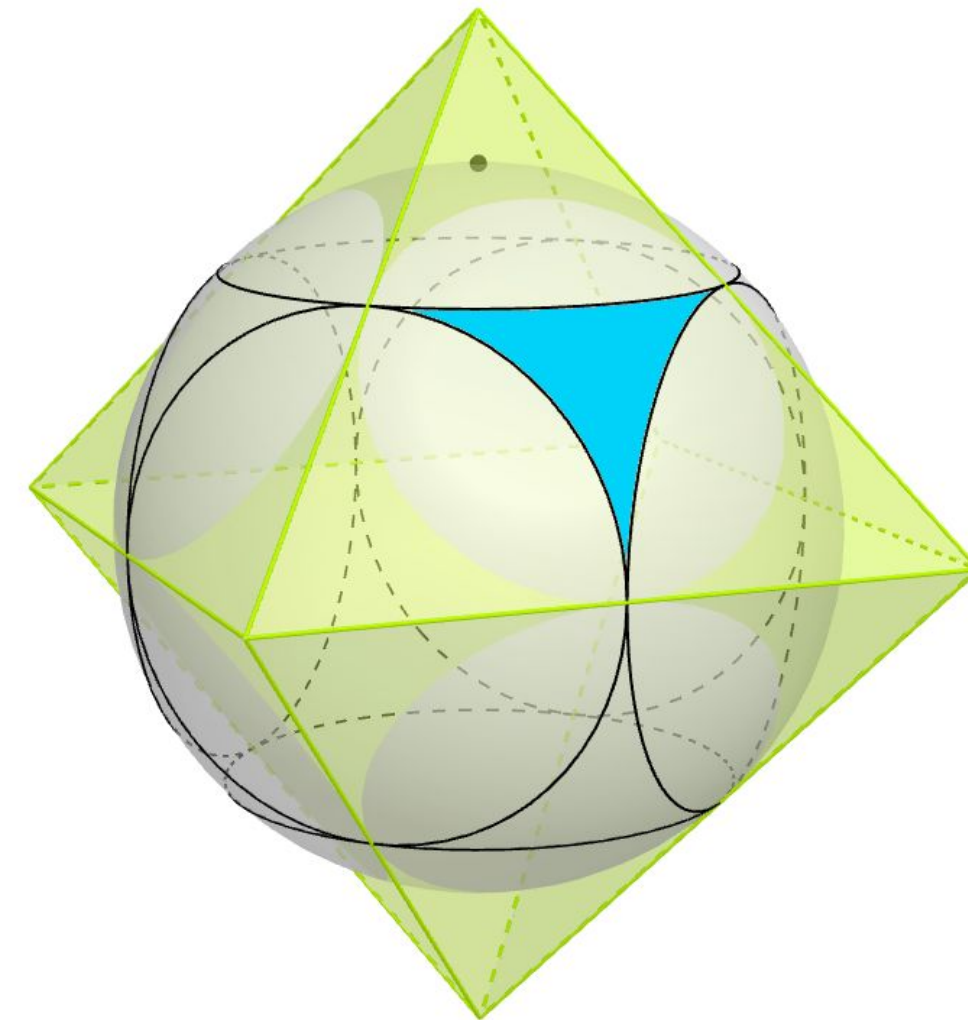
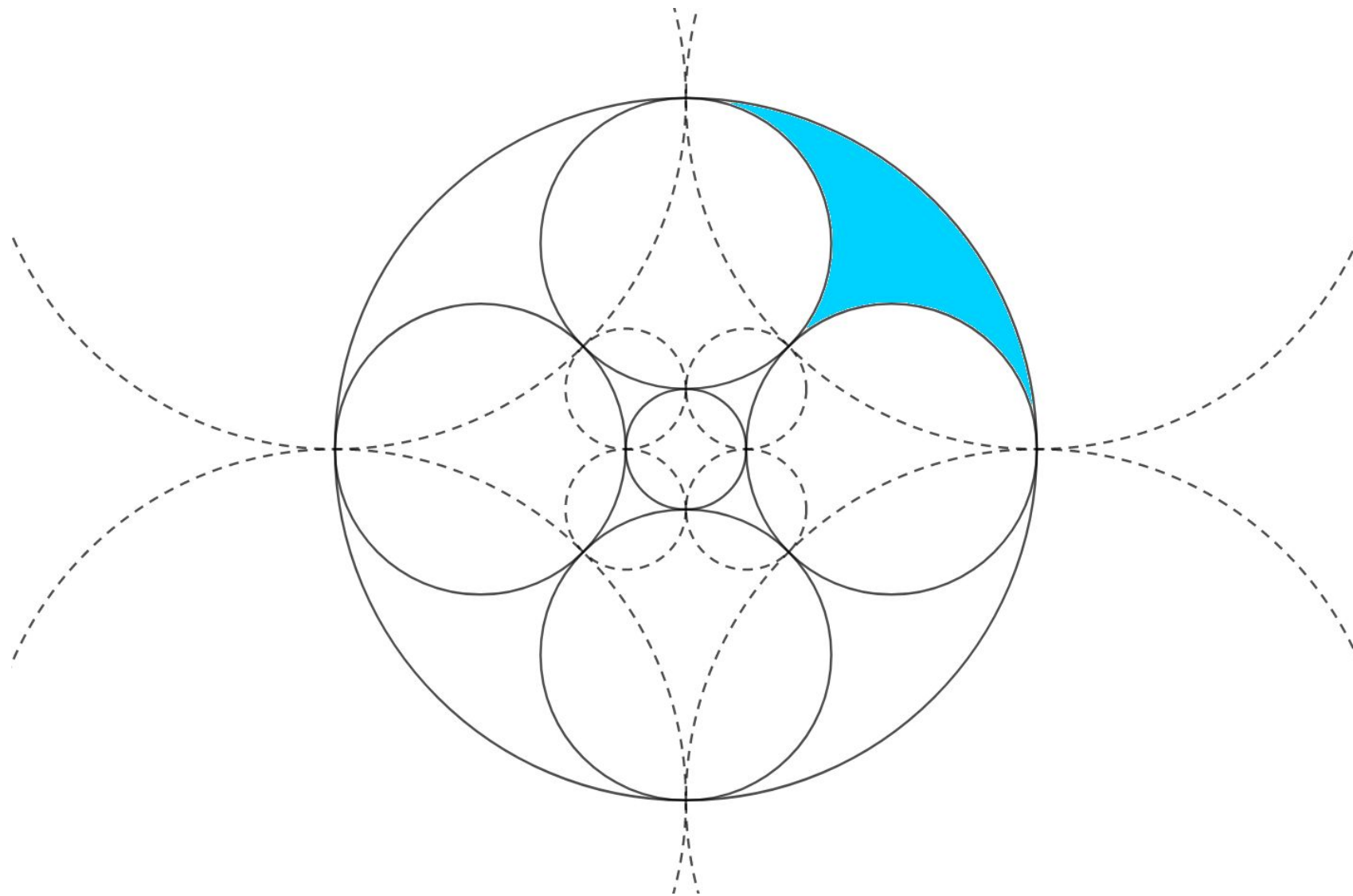
Comment sont-elles agencées ?

Adjacences des faces fractales = adjacences des faces du polyèdre



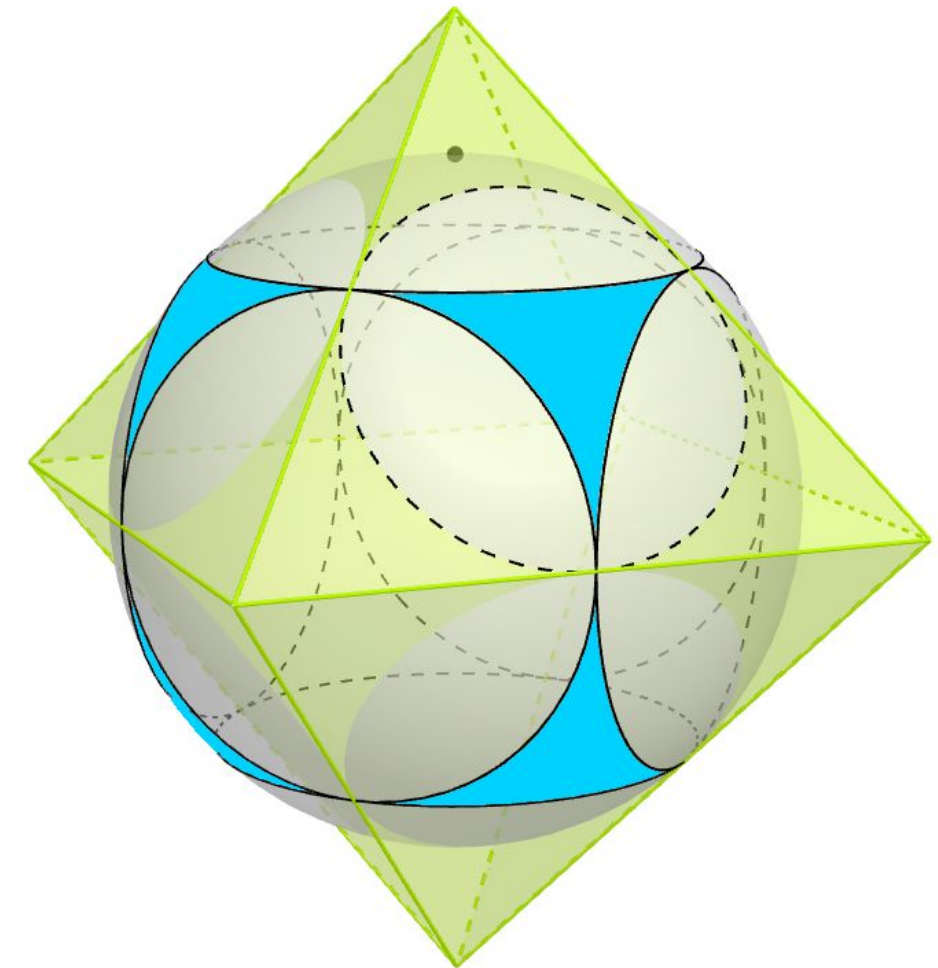
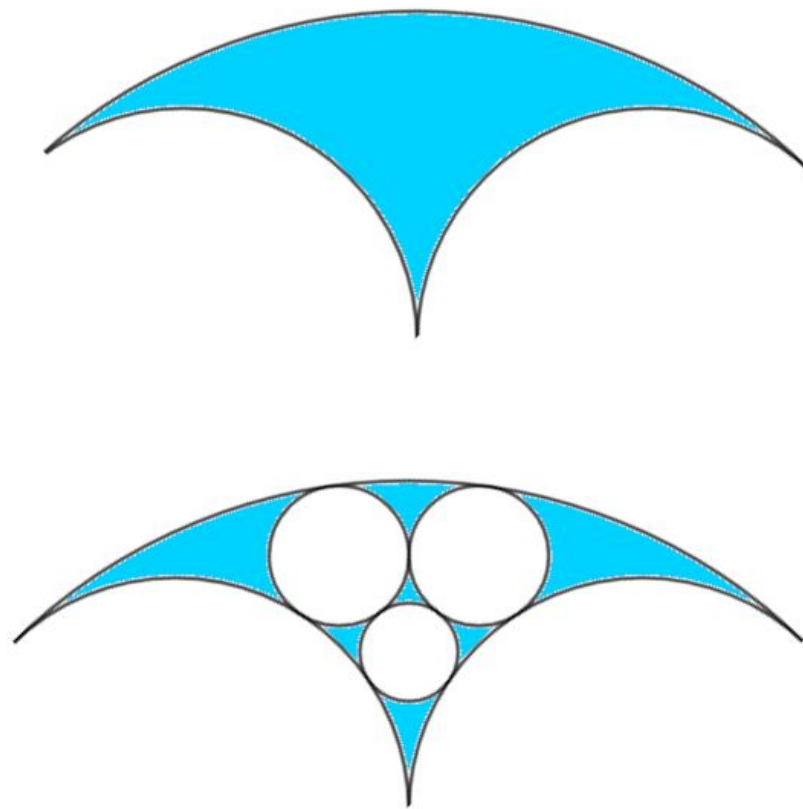
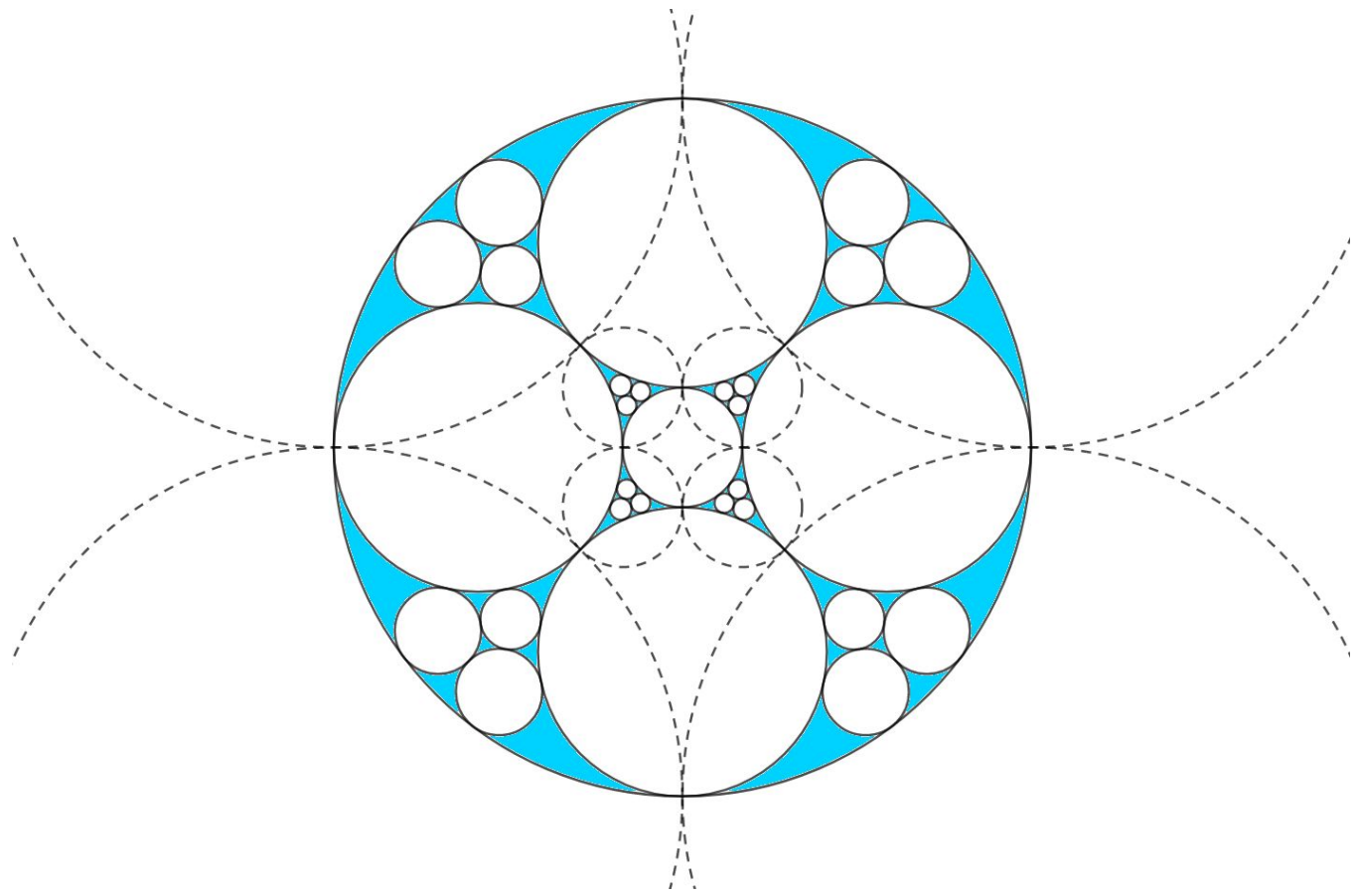
Combien de bords par face ?

Nombre de bords = nombre de sommets



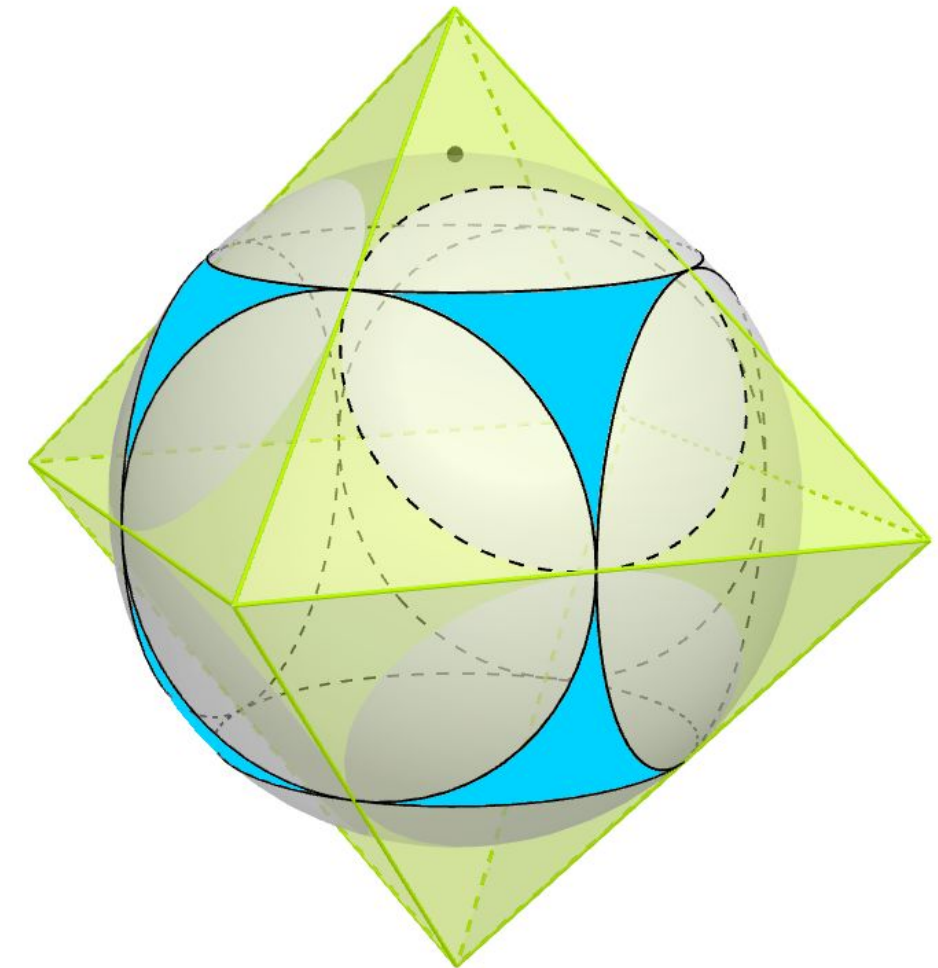
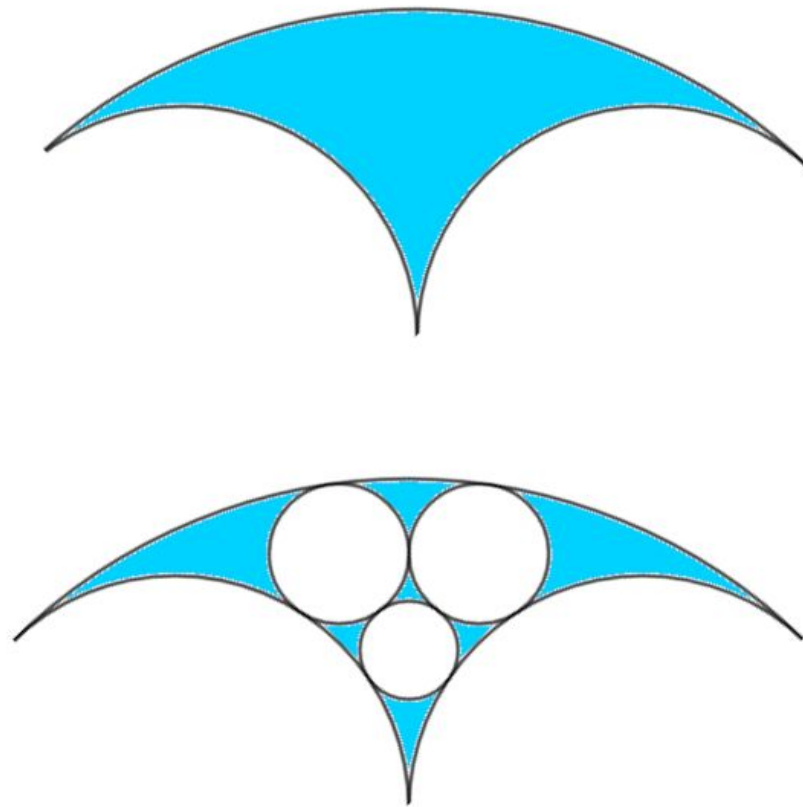
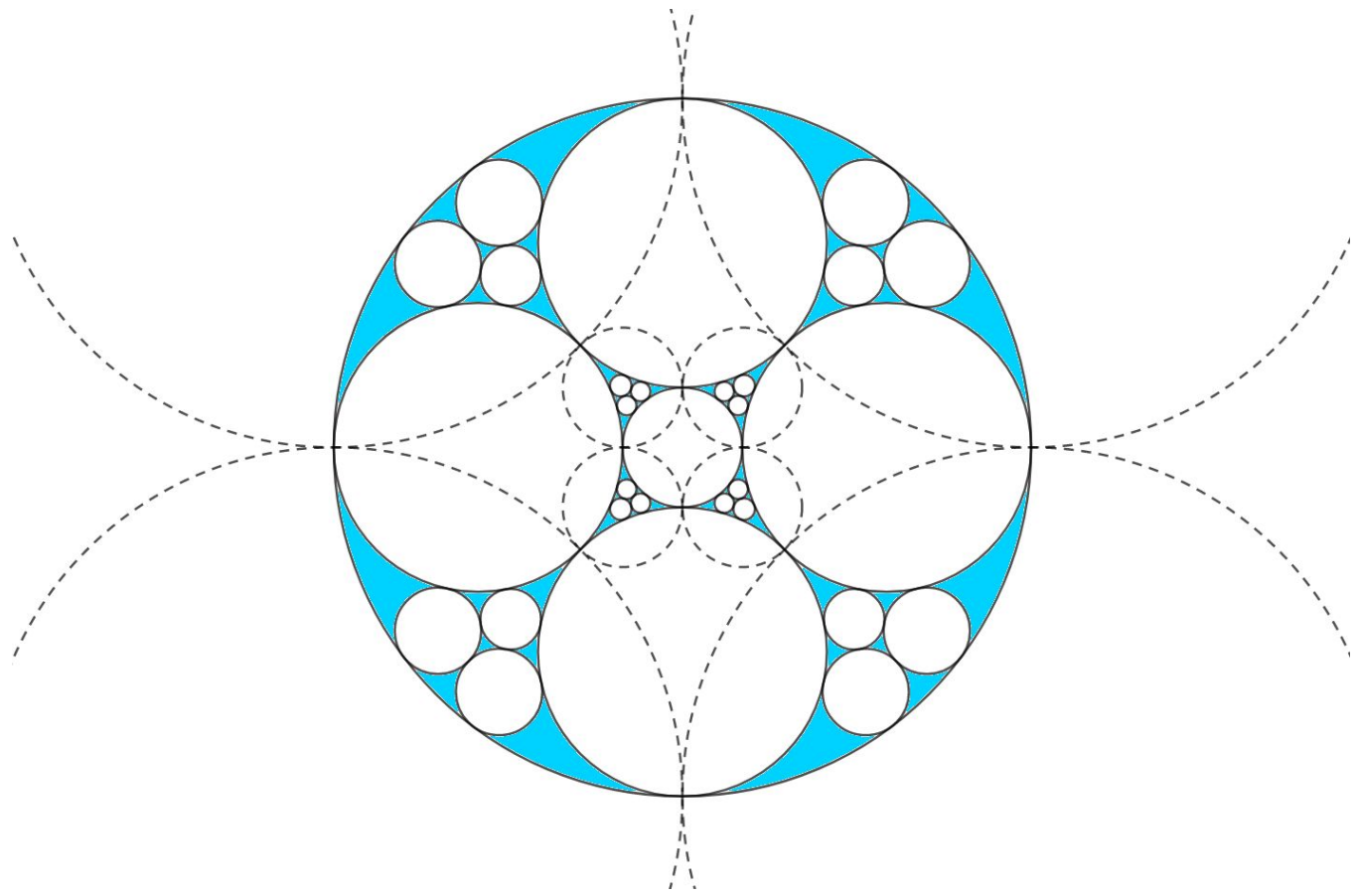
Comment se subdivisent-elles ?

Nombre de subdivisions = nombre de faces du polyèdre - 1



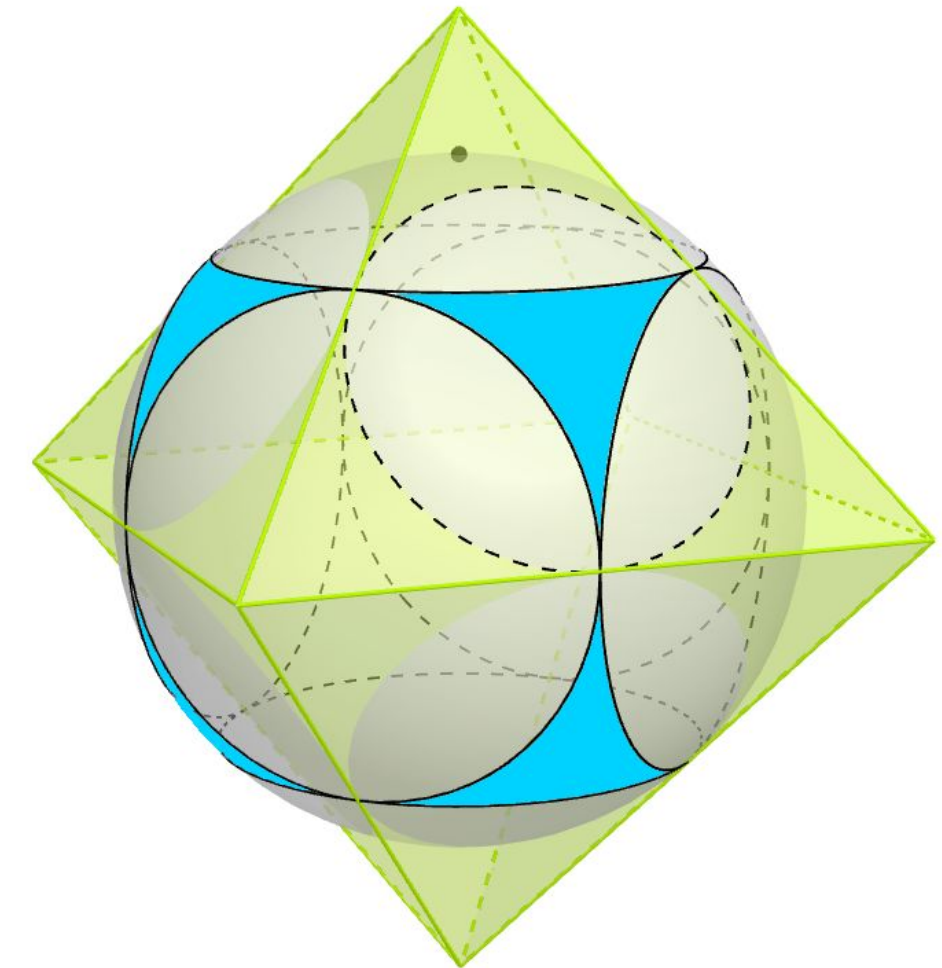
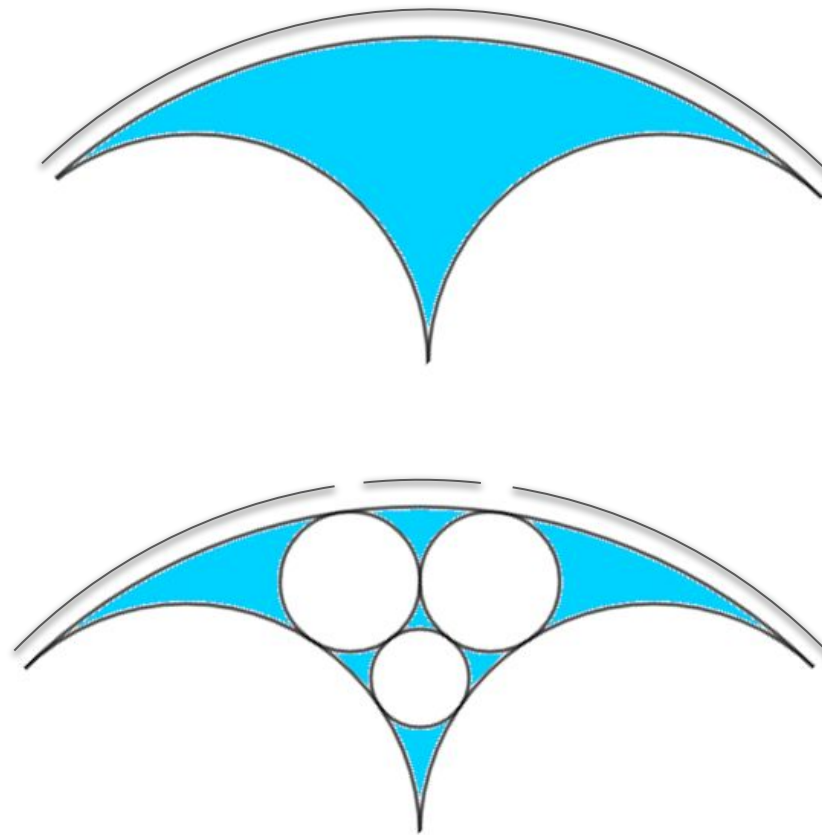
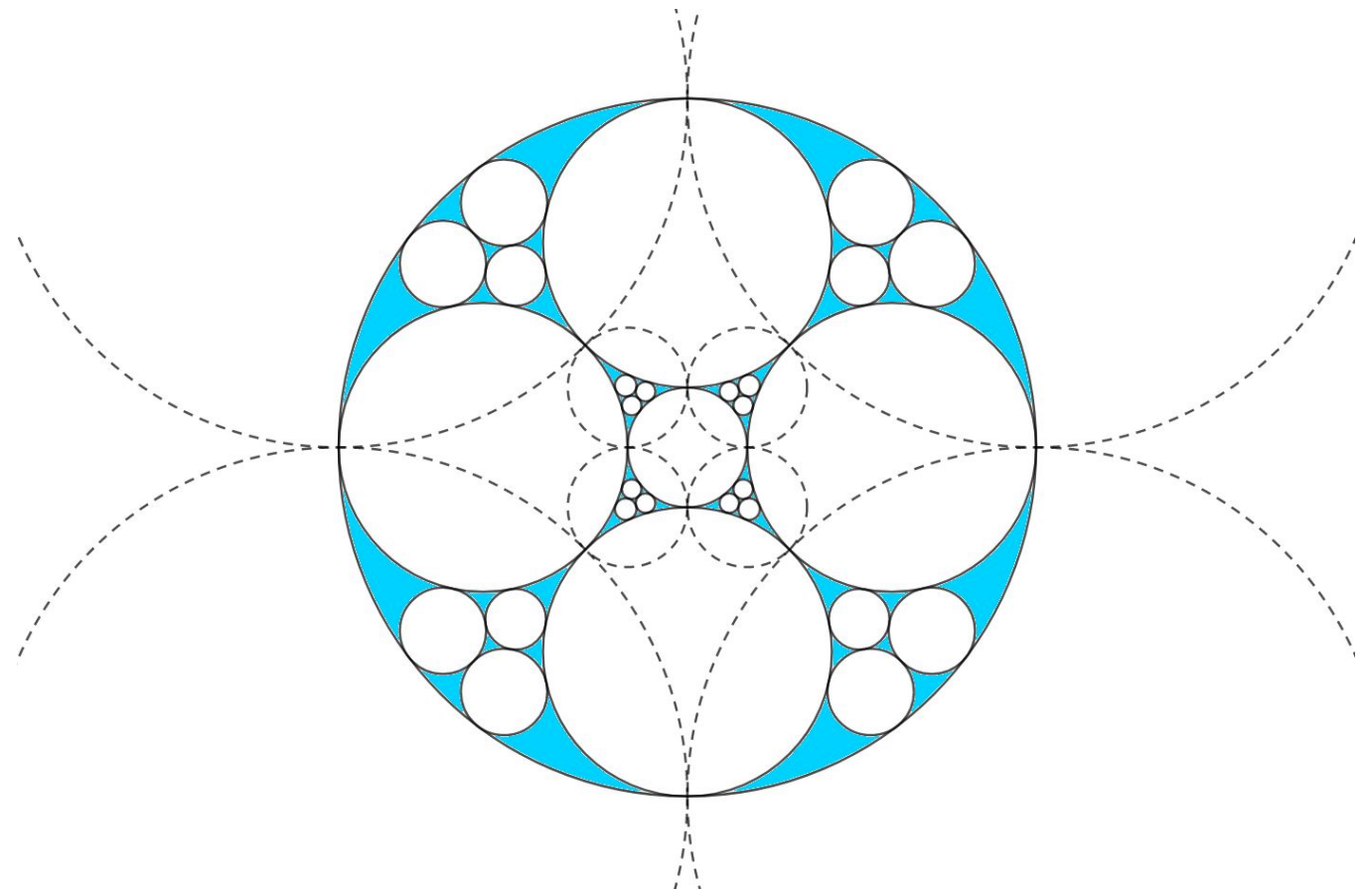
Quelles contraintes d'adjacence ?

Adjacences des subdivisions = adjacences des faces du polyèdre



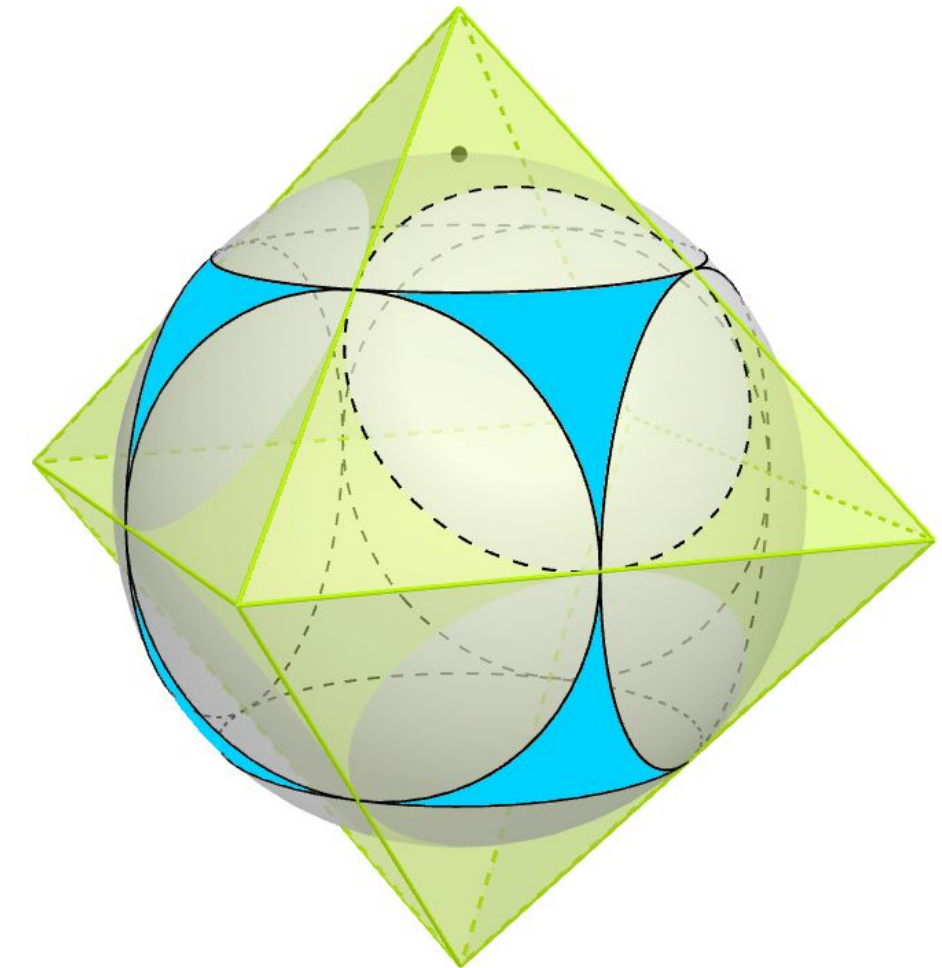
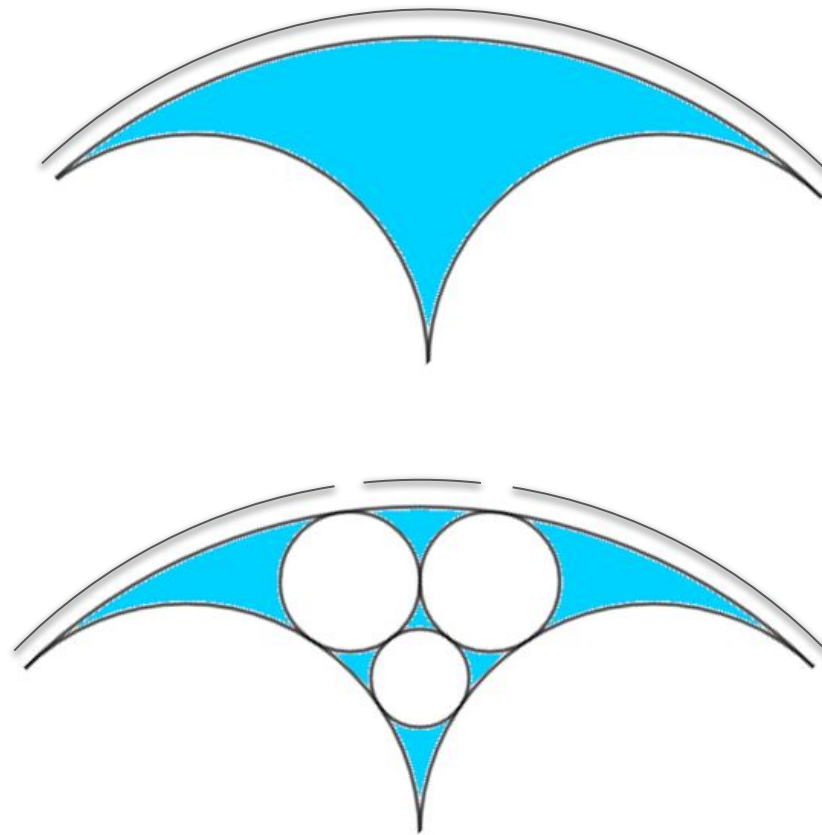
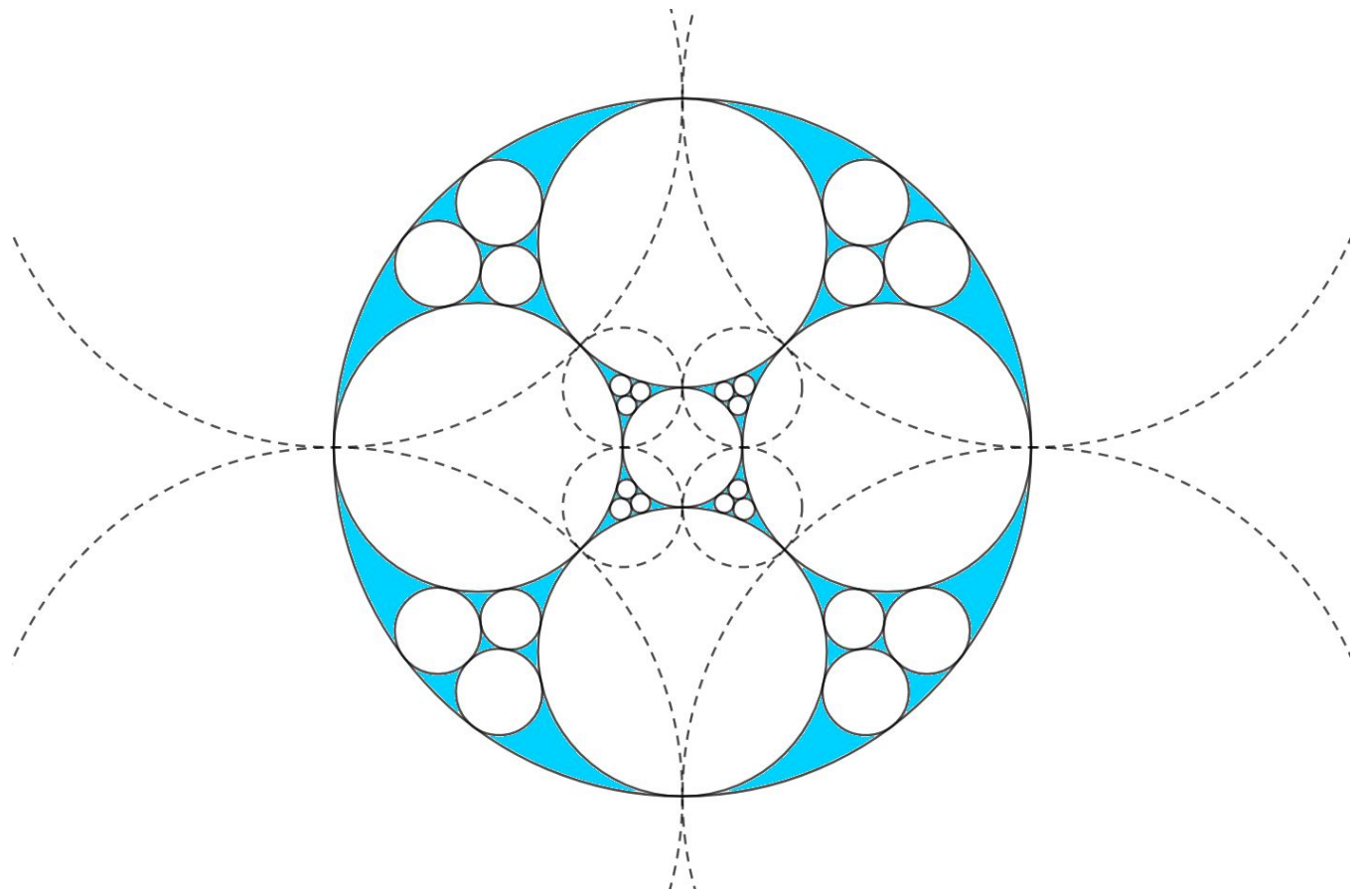
Comment se subdivisent les bords ?

Nombre de subdivisions des bords = degré du sommet associé - 1



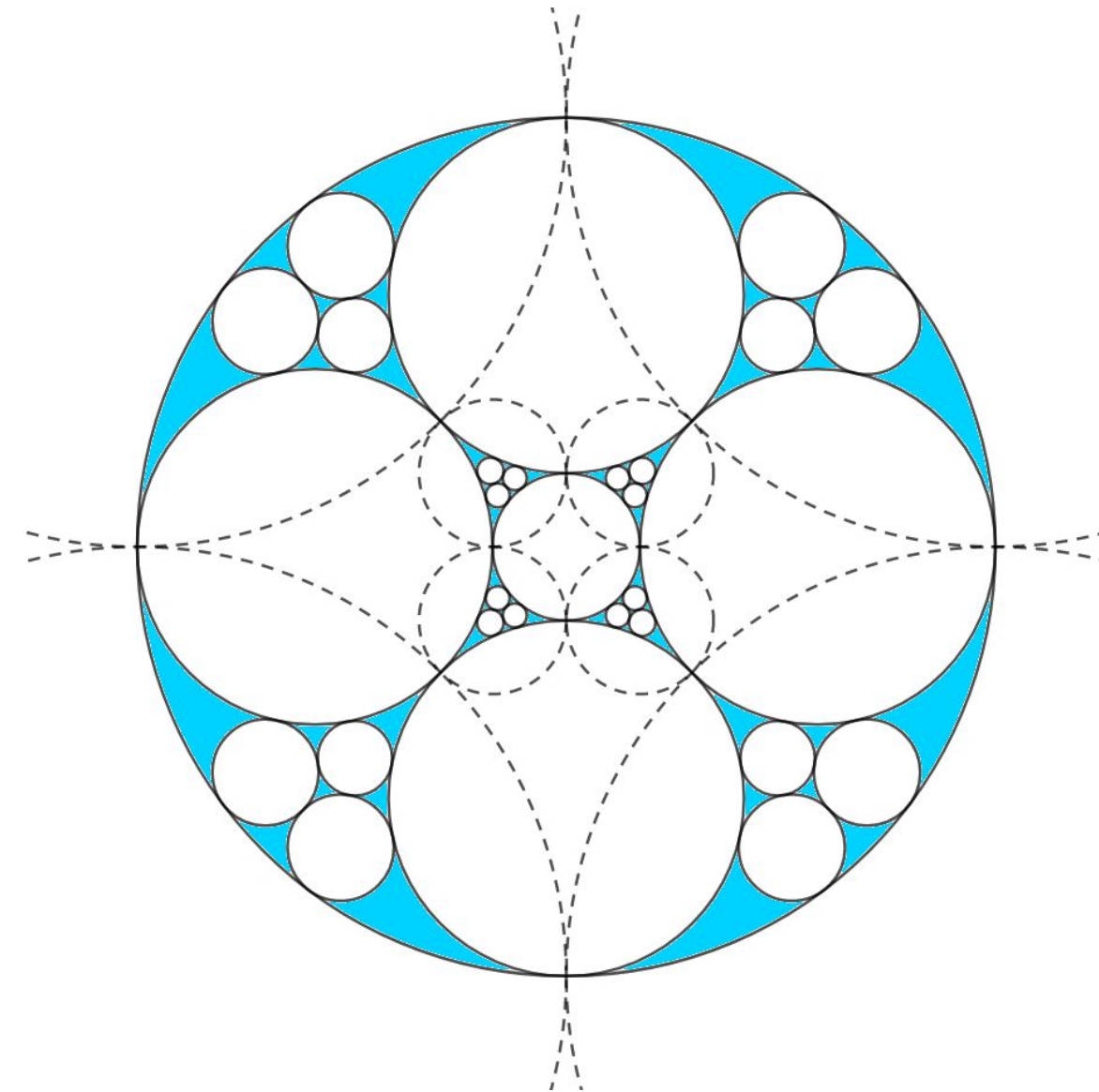
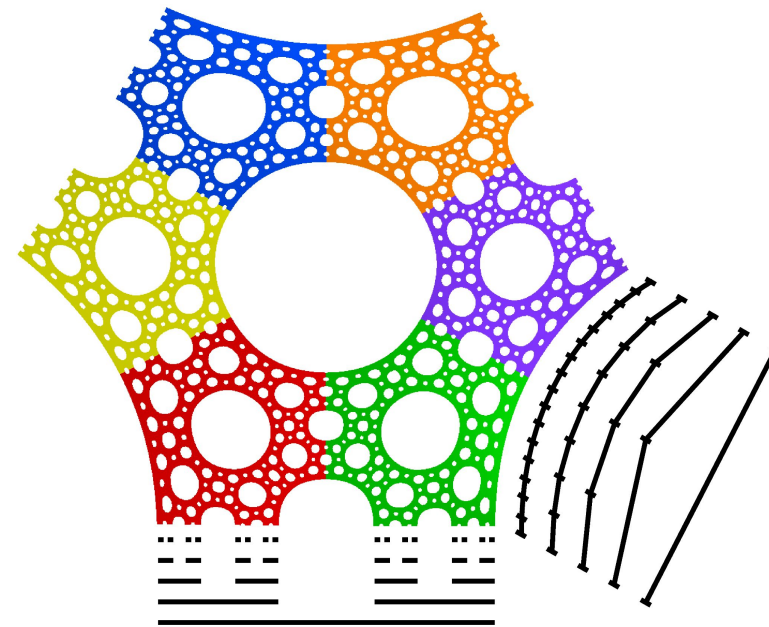
Quelles contraintes d'incidence ?

Incidences des subdivisions = mêmes sommets représentants



- Combien de faces ?
 - Une face fractale par face du polyèdre
- Comment sont-elles agencées ?
 - Adjacences des faces fractales = adjacences des faces du polyèdre
- Combien de bords par face ?
 - Nombre de bords = nombre de sommets
- Comment se subdivisent-elles ?
 - Nombre de subdivisions = nombre de faces du polyèdre - 1
- Quelles contraintes d'adjacence ?
 - Adjacences des subdivisions = adjacences des faces du polyèdre
- Comment se subdivisent les bords ?
 - Nombre de subdivisions des bords = degré du sommet associé - 1
- Quelles contraintes d'incidence ?
 - Incidences des subdivisions = mêmes sommets représentants

- Codage de la topologie des empilements de sphères en 2D
- Extraction de règles de construction
 - Construction automatique
- Travaux futurs
 - Rendre imprimable les empilements 2D
 - Polyèdres non réguliers
 - Et en 3D ?
 - Problème inverse ?



Conclusion



- Automatisation de conception de structures lacunaires
 - Algorithme de subdivision des faces 2D
 - Empilements polytopaux de sphères
- Travaux futurs
 - Passage en 3D
 - Améliorations des structures en 2D
 - Optimisation topologique

Merci de votre attention



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